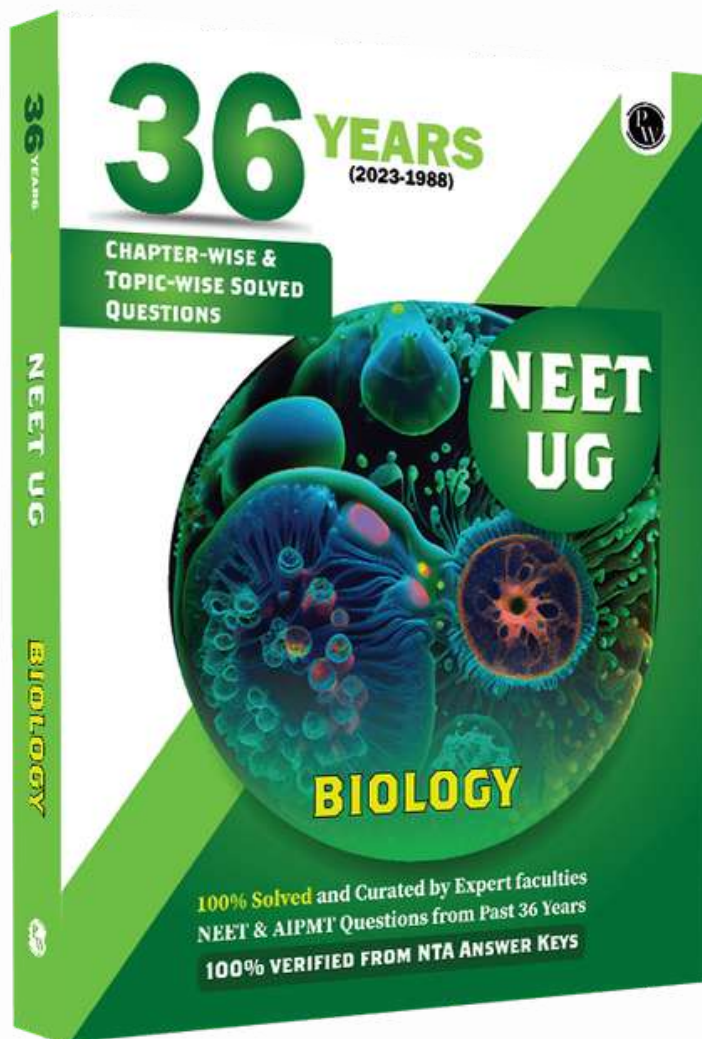




BIOLOGY 36 YEARS CHAPTERWISE & TOPIC WISE SOLVED QUESTIONS FOR NEET

**JOIN OUR TELEGRAM CHANNEL FOR ALL
COMPLETE PDFS OF PW 36 YEARS PYQS**

https://t.me/+yf6qyg_uf_NhNmJl

https://t.me/+yf6qyg_uf_NhNmJl

https://t.me/+yf6qyg_uf_NhNmJl

Click on the Above link to Join Our Channel

Class - XII

CHAPTER 1 Electric Charges and Fields

Coulomb's Law

1. The acceleration of an electron due to the mutual attraction between the electron and a proton when they are $1.6 \text{ \AA}$ apart is.

(2020-Covid)

$$(m_e \approx 9 \times 10^{-31} \text{ kg}, e = 1.6 \times 10^{-19} \text{ C})$$

$$\left(\text{Take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2 \text{C}^{-2}\right)$$

- a. 10^{23} m/s^2 b. 10^{22} m/s^2
c. 10^{25} m/s^2 d. 10^{24} m/s^2
2. Suppose the charge of a proton and an electron differ slightly. One of them is $-e$, the other is $(e + \Delta e)$. If the net of electrostatic force and gravitational force between two hydrogen atoms placed at a distance d (much greater than atomic size) apart is zero, then Δe is of the order of [Given mass of hydrogen $m_h = 1.67 \times 10^{-27} \text{ kg}$]

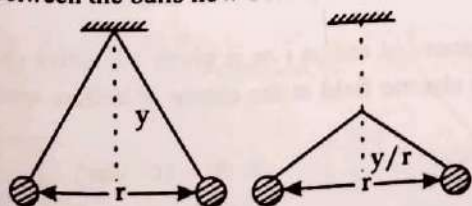
(2017-Delhi)

- a. 10^{-23} C b. 10^{-37} C
c. 10^{-47} C d. 10^{-20} C
3. Two identical charged spheres suspended from a common point by two massless strings of lengths ℓ , are initially at a distance d ($d \ll \ell$) apart because of their mutual repulsion. The charges begin to leak from both the spheres at a constant rate. As a result, the spheres approach each other with a velocity V . Then V varies as a function of the distance x between the spheres, as:

(2016 - I)

- a. $V \propto x^{\frac{1}{2}}$ b. $V \propto x$
c. $V \propto x^{-\frac{1}{2}}$ d. $V \propto x^{-1}$
4. Two pith balls carrying equal charges are suspended from a common point by strings of equal length, the equilibrium separation between them is r . Now the strings are rigidly clamped at half the height. The equilibrium separation between the balls now become:

(2013)



a. $\left(\frac{2r}{3}\right)$

b. $\left(\frac{1}{\sqrt{2}}\right)^2$

c. $\left(\frac{r}{\sqrt{2}}\right)$

d. $\left(\frac{2r}{\sqrt{3}}\right)$

5. Two positive ions, each carrying a charge q , are separated by a distance d . If F is the force of repulsion between the ions, the number of electrons missing from each ion will be (e being the charge on an electron):

(2010 Pre)

a. $\frac{4\pi\epsilon_0 F d^2}{q^2}$

b. $\frac{4\pi\epsilon_0 F d^2}{e^2}$

c. $\sqrt{\frac{4\pi\epsilon_0 F d^2}{q^2}}$

d. $\sqrt{\frac{4\pi\epsilon_0 F d^2}{e^2}}$

6. The unit of permittivity of free space ϵ_0 is:

(2004)

- a. Newton metre²/Coulomb²
b. Coulomb²/Newton metre²
c. Coulomb²/(Newton metre)²
d. Coulomb/Newton metre

7. An electron is moving round the nucleus of a hydrogen atom in a circular orbit of radius r . The Coulomb force $\vec{F}$ between the two is:

(2003)

a. $K \frac{e^2}{r^2} \hat{r}$

b. $-K \frac{e^2}{r^3} \hat{r}$

c. $K \frac{e^2}{r^3} \hat{r}$

d. $-K \frac{e^2}{r^3} \hat{r}$

$$\left(\text{where } K = \frac{1}{4\pi\epsilon_0}\right)$$

8. A charge q is placed at the centre of the line joining two exactly equal positive charges Q . The system of three charges will be in equilibrium, if q is equal to

(1995)

- a. $-Q$ b. $Q/2$
c. $-Q/4$ d. $+Q$

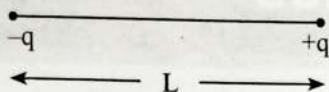
9. Point charges $+4q$, $-q$ and $+4q$ are kept on the X-axis at point $x = 0$, $x = a$ and $x = 2a$ respectively.

(1988)

- a. Only $-q$ is in stable equilibrium
b. All the charges are in stable equilibrium
c. All of the charges are in unstable equilibrium
d. None of the charges is in equilibrium

Electric Field and Relation Between Electric Intensity and Force

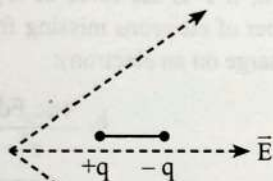
10. Two point charges $-q$ and $+q$ are placed at a distance of L , as shown in the figure



The magnitude of electric field intensity at a distance R ($R \gg L$) varies as (2022)

- a. $\frac{1}{R^6}$ b. $\frac{1}{R^2}$
c. $\frac{1}{R^3}$ d. $\frac{1}{R^4}$

11. A dipole is placed in an electric field as shown. In which direction will it move? (2021)



- a. towards the right as its potential energy will decrease.
b. towards the left as its potential energy will decrease.
c. towards the right as its potential energy will increase.
d. towards the left as its potential energy will increase.
12. A spherical conductor of radius 10 cm has a charge of 3.2×10^{-7} C distributed uniformly. What is the magnitude of electric field at a point 15 cm from the centre of the sphere? (2020)

$$\left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2 \right)$$

- a. 1.28×10^5 N/C b. 1.28×10^6 N/C
c. 1.28×10^7 N/C d. 1.28×10^4 N/C
13. A hollow metal sphere of radius R is uniformly charged. The electric field due to the sphere at a distance r from the centre (2019)
- a. Increases as r increases for $r < R$ and for $r > R$
b. Zero as r increases for $r < R$, decreases as r increases for $r > R$
c. Zero as r increases for $r < R$, increases as r increases for $r > R$
d. Decreases as r increases for $r < R$ and for $r > R$
14. Two point charges A and B, having charges $+Q$ and $-Q$ respectively, are placed at certain distance apart and force acting between them is F . If 25% charge of A is transferred to B, then force between the charges becomes : (2019)

a. F c. $\frac{16F}{9}$ b. $\frac{9F}{16}$ d. $\frac{4F}{3}$

15. An electron falls from rest through a vertical distance h in a uniform and vertically upward directed electric field E . The direction of electric field is now reversed, keeping its magnitude the same. A proton is allowed to fall from rest in it through the same vertical distance h . The time of fall of the electron, in comparison to the time of fall of the proton is (2013)

a. 10 times greater
c. Smaller

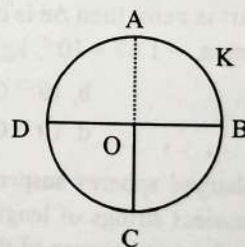
b. 5 times greater
d. Equal

16. The electric field at a distance $\frac{3R}{2}$ from the center of a charged conducting spherical shell of radius R is E . The electric field at a distance $\frac{R}{2}$ from the center of the sphere is (2010 Mains)

a. Zero

c. $\frac{E}{2}$ d. E d. $\frac{E}{3}$

17. A thin conducting ring of radius R is given a charge $+Q$. The electric field at the centre O of the ring due to the charge on the part AKB of the ring is E . The electric field at the centre due to the charge on the part ACDB of the ring is: (2008)

a. $3\vec{E}$ along OKc. $\vec{E}$ along OKb. $\vec{E}$ 3 along KOd. $\vec{E}$ along KO

18. In Millikan oil drop experiment, a charged drop falls with a terminal velocity V . If an electric field E is applied vertically upwards it moves with terminal velocity $2V$ in upward direction. If electric field reduces to $E/2$ then its terminal velocity will be: (1990)

a. $\frac{V}{2}$ c. $\frac{3V}{2}$ b. V d. $2V$

19. A hollow sphere of radius 1 m is given a positive charge of $10 \mu\text{C}$. The electric field at the centre of hollow sphere will be: (1990)

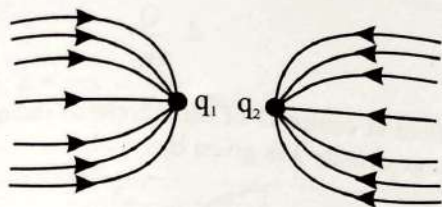
a. $60 \times 10^3 \text{ Vm}^{-1}$

c. Zero

b. $90 \times 10^3 \text{ Vm}^{-1}$

d. Infinite

20. The given figure gives electric lines of force due to two charges q_1 and q_2 . What are the signs of the two charges? (1994)



- a. q_1 is positive but q_2 is negative
 b. q_1 is negative but q_2 is positive
 c. Both are negative
 d. Both are positive

Electric Dipole, Dipole Moment

21. The electric field at a point on the equatorial plane at a distance r from the centre of a dipole having dipole moment $\vec{p}$ is given by,

($r \gg$ separation of two charges forming the dipole, ϵ_0 - permittivity of free space) (2020-Covid)

- a. $\vec{E} = \frac{2\vec{p}}{4\pi\epsilon_0 r^3}$
 b. $\vec{E} = -\frac{\vec{p}}{4\pi\epsilon_0 r^2}$
 c. $\vec{E} = -\frac{\vec{p}}{4\pi\epsilon_0 r^3}$
 d. $\vec{E} = \frac{\vec{p}}{4\pi\epsilon_0 r^3}$

22. An electric dipole is placed at an angle of 30° with an electric field intensity 2×10^5 N/C. It experiences a torque equal to 4 Nm. The charge on the dipole, if the dipole length is 2 cm, is: (2016-II)

- a. 5 mC
 b. 7 mC
 c. 8 mC
 d. 2 mC

23. Three point charges $+q$, $-2q$ and $+q$ are placed at points $(x=0, y=a, z=0)$, $(2x=0, y=0, z=0)$ and $(x=a, y=0, z=0)$ respectively. The magnitude and direction of the electric dipole moment vector of this charge assembly are: (2007)

- a. $\sqrt{2}qa$ along the line joining points $(x=0, y=0, z=0)$ and $(x=a, y=a, z=0)$
 b. qa along the line joining points $(x=0, y=0, z=0)$ and $(x=a, y=a, z=0)$
 c. $\sqrt{2}qa$ along +ve x direction
 d. $\sqrt{2}qa$ along +ve y direction

24. A dipole of dipole moment $\vec{p}$ is placed in uniform electric field $\vec{E}$ then torque acting on it is given by: (2001)

- a. $\vec{\tau} = \vec{p} \cdot \vec{E}$
 b. $\vec{\tau} = \vec{p} \times \vec{E}$
 c. $\vec{\tau} = \vec{p} + \vec{E}$
 d. $\vec{\tau} = \vec{p} - \vec{E}$

25. Electric field at the equator of a dipole is E . If strength and distance is now doubled then the electric field will be: (1998)

- a. $E/2$
 b. $E/8$
 c. $E/4$
 d. E

Potential Difference, Potential Energy and Work Done

26. A charge q is placed in an uniform electric field E . If it is released, then the K.E. of the charge after travelling distance y will be: (1998)

- a. qEy
 b. $2qEy$
 c. $\frac{eEy}{2}$
 d. $\sqrt{eEy}$

Electric Flux and Gauss's Law

27. Two parallel infinite line charges with linear charge densities $+\lambda$ C/m and $-\lambda$ C/m are placed at a distance of $2R$ in free space. What is the electric field mid-way between the two line charges? (2019)

- a. Zero
 b. $\frac{2\lambda}{\pi\epsilon_0 R}$ N/C
 c. $\frac{\lambda}{\pi\epsilon_0 R}$ N/C
 d. $\frac{\lambda}{2\pi\epsilon_0 R}$ N/C

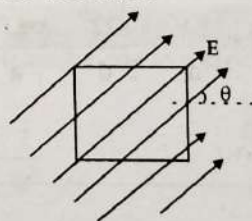
28. The electric field in a certain region is acting radially outward and is given by $E = Ar$. A charge contained in a sphere of radius 'a' centered at the origin of the field, will be given by: (2015)

- a. $A\epsilon_0 a^2$
 b. $4\pi\epsilon_0 Aa^3$
 c. $\epsilon_0 Aa^3$
 d. $4\pi\epsilon_0 Aa^2$

29. A charge Q is enclosed by a Gaussian spherical surface of radius R . If the radius is doubled, then the outward electric flux will: (2011 Pre)

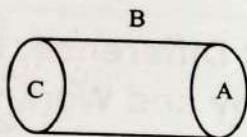
- a. Increase four times
 b. Be reduced to half
 c. Remain the same
 d. Be doubled

30. A square surface of side L metre in the plane of the paper is placed in a uniform electric field E (volt/m) acting along the same plane at an angle θ with the horizontal side of the square as shown in figure. The electric flux linked to the surface, in units of volt-m, is: (2010 Pre)



- a. Zero
 b. EL^2
 c. $EL^2 \cos \theta$
 d. $EL^2 \sin \theta$

31. A hollow cylinder has a charge q coulomb within it. If ϕ is the electric flux in units of voltmeter associated with the curved surface B, the flux linked with the plane surface A in units of voltmeter will be: (2007)



a. $\frac{q}{2\epsilon_0}$

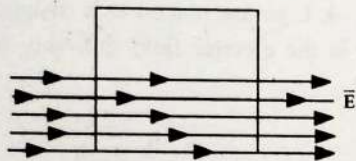
b. $\frac{\phi}{3}$

c. $\frac{q}{\epsilon_0} - \phi$

d. $\frac{1}{2} \left(\frac{q}{\epsilon_0} - \phi \right)$

32. A square surface of side L m is in the plane of the paper.

A uniform electric field $\vec{E}$ (V/m), also in the plane of the paper, is limited only to the lower half of the square surface, (see figure). The electric flux in SI unit associated with the surface is: (2006)



a. $EL^2/(2\epsilon_0)$

b. $EL^2/2$

c. Zero

d. EL^2

33. A charge q is located at the centre of a cube. The electric flux through any face is: (2003)

a. $\frac{2\pi q}{6(4\pi\epsilon_0)}$

b. $\frac{4\pi q}{6(4\pi\epsilon_0)}$

c. $\frac{\pi q}{6(4\pi\epsilon_0)}$

d. $\frac{q}{6(4\pi\epsilon_0)}$

34. A charge Q μC is placed at the centre of a cube, the flux coming out from any surface will be: (2001)

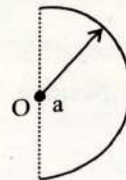
a. $\frac{Q}{6\epsilon_0} \times 10^{-6}$

b. $\frac{Q}{6\epsilon_0} \times 10^{-3}$

c. $\frac{Q}{24\epsilon_0}$

d. $\frac{Q}{8\epsilon_0}$

35. Electric field at centre O of semicircle of radius ' a ' having linear charge density λ is given by: (2000)



a. $\frac{\lambda}{\epsilon_0 a}$

b. $\frac{\lambda\pi}{\epsilon_0 a}$

c. $\frac{\lambda}{2\pi\epsilon_0 a}$

d. $\frac{\lambda}{\pi\epsilon_0 a}$

36. A charge Q is situated at the corner of a cube, the electric flux passed through all the six faces of the cube is: (2000)

a. $\frac{Q}{6\epsilon_0}$

b. $\frac{Q}{8\epsilon_0}$

c. $\frac{Q}{\epsilon_0}$

d. $\frac{Q}{2\epsilon_0}$

37. A point charge $+q$ is placed at the centre of a cube of side l . The electric flux emerging from the cube is (1996)

a. $\frac{6ql^2}{\epsilon_0}$

b. $\frac{q}{6l^2}$

c. Zero

d. $\frac{q}{\epsilon_0}$

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
b	b	c	c	d	b	d	c	c	c	a	a	b	b	c	a	c
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
a	c	c	d	d	a	b	c	a	c	b	c	a	d	c	b	a
35	36	37														
c	b	d														

Explanations

1. (b) Since, $F = ma = \frac{ke^2}{r^2}$

$$\therefore a = \frac{ke^2}{mr^2} = 9 \times 10^9 \frac{(1.6 \times 10^{-19})^2}{(9 \times 10^{-31})(1.6 \times 10^{-10})^2} = 10^{22} \text{ m/s}^2$$

2. (b) Gravitational force = Coulomb's force

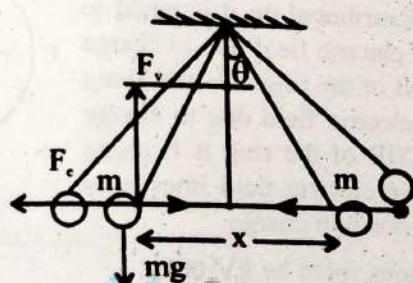
$$\frac{Gm \times m}{r^2} = \frac{k(\Delta e)^2}{r^2}$$

$$\Delta e = m \sqrt{\frac{G}{k}}$$

$$= 1.67 \times 10^{-27} \sqrt{\frac{6.67 \times 10^{-11}}{9 \times 10^9}}$$

$$= 1.427 \times 10^{-37} \text{ C}$$

3. (c)



$$F_v = F_{\text{vertical}} = mg$$

$$F_e = F_{\text{electrostatics}} = \frac{kq^2}{x^2}$$

$$\frac{F_e}{F_v} = \frac{F_e}{mg} = \frac{kq^2}{x^2 mg} = \frac{x}{2l} = \tan \theta$$

$$\Rightarrow q^2 \propto x^3$$

$$\Rightarrow q \propto x^{3/2}$$

Differentiating Eq. (1) with respect to time

$$2q \frac{dq}{dt} \propto 3x^2 \frac{dx}{dt} \quad \left(\frac{dq}{dt} = I (\text{Current}) \right)$$

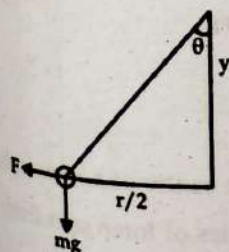
$$2x^{3/2} \frac{dq}{dt} \propto 3x^2 V \quad \left(q \propto x^{3/2} \text{ from Eq. (2)} \right)$$

$$\frac{dq}{dt} = \text{constant}$$

$$x^{3/2} \propto x^2 V$$

$$V \propto x^{-1/2}$$

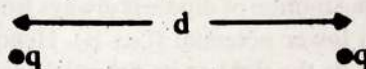
4. (c)



$$\tan \theta = \frac{F}{mg} \Rightarrow \frac{r/2}{y} = \frac{kq^2}{r^2 mg} \Rightarrow y \propto r^3$$

$$\text{Therefore } \left(\frac{r'}{r} \right)^3 = \frac{y/2}{y} \Rightarrow r' = r \left(\frac{1}{2} \right)^{1/3}$$

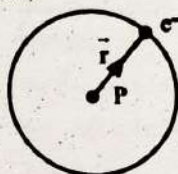
5. (d)



$$F = \frac{(ne)^2}{4\pi\epsilon_0 d^2} \Rightarrow n = \sqrt{\frac{4\pi\epsilon_0 F d^2}{e^2}}$$

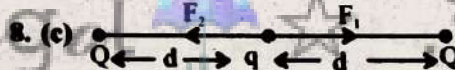
$$6. (b) F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2} \Rightarrow [\epsilon_0] = \frac{\text{Coulomb}^2}{\text{Newton Metre}^2}$$

7. (d)



$$\text{Coulomb force} = \frac{Ke^2}{r^2} (-\hat{r})$$

$$= -\frac{Ke^2}{r^3} \hat{r}$$



at equilibrium

$$\frac{KQq}{d^2} = \frac{-KQ^2}{(2d)^2}$$

$$\frac{q}{d^2} = \frac{-Q}{4d^2}$$

$$q = \frac{-Q}{4}$$

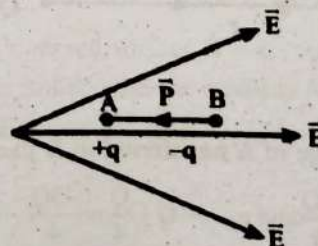
9. (c) For this arrangement net force on each of the charge due to the other charges is zero.

However, disturbance in any direction other than along the line on which the charges lie, will not make the charges return.

10. (c) Electric field intensity due to a dipole at a large distance is given by

$$E \propto \frac{P}{R^3} \Rightarrow E \propto \frac{1}{R^3}$$

11. (a)



Electric field lines at point A (i.e., +q) are denser as comparison to the field lines at point B (i.e., -q).

$$\text{So, } |\vec{E}_A| > |\vec{E}_B|$$

Potential energy of an electric dipole in external electric field

$$U = -P \cdot E$$

$$= -PE \cos \theta$$

where, $\theta = 180^\circ$

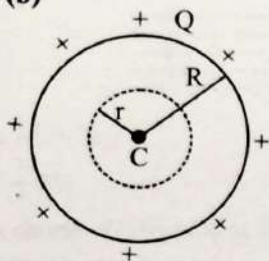
$$\therefore U = +PE$$

The potential energy at A is greater than the potential energy at B. And the motion of dipole is always from higher potential (i.e., A) to lower potential (i.e., B). Hence, the dipole will move towards the right as its potential energy will decrease.

$$12. (a) E = \frac{KQ}{r^2} = \frac{9 \times 10^9 \times 3.2 \times 10^{-7}}{(15 \times 10^{-2})^2}$$

$$E = 1.28 \times 10^5 \text{ N/C}$$

13. (b)

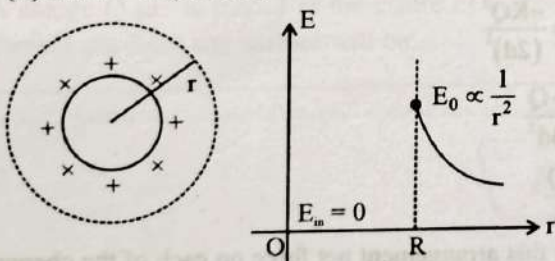


Charge Q will be distributed over the surface of hollow metal sphere.

(i) For $r < R$ (inside)

$$\text{By Gauss law, } \oint \vec{E}_{in} \cdot d\vec{s} = \frac{q_{en}}{\epsilon_0} = 0 \Rightarrow E_{in} = 0 \quad (\because q_{en} = 0)$$

(ii) For $r > R$ (outside)

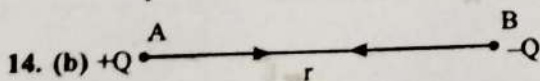


$$\oint \vec{E}_0 \cdot d\vec{s} = \frac{q_{en}}{\epsilon_0}$$

Here, $q_{en} = Q$ ($\because q_{en} = Q$)

$$\therefore E_0 4\pi r^2 = \frac{Q}{\epsilon_0}$$

$$\therefore E_0 \propto \frac{1}{r^2}$$



$$F = \frac{kQ^2}{r^2}$$

If 25% of charge of A transferred to B then

$$q_A = Q - \frac{Q}{4} = \frac{3Q}{4} \text{ and } q_B = -Q + \frac{Q}{4} = \frac{-3Q}{4}$$

$$F_1 = \frac{kq_A q_B}{r^2} \Rightarrow F_1 = \frac{k \left(\frac{3Q}{4} \right)^2}{r^2}$$

$$F_1 = \frac{9}{16} \frac{kQ^2}{r^2} \Rightarrow F_1 = \frac{9F}{16}$$

15. (c) For electron

$$a_e = \frac{eE}{m_e} + g$$

$$(\because m_e a_e = eE + m_e g)$$

for proton

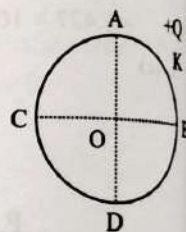
$$a_p = \frac{eE}{m_p} + g \Rightarrow \therefore t = \sqrt{\frac{2h}{a}}$$

because $a_e > a_p$

So $t_e < t_p$

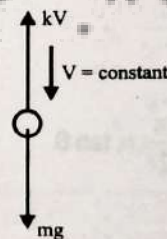
16. (a) Electric field inside a charged conductor is always zero.

17. (c) As we know that electric field at the centre of the ring due to charge uniformly distributed on it is equal to zero. So, if electric field due to charge on part AKB of the ring is E and along KO. Then electric field due to charge on part ACDB of the ring is E along OK. Because electric field lines goes away from positive charge



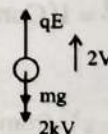
18. (a) Let viscous force be kV then

$$mg = kV$$



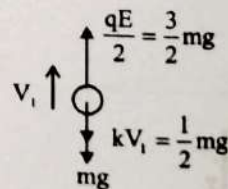
If electric field applied vertically

$$mg + 2kV = qE \Rightarrow qE = 3kV = 3mg$$



If electric field reduces to E/2 then from diagram

$$kV_1 = \frac{mg}{2}$$



$$\Rightarrow V_1 = \frac{V}{2}$$

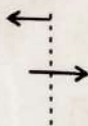
19. (c) Electric field will be zero at the centre of hollow sphere.

20. (c) Electric field lines or electric lines of force start from the positive charge and end at the negative charge.

Since the electric lines for both the charges are ending therefore both q_1 and q_2 are negative charges.

21. (d) $\vec{E} = \frac{K\vec{P}}{r^3}$

$$\vec{E} = \frac{\vec{P}}{4\pi\epsilon_0 r^3}$$

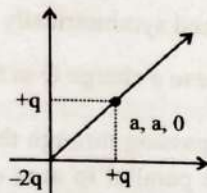


22. (d) $\tau = pE \sin\theta$

$$\tau = ql E \sin\theta$$

$$4 = q \times 2 \times 10^{-2} \times 2 \times 10^5 \sin 30^\circ \Rightarrow q = 2 \text{ mC}$$

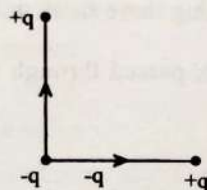
23. (a)



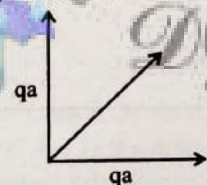
+q at (0, a, 0)

-2q at (0, 0, 0)

+q at (a, 0, 0)



Two dipoles are formed of magnitude qa and direction having along x-axis and y-axis both



Resultant magnitude qa of dipole moment vector

$$= \sqrt{(qa)^2 + (qa)^2}$$

$$= \sqrt{2}qa$$

Direction is along the line joining points. (0, 0, 0) and (a, a, 0)

24. (b) When an electric dipole is placed in a uniform electrical field $\vec{E}$, the torque on the dipole is given by $\vec{\tau} = \vec{p} \times \vec{E}$

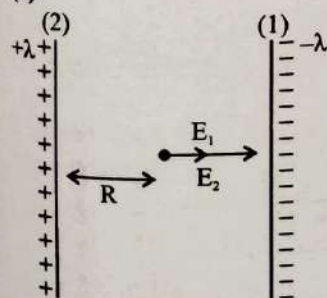
25. (c) $E = \frac{kp}{r^3} \Rightarrow E \propto \frac{p}{r^3} \Rightarrow \frac{E_1}{E} = \frac{2}{8} \Rightarrow E_1 = \frac{E}{4}$

26. (a) Work done = ΔKE

$$\Delta KE = \text{Force} \cdot \text{Displacement}$$

$$= q E \cdot y$$

27. (c)



Electric field due to line charge (1).

$$\vec{E}_1 = \frac{\lambda}{2\pi\epsilon_0 R} \hat{i} \text{ N/C}$$

Electric field due to line charge (2).

$$\vec{E}_2 = \frac{\lambda}{2\pi\epsilon_0 R} \hat{i} \text{ N/C}$$

$$\vec{E}_{\text{net}} = \vec{E}_1 + \vec{E}_2$$

$$= \frac{\lambda}{2\pi\epsilon_0 R} \hat{i} + \frac{\lambda}{2\pi\epsilon_0 R} \hat{i}$$

$$= \frac{\lambda}{\pi\epsilon_0 R} \hat{i} \text{ N/C}$$

28. (b) Flux linked with sphere = $\int \vec{E} \cdot d\vec{s}$

Since electric field radial. It is always perpendicular to the surface.

$$\text{so } \phi = Ar(4\pi a^2)$$

$$\phi = A(a)(4\pi a^2)$$

$$\phi = A4\pi a^3$$

Now according to Gauss law

$$\phi = \frac{q_{\text{in}}}{\epsilon_0} \Rightarrow q_{\text{in}} = \phi \cdot \epsilon_0$$

$$q_{\text{in}} = 4\pi\epsilon_0 Aa^3$$

29. (c) According to Gauss Law

$$\phi = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

Electric flux does not depend on area of Gaussian surface. Hence ϕ remains same.

30. (a) Electric flux

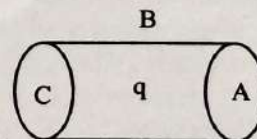
$$\phi = \vec{E} \cdot d\vec{s}$$

$$\phi = Eds \cos \theta$$

$$\phi = 0$$

Here $\vec{E} \perp \text{Area Vector}$ ($\phi = 90^\circ$)

31. (d)



$$\text{Total flux through the hollow cylinder} = \frac{q}{\epsilon_0}$$

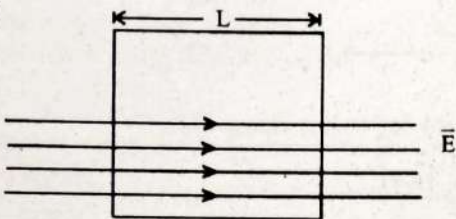
$$\text{Flux through curved surface B} = \phi$$

And the flux linked with the surfaces A and B will be equal.

So, flux linked with surface A

$$= \frac{1}{2} \left(\frac{q}{\epsilon_0} - \phi \right)$$

32. (c)



Since

$$\text{Electric flux } \phi = \vec{E} \cdot \vec{S}$$

$$= ES \cos \theta$$

Where θ is angle between $\vec{E}$ and area vector

Perpendicular to the surface

Here $\theta = 90^\circ$

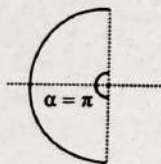
$$\phi = ES \cos 90^\circ = 0$$

33. (b) Electric flux through any face

$$= \frac{\text{Total flux}}{\text{Number of faces}} = \frac{(q/\epsilon_0)}{6} = \frac{4\pi q}{6(4\pi\epsilon_0)}$$

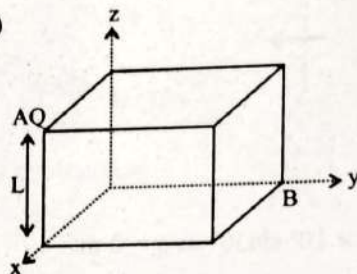
34. (a) For complete cube $\phi = \frac{Q}{\epsilon_0} \times 10^{-6}$

$$\text{For each face } \phi = \frac{1}{6} \frac{Q}{\epsilon_0} \times 10^{-6}$$

35. (c) Electric field at O = $\frac{2K\lambda}{a} \sin(\alpha/2)$ 

$$= \frac{2\lambda}{4\pi\epsilon_0 a} \sin \frac{\pi}{2} = \frac{\lambda}{2\pi\epsilon_0 a}$$

36. (b)



8 cubes can be placed symmetrically at a corner, flux linked with each cube (due to a charge Q at the corner) will be $\frac{Q}{8\epsilon_0}$.

Now for the faces passing through the edge A, electric field E at a face will be parallel to area of face and so flux for these three faces will be zero. Now as the cube has six faces and flux linked with three faces (through A) is zero, so flux linked with remaining three faces will be $\frac{Q}{8\epsilon_0}$.

Hence, electric flux passed through all the six faces of the cube is $\frac{Q}{8\epsilon_0}$.

37. (d) Electric flux emerging from the cube does not depend on size of cube.

$$\text{Total flux} = \frac{q}{\epsilon_0}$$

Electrostatic Potential and Capacitance

Electric Potential

1. Two hollow conducting sphere of radii R_1 and R_2 ($R_1 \gg R_2$) have equal charges. The potential would be: (2022)

a. dependent on the material property of the sphere
b. more on bigger sphere
c. more on smaller sphere
d. equal on both the sphere

2. Twenty seven drops of same size are charged at 220 V each. They combine to form a bigger drop. Calculate the potential of the bigger drop. (2021)

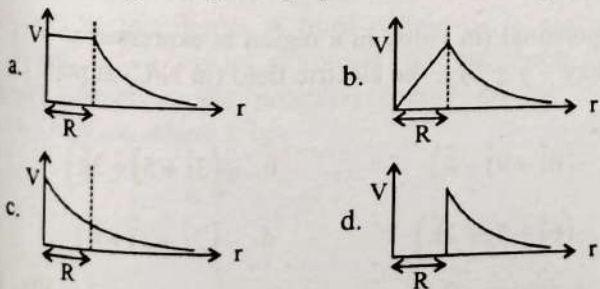
a. 1320 V
b. 1520 V
c. 1980 V
d. 660 V

3. A short electric dipole has a dipole moment of $16 \times 10^{-9} \text{ Cm}$. The electric potential due to the dipole at a point at a distance of 0.6 m from the centre of the dipole, situated on a line making an angle of 60° with the dipole axis is : (2020)

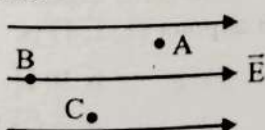
$$\left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2 \right)$$

a. 200 V
b. 400 V
c. Zero
d. 50 V

4. The variation of electrostatic potential with radial distance r from the centre of a positively charged metallic thin shell of radius R is given by the graph. (2020-Covid)



5. A, B and C are three points in a uniform electric field. The electric potential is: (2013)

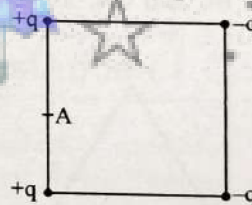


a. Same at all the three points A, B and C
b. Maximum at A
c. Maximum at B
d. Maximum at C

6. Four point charges $-Q$, $-q$, $2q$ and $2Q$ are placed, one at each corner of the square. The relation between Q and q for which the potential at the center of the square is zero is: (2012 Pre)

a. $Q = -q$
b. $Q = -\frac{1}{q}$
c. $Q = q$
d. $Q = \frac{1}{q}$

7. Four electric charges $+q$, $+q$, $-q$ and $-q$ are placed at the corners of a square of side $2L$ (see figure).



The electric potential at point A, midway between the two charges $+q$ and $+q$, is: (2011 Pre)

a. $\frac{1}{4\pi\epsilon_0} \frac{2q}{L} (1 + \sqrt{5})$
b. $\frac{1}{4\pi\epsilon_0} \frac{2q}{L} \left(1 + \frac{1}{\sqrt{5}} \right)$
c. $\frac{1}{4\pi\epsilon_0} \frac{2q}{L} \left(1 - \frac{1}{\sqrt{5}} \right)$
d. Zero

8. Three concentric spherical shells have radii a , b and c ($a < b < c$) and have surface charge densities σ , $-\sigma$ and σ respectively. If V_A , V_B and V_C denote the potentials of the three shells, then for $c = a + b$, we have: (2009)

a. $V_C = V_B \neq V_A$
b. $V_C \neq V_B \neq V_A$
c. $V_C = V_B = V_A$
d. $V_C = V_A \neq V_B$

9. Some charge is being given to a conductor. Then its potential: (2002)

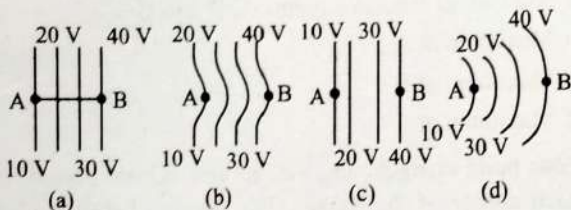
a. Is maximum at surface
b. Is maximum at centre
c. Is same throughout the conductor
d. Is maximum somewhere between surface and centre

10. Eight equally charged tiny drops are combined to form a big drop. If the potential on each drop is 10 V then potential of big drop will be: (1999)

a. 40 V
b. 10 V
c. 30 V
d. 20 V

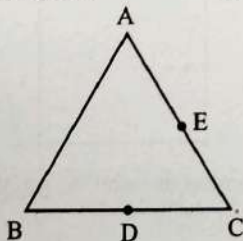
Potential Difference and Work Done

11. The diagrams below show regions of equipotentials (2017-Delhi)

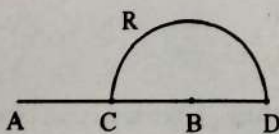


A positive charge is moved from A to B in each diagram.

- a. In all the four cases the work done is the same.
b. Minimum work is required to move q in figure (a)
c. Maximum work is required to move q in figure (b)
d. Maximum work is required to move q in figure (c)
12. Three charges, each $+q$, are placed at the corners of an isosceles triangle ABC of sides BC and AC, $2a$. D and E are the mid points of BC and CA. The work done in taking a charge Q from D to E is: (2011 Mains)

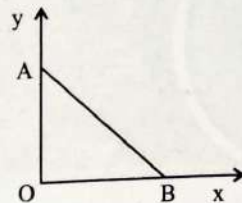


- a. $\frac{3qQ}{4\pi\epsilon_0 a}$
b. $\frac{3qQ}{8\pi\epsilon_0 a}$
c. $\frac{qQ}{4\pi\epsilon_0 a}$
d. Zero
13. Charges $+q$ and $-q$ are placed at points A and B respectively which are at distance $2L$ apart, C is the midpoint between A and B. The work done in moving a charge $+Q$ along the semicircle CRD is: (2007)



- a. $\frac{qQ}{2\pi\epsilon_0 L}$
b. $\frac{qQ}{6\pi\epsilon_0 L}$
c. $-\frac{qQ}{6\pi\epsilon_0 L}$
d. $\frac{qQ}{4\pi\epsilon_0 L}$

14. As per this diagram a point charge $+q$ is placed at the origin O. Work done in taking another point charge $-q$ from the point A [co-ordinates (0, a)] to another point B [co-ordinates (a, 0)] along the straight path AB is: (2009)



- a. Zero
b. $(-qQ/4\pi\epsilon_0)(1/a^2) \sqrt{2}a$
c. $(qQ/4\pi\epsilon_0)(1/a^2) \cdot a/\sqrt{2}$
d. $(qQ/4\pi\epsilon_0)(1/a^2) \sqrt{2}a$
15. There is an electric field E in x -direction. If the work done in moving a charge of 0.2 C through a distance of 2 m along a line making an angle 60° with x -axis is 4 J, then what is the value of E ? (1993)
- a. 5 N/C
b. 20 N/C
c. $\sqrt{3}$ N/C
d. 4 N/C
16. Charge q_2 is at the centre of a circular path with radius r . Work done in carrying charge q_1 , once around this equipotential path, would be (1994)
- a. $\frac{1}{4\pi\epsilon_0} \times \frac{q_1 q_2}{r^2}$
b. $\frac{1}{4\pi\epsilon_0} \times \frac{q_1 q_2}{r}$
c. Zero
d. Infinite

Potential Gradient and Electric Field

17. The angle between the electric lines of force and the equipotential surface is: (2012)
- a. 180°
b. 0°
c. 45°
d. 90°
18. In a certain region of space with volume 0.2 m³, the electric potential is found to be 5 V throughout. The magnitude of electric field in this region is: (2013)
- a. 0.5 N/C
b. 1 N/C
c. 5 N/C
d. Zero
19. If potential (in volts) in a region is expressed as $V(x, y, z) = 6xy - y + 2yz$, the electric field (in N/C) at point (1, 1, 1) is: (2015)
- a. $-(6\hat{i} + 9\hat{j} + \hat{k})$
b. $-(3\hat{i} + 5\hat{j} + 3\hat{k})$
c. $-(6\hat{i} + 5\hat{j} + 2\hat{k})$
d. $-(2\hat{i} + 3\hat{j} + \hat{k})$
20. In a region, the potential is represented by $V(x, y, z) = 6x - 8xy - 8y + 6yz$, where V is in volts and x, y, z are in meters. The electric force experienced by a charge of 1 coulomb situated at point (1, 1, 1) is: (2014)
- a. $6\sqrt{5}$ N
b. 30 N
c. 24 N
d. $4\sqrt{35}$ N

33. Two metallic spheres of radii 1 cm and 2 cm are given charges 10^{-2} C and 5×10^{-2} C respectively. If they are connected by a conducting wire, the final charge on the smaller sphere is (1995)

a. 3×10^{-2} C b. 4×10^{-2} C
c. 1×10^{-2} C d. 2×10^{-2} C

34. A hollow metallic sphere of radius 10 cm is charged such that potential of its surface is 80 V. The potential at the centre of the sphere would be (1994)

a. 80 V b. 800 V
c. Zero d. 8 V

Capacitance and Parallel Plate Capacitor

35. The electrostatic force between the metal plates of an isolated parallel plate capacitor C having a charge Q and area A, is (2018)

a. Proportional to the square root of the distance between the plates.
b. Linearly proportional to the distance between the plates.
c. Independent of the distance between the plates.
d. Inversely proportional to the distance between the plates.

36. A parallel plate air capacitor has capacity 'C' distance of separation between plates is 'd' and potential difference 'V' is applied between the plates force of attraction between the plates of the parallel plate air capacitor is: (2015 Pre)

a. $\frac{C^2 V^2}{2d^2}$ b. $\frac{C^2 V^2}{2d}$
c. $\frac{CV^2}{2d}$ d. $\frac{CV^2}{d}$

37. A parallel plate air capacitor is charged to a potential difference of V volts. After disconnecting the charging battery the distance between the plates of the capacitor is increased using an insulating handle. As a result the potential difference between the plates: (2006)

a. Decreases b. Does not change
c. Becomes zero d. Increases

38. A capacitor of capacity C_1 charged upto V volt and then connected to an uncharged capacitor of capacity C_2 . Then final P.D. across each will be: (2002)

a. $\frac{C_2 V}{C_1 + C_2}$ b. $\frac{C_1 V}{C_1 + C_2}$
c. $\left(1 + \frac{C_2}{C_1}\right)$ d. $\left(1 - \frac{C_2}{C_1}\right)V$

Effect of Dielectric on Capacitance and Polarization

39. Polar molecules are the molecules: (2021)

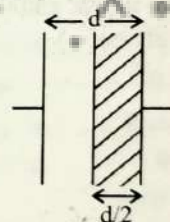
a. Acquire a dipole moment only in the presence of electric field due to displacement of charges.
b. Acquire a dipole moment only when magnetic field is absent.
c. Having a permanent electric dipole moment.
d. Having zero dipole moment.

40. The capacitance of a parallel plate capacitor with air as medium is 6mF. With the introduction of a dielectric medium, the capacitance becomes 30 mF. The permittivity of the medium is : (2020)

$$(\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{ m}^{-2})$$

a. $1.77 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{ m}^{-2}$ b. $0.44 \times 10^{-10} \text{ C}^2 \text{N}^{-1} \text{ m}^{-2}$
c. $5.00 \text{ C}^2 \text{N}^{-1} \text{ m}^{-2}$ d. $0.44 \times 10^{-13} \text{ C}^2 \text{N}^{-1} \text{ m}^{-2}$

41. A parallel plate capacitor having cross-sectional area A and separation d has air in between the plates. Now an insulating slab of same area but thickness d/2 is inserted between the plates as shown in figure having dielectric constant K (= 4). The ratio of new capacitance to its original capacitance will be, (2020-Covid)



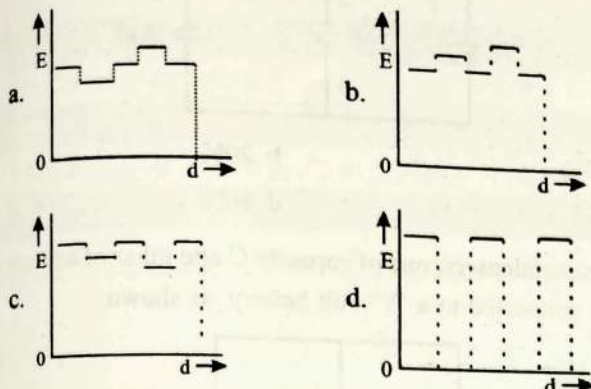
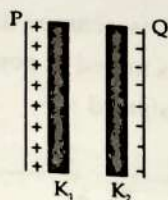
a. 8 : 5 b. 6 : 5
c. 4 : 1 d. 2 : 1

42. A parallel plate air capacitor of capacitance C is connected to a cell of emf V and then disconnected from it. A dielectric slab of dielectric constant K, which can just fill the air gap of the capacitor, is now inserted in it. Which of the following is incorrect? (2015)

a. The energy stored in the capacitor decreases K times
b. The change in energy stored is $\frac{1}{2} CV^2 \left(\frac{1}{K} - 1 \right)$

c. The charge on the capacitor is not conserved
d. The potential difference between the plates decreases K times

43. Two thin dielectric slabs of dielectric constants K_1 and K_2 ($K_1 < K_2$) are inserted between plates of a parallel plate capacitor, as shown in the figure. The variation of electric field E between the plates with distance d as measured from plate P is correctly shown by: (2014)



44. Two parallel metal plates having charges $+Q$ and $-Q$ face each other at a certain distance between them. If the plates are now dipped in kerosene oil tank, the electric field between the plates will: (2010 Mains)

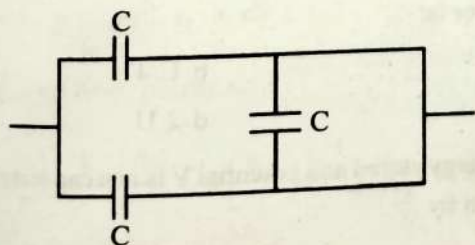
a. Become zero b. Increase
c. Decrease d. Remain same

45. The energy and capacity of a charged parallel plate capacitor are E and C respectively. Now a dielectric slab of $\epsilon_r = 6$ is inserted in it then energy and capacity becomes (Assuming charge on plates remains constant): (1999)

a. $6E, 6C$ b. E, C
c. $\frac{E}{6}, 6C$ d. $E, 6C$

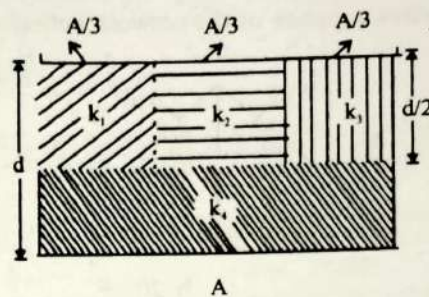
Grouping of Capacitors

46. The equivalent capacitance of the combination shown in the figure is: (2021)



a. $2C$ b. $C/2$
c. $3C/2$ d. $3C$

47. A parallel-plate capacitor of area A , plate separation d and capacitance C is filled with four dielectric materials having dielectric constants k_1, k_2, k_3 and k_4 as shown in the figure below. If a single dielectric material is to be used to have the same capacitance C in this capacitor, then its dielectric constant k is given by: (2016 - II)



a. $\frac{2}{k} = \frac{3}{k_1 + k_2 + k_3} + \frac{1}{k_4}$ b. $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} + \frac{3}{2k_4}$
c. $k = k_1 + k_2 + k_3 + 3k_4$ d. $k = \frac{2}{3}(k_1 + k_2 + k_3) + 2k_4$

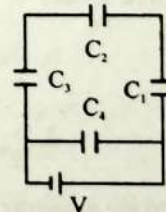
48. A series combination of n_1 capacitors, each of value C_1 , is charged by a source of potential difference $4V$. When another parallel combination of n_2 capacitors, each of value C_2 , is charged by a source of potential difference V , it has the same (total) energy stored in it, as the first combination has. The value of C_2 , in terms of C_1 , is then: (2010 Pre)

a. $\frac{16C_1}{n_1 n_2}$ b. $\frac{2C_1}{n_1 n_2}$
c. $16 \frac{n_2}{n_1} C_1$ d. $2 \frac{n_2}{n_1} C_1$

49. Three capacitors each of capacitance C and of breakdown voltage V are joined in series. The capacitance and breakdown voltage of the combination will be: (2009)

a. $3C, \frac{V}{3}$ b. $\frac{C}{3}, 3V$
c. $3C, 3V$ d. $\frac{C}{3}, \frac{V}{3}$

50. A network of four capacitors of capacity equal to $C_1 = C$, $C_2 = 2C$, $C_3 = 3C$ and $C_4 = 4C$ are connected to a battery as shown in the figure. The ratio of the charges on C_2 and C_4 is (2005)

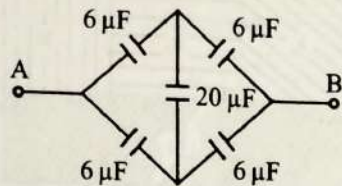


a. $4/7$ b. $3/22$
c. $7/4$ d. $22/3$

51. Three capacitors each of capacity $4 \mu F$ are to be connected in such a way that the effective capacitance of $6 \mu F$. This can be done by: (2003)

a. Connecting all of them in series
b. Connecting them in parallel
c. Connecting two in series and one in parallel
d. Connecting two in parallel and one in series

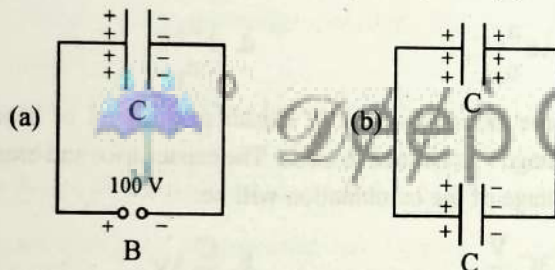
52. The effective capacity of the network between terminals A and B is: (1999)



- a. $6 \mu\text{F}$
b. $20 \mu\text{F}$
c. $3 \mu\text{F}$
d. $10 \mu\text{F}$

Energy Stored in a Capacitor and Loss of Energy on Sharing Charges

53. A capacitor of capacitance $C = 900 \text{ pF}$ is charged fully by 100 V battery B as shown in figure (a). Then it is disconnected from the battery and connected to another uncharged capacitor of capacitance $C = 900 \text{ pF}$ as shown in figure (b). The electrostatic energy stored by the system (b) is (2022)



- a. $1.5 \times 10^{-6} \text{ J}$
b. $4.5 \times 10^{-6} \text{ J}$
c. $3.25 \times 10^{-6} \text{ J}$
d. $2.25 \times 10^{-6} \text{ J}$

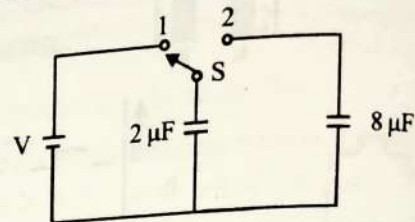
54. A parallel plate capacitor has a uniform electric field ' $\vec{E}$ ' in the space between the plates. If the distance between the plates is ' d ' and the area of each plate is ' A ', the energy stored in the capacitor is: (2021, 2012 Mains, 2011 pre, 2008) (ϵ_0 = permittivity of free space)

- a. $\epsilon_0 E A d$
b. $\frac{1}{2} \epsilon_0 E^2 A d$
c. $\frac{E A d}{\epsilon}$
d. $\frac{1}{2} \epsilon_0 E^2$

55. A capacitor is charged by a battery. The battery is removed and another identical uncharged capacitor is connected in parallel. The total electrostatic energy of resulting system: (2017-Delhi)

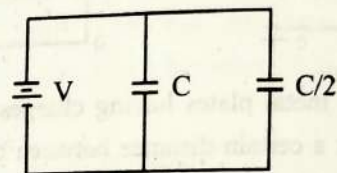
- a. Decreases by a factor of 2
b. Remains the same
c. Increases by a factor of 2
d. Increases by a factor of 4

56. A capacitor of $2 \mu\text{F}$ is charged as shown in the diagram. When the switch S is turned to position 2, the percentage of its stored energy dissipated is: (2016-J)



- a. 0%
b. 20%
c. 75%
d. 80%

57. Two condensers, one of capacity C and other of capacity $C/2$ are connected to a ' V ' volt battery, as shown.



The work done in charging fully both the condensers is: (2007)

- a. $\frac{1}{4} C V^2$
b. $\frac{3}{4} C V^2$
c. $\frac{1}{2} C V^2$
d. $2 C V^2$

58. Energy per unit volume for a capacitor having area A and separation d kept at potential difference V is given by: (2001)

- a. $\frac{1}{2} \epsilon_0 \frac{V^2}{d^2}$
b. $\frac{1}{2 \epsilon_0} \frac{V^2}{d^2}$
c. $\frac{1}{2} C V^2$
d. $\frac{Q^2}{2 C}$

59. A capacitor is charged with a battery and energy stored is U . After disconnecting battery another capacitor of same capacity is connected in parallel with it. Then energy stored in each capacitor is: (2000)

- a. $U/2$
b. $U/4$
c. $4 U$
d. $2 U$

60. The energy stored and potential V is in a capacitor of capacity C given by (1996)

- a. $\frac{C V}{2}$
b. $\frac{C^2 V^2}{2}$
c. $\frac{C^2 V}{2}$
d. $\frac{C V^2}{2}$

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
c	c	a	a	c	a	c	d	c	a	a	d	c	a	b	c	d
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
d	c	d	d	b	a	d	a	b	a	c	b	a	b	b	d	a
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
c	c	d	b	c	b	a	c	c	c	c	a	a	a	b	b	c
52	53	54	55	56	57	58	59	60								
a	d	b	a	d	b	a	b	d								

Explanations

1. (c) Potential for conducting sphere of radius R and charge q is:

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{R} \Rightarrow V \propto \frac{q}{R}$$

For two spheres:

$$\frac{V_1}{V_2} = \left(\frac{q_1}{q_2}\right) \left(\frac{R_2}{R_1}\right)$$

$$\because q_1 = q_2 \text{ and } R_1 > R_2$$

$$\therefore V_2 > V_1 \text{ (Potential would be more on smaller sphere)}$$

2. (c) For a sphere, electric potential is given as:

$$V = \frac{KQ}{r}$$

For small drops, electric potential is given as:

$$V_s = \frac{Kq}{r} = 220 \text{ V} \quad \dots(i)$$

According to the question, volume remains same.

$$\Rightarrow 27 V'_s = V'_B \Rightarrow 27 \left(\frac{4}{3}\pi r^3\right) = \frac{4}{3}\pi R^3$$

$$\Rightarrow 27r^3 = R^3 \Rightarrow R = 3r$$

For bigger drop, potential: $\begin{cases} V'_s : \text{Value of small drops} \\ V'_B : \text{Value of bigger drops} \end{cases}$

$$V_B = \frac{KQ}{R} = \frac{K(27q)}{3r} = q \left(\frac{Kg}{r}\right)$$

$$\Rightarrow V_B = 9 \times 220$$

$$\Rightarrow V_B = 1980 \text{ V}$$

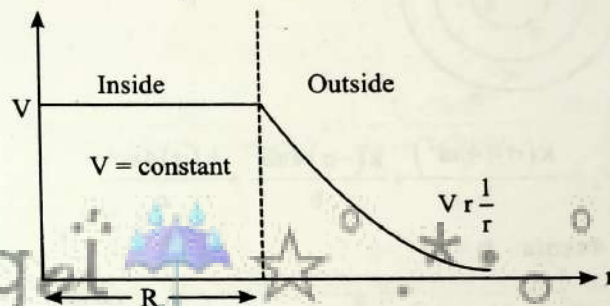
3. (a) The electric potential due to the dipole at a point at a distance of 0.6 m from the centre of the dipole is

$$V = \frac{Kp \cos \theta}{r^2}$$

$$V = \frac{9 \times 10^9 \times 16 \times 10^{-9} \times \cos 60}{0.36}$$

$$V = 200 \text{ V}$$

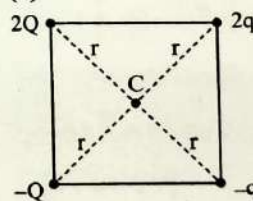
4. (a) $V_{in} = V_s = \frac{KQ}{R}$ and $V_{out} = \frac{KQ}{r} (r > R)$



5. (c) Electric field lines point from high to low potential. Electric potential decreases in the direction of electric field. Therefore, potential is maximum at B.

$$\therefore \text{Potential } V_B > V_C > V_A$$

6. (a)

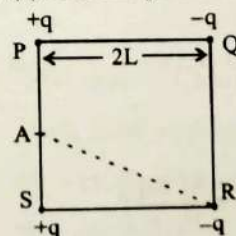


$\therefore$ Distances from charges = r
and potential at centre $V_C = 0$

$$\therefore V_C = \frac{k(-Q)}{r} + \frac{k(-q)}{r} + \frac{k(2q)}{r} + \frac{k(2Q)}{r} = 0$$

$$\therefore -Q - q + 2q + 2Q = 0 \Rightarrow q = -Q$$

7. (c) A is midpoint of PS



$$AR = AQ = \sqrt{(SR)^2 + (AS)^2} = \sqrt{(2L)^2 + (L)^2} = \sqrt{5}L$$

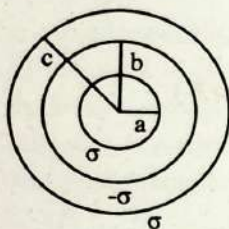
Electrostatic potential at A due to given configuration,

$$\begin{aligned} V_A &= \frac{1}{4\pi\epsilon_0} \left[\frac{q}{PA} + \frac{q}{AS} + \frac{(-q)}{AQ} + \frac{(-q)}{AR} \right] \\ &= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{L} + \frac{1}{L} - \frac{1}{L\sqrt{5}} - \frac{1}{L\sqrt{5}} \right] \\ &= \frac{q}{4\pi\epsilon_0} \left[\frac{2}{L} - \frac{2}{L\sqrt{5}} \right] = \frac{1}{4\pi\epsilon_0} \frac{2q}{L} \left[1 - \frac{1}{\sqrt{5}} \right] \end{aligned}$$

8. (d) Potential of spherical shell

$$V = \frac{kq}{R} = \frac{k(\sigma)(4\pi R^2)}{R}$$

So, let check the potential V_A , V_B and V_C



$$V_A = \frac{k(\sigma)(4\pi a^2)}{a} + \frac{k(-\sigma)4\pi b^2}{b} + \frac{k(\sigma)4\pi c^2}{c}$$

$$= 4\pi k\sigma(a - b + c)$$

$$\text{Similarly } V_B = 4\pi k\sigma \left(\frac{a^2}{b} - b + c \right)$$

$$V_C = 4\pi k\sigma \left(\frac{a^2}{c} - \frac{b^2}{c} + c \right)$$

$$= 4\pi k\sigma \left(\frac{a^2 - b^2}{c} + c \right)$$

$$= 4\pi k\sigma \left(\frac{(a+b)(a-b)}{c} + c \right)$$

$$= 4\pi k\sigma [(a-b) + c] \quad \{\because c = a+b\}$$

$$= 4\pi k\sigma(a - b + c)$$

So, $V_A = V_C \neq V_B$

9. (c) Inside the conductor $E = 0$ so potential remains same.

Hence, potential is same throughout the conductor.

10. (a) 8 equals charged combined to form big drops and volume will same.

$$\therefore nV_{\text{small}} = V_{\text{big}}$$

$$8 \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3 \Rightarrow R = 2r$$

$$\text{Now, potential, } V'_{\text{big}} = \frac{KQ}{R} = \frac{K(8q)}{2r} = \frac{8}{2} \left(\frac{Kq}{r} \right)$$

$$\therefore V'_{\text{small}} = \frac{Kq}{r} = 10V$$

$$\therefore V'_{\text{big}} = \frac{8}{2} V_{\text{small}}$$

$$V'_{\text{big}} = 4 \times 10$$

$$\Rightarrow V'_{\text{big}} = 40 \text{ volt}$$

11. (a) Work done, $W = q\Delta V$

(ΔV is potential difference between A and B)

For all cases potential difference (ΔV) remains same hence work done is same for all cases.

12. (d) As this triangle is isosceles,

$$AD = BE$$

Hence, potential due to given charge configuration is same at D and E.

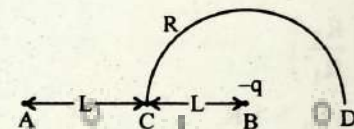
$$\therefore V_D = V_E$$

Work done in moving charge Q from D to E.

$$W = q\Delta V = Q(V_E - V_D) = 0$$

There will be no work done in moving charge Q from D to E.

13. (c)



$$\text{Potential at point C} = \frac{kq}{L} - \frac{kq}{L} = 0$$

$$\text{Potential at point D} = \frac{kq}{3L} - \frac{kq}{L}$$

$$= \frac{kq - 3kq}{3L} = \frac{-2kq}{3L}$$

Work done in moving a charge +Q along the semicircle CRD

$$W = \Delta U$$

$$W = +Q(V_D - V_C)$$

$$W = +Q \left(\frac{-2kq}{3L} - 0 \right) \Rightarrow \frac{-2kQq}{3L} = -2 \times \frac{1}{4\pi\epsilon_0} \times \frac{Qq}{3L}$$

$$W = \frac{-Qq}{6\pi\epsilon_0 L}$$

14. (a)

$$\text{Potential at point A, } V_A = \frac{kq}{a}$$

$$\text{Potential at point B, } V_B = \frac{kq}{a}$$

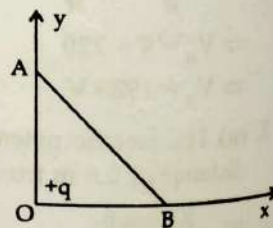
$$\therefore V_A = V_B$$

$$\therefore \Delta V = 0$$

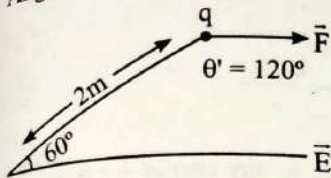
Work done = Change in potential energy

$$W = q(\Delta V) = 0$$

So, work done is equal to zero.



15. (b) Charge (q) = 0.2 C; Distance $d = 2$ m;
Angle $\theta = 60^\circ$ and work done (W) = 4 J.



Work done in moving the charge (W)

$$W = \vec{F} \cdot \vec{d} = Fd \cos \theta' = qEd \cos \theta'$$

$$\text{Or, } E = \frac{W}{qd \cos \theta'} = \frac{4}{0.2 \times 2 \times \cos 120^\circ} = \frac{4}{0.4 \times 0.5} = 20 \text{ N/C}$$

16. (c) For equipotential surface,

$$\Delta V = 0$$

$$\therefore \text{Work, } W = q\Delta V = 0$$

Total work done on carrying a charge from one place to another on an equipotential surface is zero.

17. (d) When the potential becomes constant, the negative potential gradient also becomes zero, which is necessary for the need of the electric field to be always normal with the equipotential surface. Thus, the angle between electric lines of force and equipotential surface is 90° .

18. (d) Since, electric potential (V) is found to be constant, throughout

$$V = \text{Constant}$$

$$\therefore \frac{dV}{dr} = 0$$

$$\text{Hence electric field, } E = -\frac{dV}{dr} = 0$$

19. (c) Potential $V(x, y, z) = 6xy - y + 2yz$

$$\therefore \text{Electric field } \vec{E} = -\vec{\nabla} V$$

$$\vec{E} = -\frac{\partial V}{\partial x} \hat{i} - \frac{\partial V}{\partial y} \hat{j} - \frac{\partial V}{\partial z} \hat{k}$$

$$\vec{E} = -(6y) \hat{i} - (6x - 1 + 2z) \hat{j} - (2y) \hat{k}$$

at point (1, 1, 0)

$$\vec{E} = -6\hat{i} - 5\hat{j} - 2\hat{k} = -(6\hat{i} + 5\hat{j} + 2\hat{k})$$

20. (d) Potential $V(x, y, z) = 6x - 8xy - 8y + 6yz$

$$\therefore \text{Electric field } \vec{E} = -\vec{\nabla} V$$

$$\vec{E} = -\frac{\partial V}{\partial x} \hat{i} - \frac{\partial V}{\partial y} \hat{j} - \frac{\partial V}{\partial z} \hat{k}$$

$$\vec{E} = -[(6 - 8y) \hat{i} + (-8x - 8 + 6z) \hat{j} + (6y) \hat{k}]$$

$$\text{At } (1, 1, 1), \vec{E} = 2\hat{i} + 10\hat{j} - 6\hat{k}$$

$$\Rightarrow |\vec{E}| = \sqrt{2^2 + 10^2 + 6^2} = \sqrt{140} = 2\sqrt{35}$$

$$\text{Force} = qE = 2 \times 2\sqrt{35} = 4\sqrt{35} \text{ N}$$

21. (d) Potential, $V = -x^2y - xz^3 + 4$

$$\therefore \text{Electric field } E = -\vec{\nabla} V$$

$$\vec{E} = -\frac{\partial V}{\partial x} \hat{i} - \frac{\partial V}{\partial y} \hat{j} - \frac{\partial V}{\partial z} \hat{k}$$

$$\vec{E}_x = -\frac{\partial V}{\partial x} \hat{i} = -\left[\frac{\partial}{\partial x}(-x^2y - xz^3 + 4)\right] \hat{i} = (2xy + z^3) \hat{i}$$

$$\vec{E}_y = -\frac{\partial V}{\partial y} \hat{j} = -\frac{\partial}{\partial y}(-x^2y - xz^3 + 4) \hat{j} = x^2 \hat{j}$$

$$\vec{E}_z = -\frac{\partial V}{\partial z} \hat{k} = -\frac{\partial}{\partial z}(-x^2y - xz^3 + 4) \hat{k} = 3xz^2 \hat{k}$$

$$\text{So, } \vec{E} = (2xy + z^3) \hat{i} + x^2 \hat{j} + 3xz^2 \hat{k}$$

22. (b) Electric potential at a point

$$\Rightarrow \frac{Q}{4\pi\epsilon_0 r} = Q \times 10^{11} \Rightarrow \frac{Q \times 9 \times 10^9}{r} = Q \times 10^{11}$$

$$\Rightarrow r = 9 \times 10^{-2} \text{ m}$$

Electric field at that point

$$E = \frac{Q}{4\pi\epsilon_0 r^2} = \frac{Q}{4\pi\epsilon_0 \times (9 \times 10^{-2})^2}$$

$$E = \frac{Q}{4\pi\epsilon_0 \times 9 \times 10^{-2} \times 9 \times 10^{-2}} \Rightarrow \frac{Q \times 9 \times 10^9}{9 \times 10^{-2} \times 9 \times 10^{-2}} = \frac{Q \times 10^{13}}{9}$$

$$\text{As } \left(4\pi\epsilon_0 = \frac{1}{9 \times 10^9}\right)$$

Multiplying $4\pi\epsilon_0$ to both numerator and denominator.

$$E = \frac{Q \times 10^{13}}{9} \times \frac{4\pi\epsilon_0}{4\pi\epsilon_0}$$

$$E = (4\pi\epsilon_0 Q) \times \frac{10^{13}}{9} \times 9 \times 10^9 \quad \left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9\right)$$

$$E = (4\pi\epsilon_0 Q) \times 10^{22} \text{ volt/m}$$

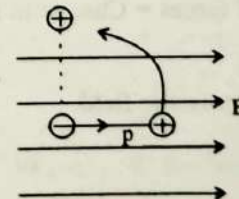
23. (a) Torque of dipole,

$$\vec{\tau} = \vec{p} \times \vec{E} = pE \sin \theta$$

Potential energy of dipole,

$$U = -\vec{p} \cdot \vec{E} = -pE \cos \theta$$

24. (d)



Work done = Change in Energy of dipole

$$\text{Work done} = pE (\cos \theta_1 - \cos \theta_2)$$

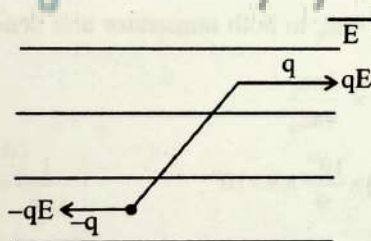
$$W = pE (\cos 0^\circ - \cos 90^\circ)$$

$$(\because \theta_1 = 0^\circ, \theta_2 = 90^\circ)$$

$$W = pE$$

So, $BD = 10$ cm

26. (b) In uniform electric field net force experienced by the electric dipole is zero whatever be its orientation.



$$PE = -pE \text{ [minimum]}$$

There is only electrostatic field.

$$2 \times 10^{-6} \times \Delta V = 10^{-1}$$

$$\left\{ \begin{array}{l} \text{Change (q)} = 2 \times 10^{-6} \text{ C} \\ \text{Mass (m)} = 2 \times 10^{-3} \text{ kg} \\ \text{Velocity (v)} = 10 \text{ m/sec} \end{array} \right.$$

$$\therefore q_1 = 3q_2 = 3 \times 10^{-2} \text{ C}$$

33. (d) Radii of sphere

$$R_1 = 1 \text{ cm} = 1 \times 10^{-2} \text{ m};$$

$$R_2 = 2 \text{ cm} = 2 \times 10^{-2} \text{ m}$$

Charges on sphere;

$$Q_1 = 10^{-2} \text{ C and } Q_2 = 5 \times 10^{-2} \text{ C.}$$

Capacitance of sphere:

$$C_1 = 4\pi\epsilon_0(R_1) = 4\pi\epsilon_0(10^{-2})$$

$$C_2 = 4\pi\epsilon_0(R_2) = 4\pi\epsilon_0(2 \times 10^{-2})$$

$$\text{Common potential (V)} = \frac{\text{Total charge}}{\text{Total capacity}}$$

$$V_{\text{common}} = \frac{Q_1 + Q_2}{C_1 + C_2} = \frac{(1 \times 10^{-2}) + (5 \times 10^{-2})}{4\pi\epsilon_0 10^{-2} + 4\pi\epsilon_0(2 \times 10^{-2})} = \frac{6 \times 10^{-2}}{4\pi\epsilon_0(3 \times 10^{-2})}$$

Therefore final charge on smaller sphere ($C_1 V_{\text{common}}$)

$$= 4\pi\epsilon_0 \times 10^{-2} \times \frac{6 \times 10^{-2}}{4\pi\epsilon_0 \times 3 \times 10^{-2}} = 2 \times 10^{-2} \text{ C.}$$

34. (a) Electric field is zero inside a hollow spherical shell, and potential remains constant at all points inside the sphere and equal to potential at the surface.

$$\therefore V_{\text{surface}} = V_{\text{centre}} = 80 \text{ V}$$

35. (c) For on isolated capacitor

$$Q = \text{constant and Force} = \frac{Q^2}{2A\epsilon_0}$$

Force is independent of separation d .

36. (c) Force of attraction between the plates, $F = \frac{Q^2}{2\epsilon_0 A}$

$$\therefore Q = CV \text{ and } C = \frac{\epsilon_0 A}{d} \Rightarrow \epsilon_0 A = Cd$$

$$\text{So } F = \frac{C^2 V^2}{2Cd} = \frac{CV^2}{2d} \Rightarrow F = \frac{CV^2}{2d}$$

37. (d) When capacitor is connected to battery, It gets charged to $Q = CV$.

After disconnecting the battery, charge of capacitor will remain constant.

As we are increasing the distance between the plates of the capacitor, capacitance $\left[C = \frac{A\epsilon_0}{d} \right]$ is decreasing.

$$\therefore Q = CV$$

Since Q is constant and C is decreasing, V will increase.

38. (b) When two capacitors are connected then common potential,

$$V_{\text{common}} = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} \quad (\because V_2 = 0, V_1 = V)$$

$$\therefore V_{\text{common}} = \frac{C_1 V}{C_1 + C_2}$$

39. (c) Polar molecules have centres of positive and negative charges apart by a certain distance which means positive and negative polarity arises. So, the polar molecules have permanent dipole moment.

40. (b) Capacitance of capacitor with air, $C_o = \frac{\epsilon_0 A}{d}$ Capacitance

$$\text{Capacitance of capacitor with dielectric } (\epsilon_r) C_m = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$C_m = \epsilon_r C_o$$

$$\epsilon_r = \frac{30}{6} = 5$$

$$\therefore \text{Permittivity } \epsilon = \epsilon_0 \epsilon_r = 8.85 \times 10^{-12} \times 5$$

$$\Rightarrow \epsilon = 0.44 \times 10^{-10} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$$

41. (a) Capacitance of parallel plate capacitor with dielectric (K) and slab of thickness t .

$$C_K = \frac{\epsilon_0 A}{d - t + \frac{t}{K}} = \frac{\epsilon_0 A}{d - \frac{d}{2} + \frac{d}{8}} = \frac{8 \epsilon_0 A}{5 d}$$

$$\text{Capacitance without dielectric } C_a = \frac{\epsilon_0 A}{d}$$

$$\therefore C_K = \frac{8}{5} C_a \Rightarrow \frac{C_K}{C_a} = \frac{8}{5}$$

42. (c) When dielectric slab is inserted

$$\text{Capacitance: } C = \frac{\epsilon_0 A}{d} \Rightarrow C_{\text{New}} = KC$$

$$\text{Energy: } E = \frac{Q^2}{2C} \Rightarrow E_{\text{New}} = \frac{1}{K} E_{\text{initial}}$$

$$\text{Voltage: } V = \frac{Q}{C} \Rightarrow V_{\text{New}} = \frac{1}{K} V$$

$$\text{Charge: } Q = CV \Rightarrow Q_{\text{New}} = C_{\text{New}} V_{\text{New}}$$

$$Q_{\text{New}} = (KC) \left(\frac{V}{K} \right) = CV$$

Hence charge on capacitor remains conserved.

43. (c) Electric field between plates of parallel plate capacitor:

$$E = \frac{\sigma}{K \epsilon_0} \quad (\text{Where } \sigma \text{ is charge density and } K \text{ is dielectric constant})$$

$$\text{For } K = 1 \text{ (Air), } E = \frac{\sigma}{\epsilon_0}$$

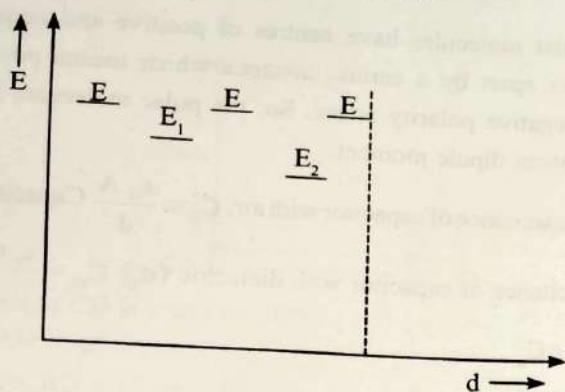
$$\text{For } K = K_1, E_1 = \frac{\sigma}{K_1 \epsilon_0} = \frac{E}{K_1}$$

$$\text{For } K = K_2, E_2 = \frac{\sigma}{K_2 \epsilon_0} = \frac{E}{K_2}$$

$$\therefore K_1 < K_2 \Rightarrow E_1 > E_2$$

$$\therefore E > E_1 > E_2$$

So, variation of electric field with distance-



44. (c) Electric field between plates of parallel plate capacitor

$$E = \frac{\sigma}{K \epsilon_0}$$

$$\therefore E \propto \frac{1}{K}$$

Dielectric constant; $K_{\text{Kerosene}} > K_{\text{air}}$

$\therefore$ Electric field between plates decreases.

45. (c) Energy and capacitance for capacitor without dielectric are E and C .

Capacitance of capacitor with dielectric:-

$$C' = \frac{\epsilon_0 \epsilon_r A}{d}$$

Where A : Area of plate

B : Space between plates

ϵ_r : Permittivity of dielectric slab

$$\therefore \epsilon_r = 6$$

$$\therefore C' = 6 \frac{\epsilon_0 A}{d} = 6C$$

$$\therefore Q \text{ is constant}$$

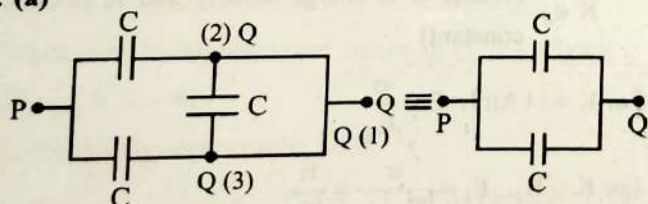
$$\therefore Q \text{ is constant}$$

$$\therefore V' = \frac{V}{6}$$

$$\text{Energy } E' = \frac{1}{2} C' V'^2$$

$$E' = \frac{1}{2} (6C) \left(\frac{V}{6} \right)^2 = \frac{CV^2}{12} = \frac{1}{6} \left(\frac{CV^2}{2} \right) = \frac{E}{6}$$

46. (a)



At 1, 2, 3, the potential is same. So, the capacitor connected in between point (2) and (3) will get short circuited as it is not storing any charge.

Now, equivalent capacitance for the combination of parallel capacitors.

$$\Rightarrow C_{eq} = C + C = 2C$$

47. (a) Total capacitance of capacitor with dielectric k ,

$$C = \frac{k \epsilon_0 A}{d}$$

When A : Area of plate

d : Space between plates

$$\text{For, Capacitor } C_1 \quad A_1 = A/3 \quad d_1 = d/2$$

$$\text{Capacitor } C_2 \quad A_2 = A/3 \quad d_2 = d/2$$

$$\text{Capacitor } C_3 \quad A_3 = A/3 \quad d_3 = d/2$$

$$\text{Capacitor } C_4 \quad A_4 = A \quad d_4 = d/2$$

$$\therefore C_1 = \frac{2k_1 \epsilon_0 A}{3d}, \quad C_2 = \frac{2k_2 \epsilon_0 A}{3d}$$

$$C_3 = \frac{2k_3 \epsilon_0 A}{3d}, \quad C_4 = \frac{k_4 2 \epsilon_0 A}{d}$$

Three capacitors C_1 , C_2 and C_3 are in parallel and it is series with C_4 .

$$\therefore \frac{1}{C} = \frac{1}{C_1 + C_2 + C_3} + \frac{1}{C_4}$$

Solving, we get

$$C = \frac{k \epsilon_0 A}{d}$$

$$\therefore \frac{d}{k \epsilon_0 A} = \frac{d}{2 \epsilon_0 A} \left[\frac{3}{k_1 + k_2 + k_3} + \frac{1}{k_4} \right]$$

$$\Rightarrow \frac{2}{k} = \left[\frac{3}{k_1 + k_2 + k_3} + \frac{1}{k_4} \right]$$

48. (a) Ist case

$$C_{eq} = \frac{C_1}{n_1} \text{ (Series combination)}$$

$$\therefore \text{Energy } E = \frac{1}{2} C V^2$$

$$\therefore E_1 = \frac{1}{2} \frac{C_1}{n_1} (4)^2$$

$$E_1 = \frac{1}{2} \frac{C_1}{n_1} 16$$

IInd case

$$C_{eq} = n_2 C_2 \text{ (Parallel combination)}$$

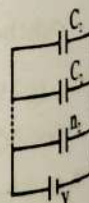
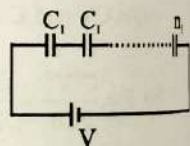
$$\therefore \text{Energy } E = \frac{1}{2} C V^2$$

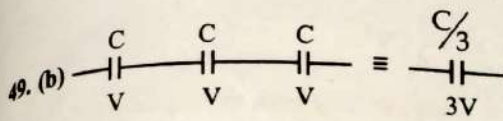
$$\therefore E_2 = \frac{1}{2} n_2 C_2 V^2$$

Given, energy is same

$$\therefore E_1 = E_2$$

$$\frac{1}{2} \frac{C_1}{n_1} 16 V^2 = \frac{1}{2} n_2 C_2 V^2 \Rightarrow C_2 = \frac{16 C_1}{n_1 n_2}$$

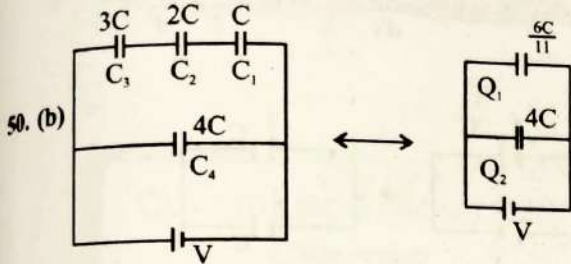




$$C_{eq} = \frac{C}{3}$$

$$V_{eq} = 3V$$

Breakdown voltage gets added in series.



$$C_{eq} = \frac{6C}{11} + 4C = \frac{50C}{11}$$

Net Charge:

$$Q_{net} = \frac{50CV}{11}$$

Charge in row-1,

$$Q_1 = \frac{Q_{net} C_1}{C_1 + C_2}$$

$$Q_1 = \frac{\left(\frac{50CV}{11}\right)\left(\frac{6C}{11}\right)}{\frac{6C}{11} + 4C}$$

$$\therefore Q_1 = \frac{6CV}{11}$$

In series combination, charge is same on all capacitors (C_1, C_2 and C_3).

Charge in row - 2,

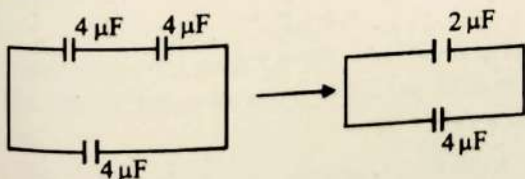
$$Q_2 = Q_{net} - Q_1 = \frac{50CV}{11} - \frac{6CV}{11} = \frac{44CV}{11}$$

$$Q_2 = 4CV$$

Ratio of charge on C_2 and C_4

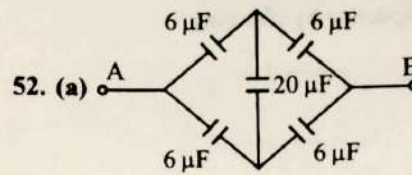
$$\frac{Q_1}{Q_2} \Rightarrow \frac{6CV}{11(4CV)} = \frac{3}{22}$$

51. (c) Let two capacitors are in series and it is parallel with third capacitor.

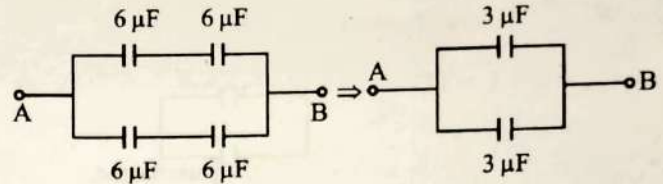


$$C_{eq} = 2 + 4 = 6 \mu F.$$

Hence, 2 capacitors in series, while 1 capacitors in parallel

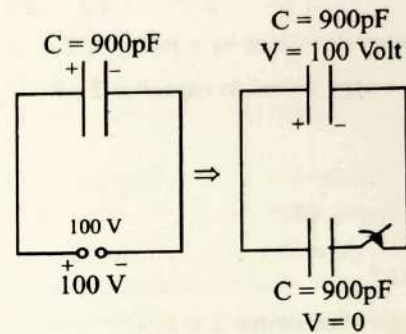


This is balanced wheatstone bridge so it is equivalent to



$$\text{So, } C_{AB} = 3 + 3 = 6 \mu F$$

53. (d)



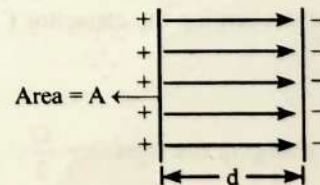
Common potential across the combination of the capacitors

$$V_c = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} = \frac{C \times 100 + C \times 0}{C + C} = 50 \text{ Volt}$$

Electrostatic energy stored by the system (b)

$$\begin{aligned} &= \frac{1}{2} CV^2 + \frac{1}{2} CV^2 = CV^2 \\ &= 900 \times 10^{-12} \times 50 \times 50 \\ &= 2.25 \times 10^{-6} \text{ J} \end{aligned}$$

54. (b)



Energy density is given by;

$$\frac{dU}{dV} = \frac{1}{2} \epsilon_0 E^2 \Rightarrow dU = \frac{1}{2} \epsilon_0 E^2 dV$$

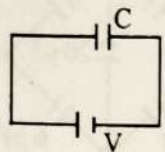
For total energy stored in capacitor,

$$\begin{aligned} U &= \int dU = \int \frac{1}{2} \epsilon_0 E^2 dV \\ &= \frac{1}{2} \epsilon_0 E^2 \int dV = \frac{1}{2} \epsilon_0 E^2 V \end{aligned}$$

$$U = \frac{1}{2} \epsilon_0 E^2 Ad$$

55. (a) Energy stored in capacitor by charging

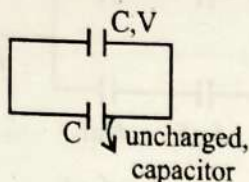
$$U_1 = \frac{1}{2} CV^2$$



When it is connected parallel to uncharged capacitor

$$V_1 = \frac{V}{2} \text{ [potential equal]}$$

$$C_{\text{eff}} = C + C = 2C$$



$$\text{Now Energy stored, } U_2 = \frac{1}{2} \times (2C) \left(\frac{V}{2} \right)^2 = \frac{U}{2}$$

Hence Energy decreases by a factor of 2.

56. (d) Initial energy stored in capacitor $2 \mu\text{F}$

$$U_i = \frac{1}{2} CV^2$$

$$U_i = \frac{1}{2} \times 2 \times 10^{-6} \times V^2$$

$$U_i = V^2 \times 10^{-6}$$

Final voltage after switch 2 is ON

$$V_f = \frac{C_1 V_1}{C_1 + C_2} = \frac{2V}{10} = 0.2V$$

Final energy in both capacitors

$$U_f = \frac{1}{2} (C_1 + C_2) V_f^2 = \frac{1}{2} \times 10 \times 10^{-6} \times \left(\frac{2V}{10} \right)^2$$

$$U_f = 0.2V^2 \times 10^{-6}$$

$$\text{So energy dissipated} = \frac{U_i - U_f}{U_i} \times 100$$

$$\text{Energy dissipated} = \frac{V^2 - 0.2V^2}{V^2} \times 100 = 80\%$$

57. (b) Work done is charging the capacitor C,

$$W_1 = \frac{1}{2} CV^2$$

Work done in charging the capacitor $\frac{C}{2}$,

$$W_2 = \frac{1}{2} \left(\frac{C}{2} \right) \times V^2 = \frac{1}{4} CV^2$$

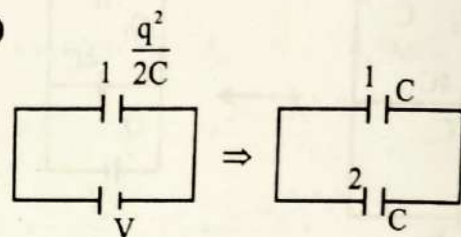
Work done in charging fully both the capacitors

$$W = W_1 + W_2$$

$$W = \frac{1}{2} CV^2 + \frac{1}{2} \left(\frac{C}{2} \right) V^2 = \frac{3}{4} CV^2$$

$$58. (a) \text{ Energy density} = \frac{dU}{dV} = \frac{1}{2} \epsilon_0 E^2 = \frac{1}{2} \epsilon_0 \frac{V^2}{d^2}$$

59. (b)



$$\text{Energy stored (U)} = \frac{q^2}{2C}$$

After connecting with another capacitor

$$V_{\text{common}} = \frac{q_1 + q_2}{C_1 + C_2} = \frac{q + 0}{C + C} = \frac{q}{2C}$$

$\therefore$ Energy on each capacitor

$$= \frac{1}{2} C V_{\text{common}}^2 = \frac{C}{2} \left[\frac{q}{2C} \right]^2$$

$$= \frac{q^2}{8C} = \frac{U}{4}$$

60. (d) Potential of the conductor $V = \frac{Q}{C}$,

Work done to bring a small charge dQ at this potential is,

$$dW = V dQ$$

Total work done-

$$W = \int_0^Q dW = \int_0^Q V dQ = \int_0^Q \frac{Q}{C} dQ$$

$$W = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} CV^2$$

$$(\because Q = CV)$$

The energy stored in a capacitor (work done) of capacity C and potential V is $\frac{CV^2}{2}$.

Current Electricity

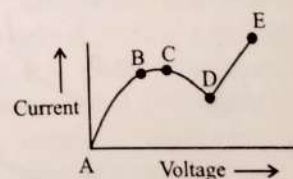
Current and Motion of Charges

- Across a metallic conductor of non-uniform cross section a constant potential difference is applied. The quantity which remains constant along the conductor is: (2015)
 - Current
 - Drift velocity
 - Electric field
 - Current density
- A flow of 10^7 electrons per second in a conducting wire constitutes a current of (1994)
 - 1.6×10^{-12} A
 - 1.6×10^{26} A
 - 1.6×10^{-26} A
 - 1.6×10^{12} A
- The velocity of charge carriers of current (about 1 ampere) in a metal under normal conditions is of the order of (1991)
 - A fraction of mm/sec
 - Velocity of light
 - Several thousand metres/second
 - A few hundred metres per second

Ohm's Law Resistance and Resistivity

- Two solid conductors are made up of same material have same length and same resistance. One of them has a circular cross section of area A_1 and the other one has a square cross section of area A_2 . The ratio A_1/A_2 is (2020-Covid)
 - 1
 - 0.8
 - 2
 - 1.5
- The resistance of a wire is 'R' ohm. If it is melted and stretched to 'n' times its original length, its new resistance will be: (2017-Delhi)
 - $\frac{R}{n}$
 - $n^2 R$
 - $\frac{R}{n^2}$
 - nR

- A wire of resistance 4Ω is stretched to twice its original length. The resistance of stretched wire would be: (2013)
 - 16Ω
 - 2Ω
 - 4Ω
 - 8Ω
- A wire of a certain material is stretched slowly by ten percent. Its new resistance and specific resistance become respectively: (2008)
 - 1.1 times, 1.1 times
 - 1.2 times, 1.1 times
 - 1.21 times, same
 - Both remain the same
- The electric resistance of a certain wire of iron is R. If its length and radius are both doubled, then: (2004)
 - The resistance will be halved and the specific resistance will remain unchanged
 - The resistance will be halved and the specific resistance will be doubled
 - The resistance and the specific resistance, will both remain unchanged
 - The resistance will be doubled and the specific resistance will be halved
- Copper and silicon is cooled from 300 K to 60 K, the specific resistance: (2001)
 - Decrease in copper but increase in silicon
 - Increase in copper but decrease in silicon
 - Increase in both
 - Decrease in both
- Three copper wires of lengths and cross-sectional areas are (l, A), ($2l, A/2$) and ($l/2, 2A$). Resistance is minimum in (1997)
 - Wire of cross-sectional area $2A$
 - Wire of cross-sectional area $\frac{A}{2}$
 - Wire of cross-sectional area A
 - Same in all three cases
- From the graph between current (I) and voltage (V) is shown below. Identify the portion corresponding to negative resistance (1997)



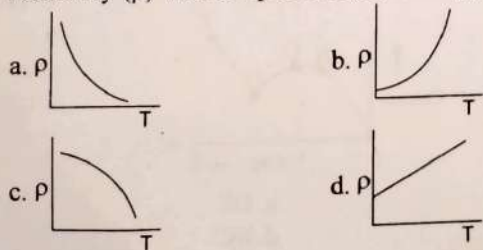
- CD
- DE
- AB
- BC

12. A wire 50 cm long and 1 mm² in cross-section carries a current of 4 A when connected to a 2 V battery. The resistivity of the wire is: (1994)
- $4 \times 10^{-6} \Omega \cdot \text{m}$
 - $1 \times 10^{-6} \Omega \cdot \text{m}$
 - $2 \times 10^{-7} \Omega \cdot \text{m}$
 - $5 \times 10^{-7} \Omega \cdot \text{m}$
13. Identify the set in which all the three materials are good conductors of electricity (1994)
- Cu, Hg and NaCl
 - Cu, Ge and Hg
 - Cu, Ag and Au
 - Cu, Si and diamond.
14. The masses of the wires of copper is in the ratio of 1 : 3 : 5 and their lengths are in the ratio of 5 : 3 : 1. The ratio of their electrical resistance is: (1988)
- 1 : 3 : 5
 - 5 : 3 : 1
 - 1 : 25 : 125
 - 125 : 15 : 1

Resistance and Conductance

15. A copper wire of length 10 m and radius $(10^{-2}/\sqrt{\pi})$ m has electrical resistance of 10 Ω . The current density in the wire for an electric field strength of 10 V/m is: (2022)
- 10^5 A/m^2
 - 10^4 A/m^2
 - 10^6 A/m^2
 - 10^{-5} A/m^2
16. Column-I gives certain physical terms associated with flow of current through a metallic conductor. Column-II gives some mathematical relations involving electrical quantities. Match Column-I and Column-II with appropriate relations. (2021)
- | Column-I | Column-II |
|----------------------------|--------------------------|
| (A) Drift Velocity | (P) $\frac{m}{ne^2\rho}$ |
| (B) Electrical Resistivity | (Q) nev_d |
| (C) Relaxation Period | (R) $\frac{eE}{m}\tau$ |
| (D) Current Density | (S) $\frac{E}{J}$ |
- (A) - (R), (B) - (S), (C) - (Q), (D) - (P)
 - (A) - (R), (B) - (P), (C) - (S), (D) - (Q)
 - (A) - (R), (B) - (Q), (C) - (S), (D) - (P)
 - (A) - (R), (B) - (S), (C) - (P), (D) - (Q)

17. A charged particle having drift velocity of $7.5 \times 10^{-4} \text{ m S}^{-1}$ in an electric field of $3 \times 10^{-10} \text{ Vm}^{-1}$, has a mobility in $\text{m}^2 \text{ V}^{-1} \text{ S}^{-1}$ of: (2020)
- 2.5×10^6
 - 2.5×10^{-6}
 - 2.25×10^{-15}
 - 2.25×10^{15}
18. Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for copper (2020)



19. Two metal wires of identical dimensions are connected in series. If σ_1 and σ_2 are the conductivities of the metal wires respectively, the effective conductivity of the combination is: (2015 Re)

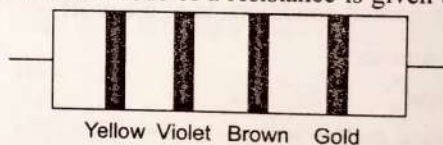
- $\frac{\sigma_1\sigma_2}{\sigma_1 + \sigma_2}$
- $\frac{2\sigma_1\sigma_2}{\sigma_1 + \sigma_2}$
- $\frac{\sigma_1 + \sigma_2}{2\sigma_1\sigma_2}$
- $\frac{\sigma_1 + \sigma_2}{\sigma_1\sigma_2}$

20. The mean free path of electrons in a metal is $4 \times 10^{-8} \text{ m}$. The electric field which can give on an average 2 eV energy to an electron in the metal will be in units V/m. (2009)
- 5×10^{-11}
 - 8×10^{-11}
 - 5×10^7
 - 8×10^7

21. Specific resistance of a conductor increases with: (2002)
- Increase in temperature
 - Increase in cross section area
 - Increase in cross section and decrease in length
 - Decrease in cross section area

Colour Code of Carbon Resistance

22. The color code of a resistance is given below: (2020)



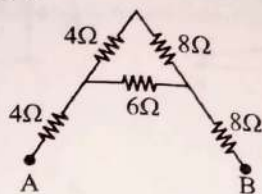
The values of resistance and tolerance, respectively, are:

- 47 k Ω , 10%
 - 4.7 k Ω , 5%
 - 470 Ω , 5%
 - 470 k Ω , 5%
23. A carbon resistor of $(47 \pm 4.7) \text{ k}\Omega$ is to be marked with rings of different colours for its identification. The colour code sequence will be (2018)
- Yellow - Green - Violet - Gold
 - Yellow - Violet - Orange - Silver
 - Violet - Yellow - Orange - Silver
 - Green - Orange - Violet - Gold

Combination of Resistance

24. The effective resistance of a parallel connection that consists of four wires of equal length, equal area of cross-section and same material is 0.25 Ω . What will be the effective resistance if they are connected in series? (2021)
- 0.5 Ω
 - 1 Ω
 - 4 Ω
 - 0.25 Ω

25. The equivalent resistance between A and B for the mesh shown in the figure is (2020-Covid)

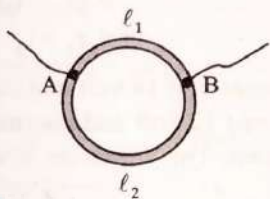


- a. $16\ \Omega$
b. $30\ \Omega$
c. $4.8\ \Omega$
d. $7.2\ \Omega$
26. A circuit contains an ammeter, a battery of 30 V and a resistance 40.8 ohm all connected in series. If the ammeter has a coil of resistance 480 ohm and a shunt of 20 ohm, the reading in the ammeter will be: (2015 Re)

- a. 1 A
b. 0.5 A
c. 0.25 A
d. 2 A
27. A ring is made of a wire having a resistance $R_0 = 12\ \Omega$. Find the points A and B, as shown in the figure, at which a current carrying conductor should be connected so that the resistance R of the sub circuit between these points is equal to

$$\frac{8}{3}\ \Omega$$

(2012 Pre)



- a. $\frac{l_1}{l_2} = \frac{5}{8}$
b. $\frac{l_1}{l_2} = \frac{1}{3}$
c. $\frac{l_1}{l_2} = \frac{3}{8}$
d. $\frac{l_1}{l_2} = \frac{1}{2}$

28. A wire of resistance 12 ohms per meter is bent to form a complete circle of radius 10 cm. The resistance between its two diametrically opposite points, A and B as shown in the figure, is: (2009)

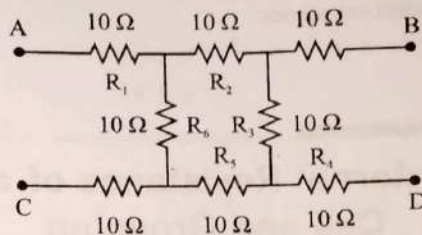


- a. $3\ \Omega$
b. $6\pi\ \Omega$
c. $6\ \Omega$
d. $0.6\pi\ \Omega$
29. When a wire of uniform cross-section a , length l and resistance R is bent into a complete circle, resistance between two of diametrically opposite points will be: (2005)

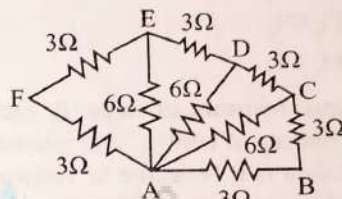
- a. $R/4$
b. $2R$
c. $4R$
d. $R/2$
30. Resistances n , each of r ohm, when connected in parallel give an equivalent resistance of R ohm. If these resistances were connected in series, the combination would have a resistance in ohms, equal to: (2004)

- a. $\frac{R}{n^2}$
b. $\frac{R}{n}$
c. nR
d. n^2R

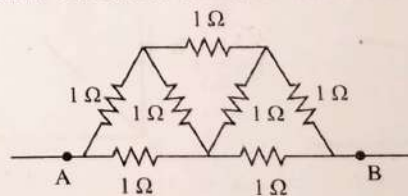
31. What will be the equivalent resistance between the two points A and D? (1996)



- a. $30\ \Omega$
b. $40\ \Omega$
c. $10\ \Omega$
d. $10\ \Omega$
32. Two wires of the same metal have same length, but their cross-sections are in the ratio 3 : 1. They are joined in series. The resistance of thicker wire is $10\ \Omega$. The total resistance of the combination will be (1995)
- a. $40\ \Omega$
b. $100\ \Omega$
c. $(5/2)\ \Omega$
d. $(40/3)\ \Omega$
33. Six resistors of $3\ \Omega$ each are connected along the sides of a hexagon and three resistors of $6\ \Omega$ each are connected along AC, AD and AE as shown in the figure. The equivalent resistance between A and B is equal to (1994)



- a. $2\ \Omega$
b. $6\ \Omega$
c. $3\ \Omega$
d. $9\ \Omega$
34. Three resistances each of $4\ \Omega$ are connected to form a triangle. The resistance between any two terminals is (1993)
- a. $12\ \Omega$
b. $2\ \Omega$
c. $6\ \Omega$
d. $8/3\ \Omega$
35. In the network shown in figure each resistance is $1\ \Omega$. The effective resistance between A and B is (1990)



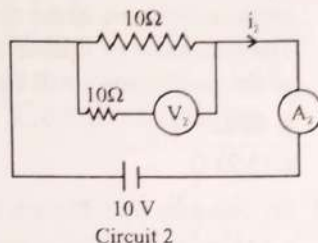
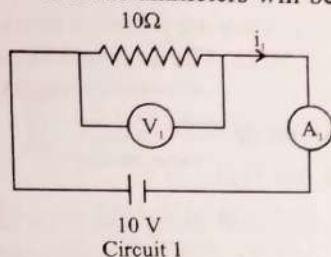
- a. $\frac{4}{3}\ \Omega$
b. $\frac{3}{2}\ \Omega$
c. $7\ \Omega$
d. $\frac{8}{7}\ \Omega$
36. You are given several identical resistances each of value $R = 10\ \Omega$ and each capable of carrying a maximum current of one ampere. It is required to make a suitable combination of these resistances of $5\ \Omega$ which can carry a current of 4 ampere. The minimum number of resistances of the type R that will be required for this job is: (1990)
- a. 4
b. 10
c. 8
d. 20

37. n equal resistors are first connected in series and then connected in parallel, what is the ratio of the maximum to the minimum resistance? (1989)

a. n b. $1/n^2$
c. n^2 d. $1/n$

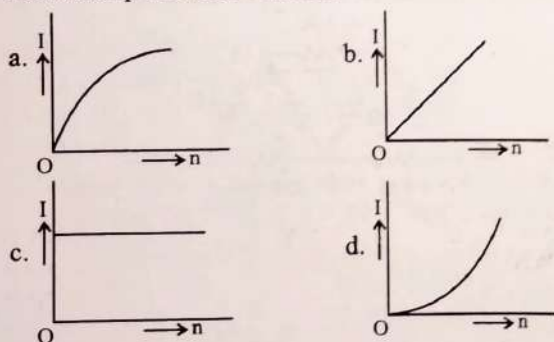
Internal Resistance of a Cell and Grouping

38. In the circuits shown below, the readings of the voltmeters and the ammeters will be (2019)



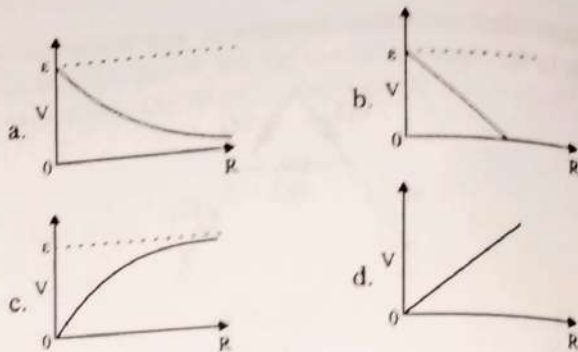
- a. $V_2 > V_1$ and $i_1 = i_2$
b. $V_1 = V_2$ and $i_1 > i_2$
c. $V_1 = V_2$ and $i_1 = i_2$
d. $V_2 > V_1$ and $i_1 > i_2$
39. A set of ' n ' equal resistors, of value ' R ' each, are connected in series to a battery of emf ' E ' and internal resistance ' R '. The current drawn is I . Now, the ' n ' resistors are connected in parallel to the same battery. Then the current drawn from battery becomes $10 I$. The value of ' n ' is (2018)
- a. 20 b. 11
c. 10 d. 9

40. A battery consists of a variable number ' n ' of identical cells (having internal resistance ' r ' each) which are connected in series. The terminals of the battery are short-circuited and the current I is measured. Which of the graphs shows the correct relationship between I and n ? (2018)



41. The internal resistance of a 2.1 V cell which gives a current of 0.2 A through a resistance of 10Ω is: (2013)
- a. 1.0Ω b. 0.2Ω
c. 0.5Ω d. 0.8Ω

42. A cell having an emf ϵ and internal resistance r is connected across a variable external resistance R . As the resistance R is increased, the plot of potential difference V across R is given by: (2012 Mains)



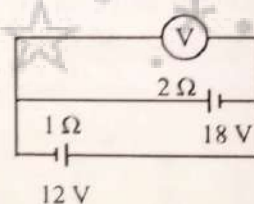
43. A student measures the terminal potential difference (V) of a cell (of emf ϵ and internal resistance r) as a function of the current (I) flowing through it. The slope, and intercept, of the graph between V and I , then, respectively, equal: (2009)

a. $-r$ and ϵ b. r and $-\epsilon$
c. $-\epsilon$ and r d. ϵ and $-r$

44. Two cells, having the same emf, are connected in series through an external resistance R . Cells have internal resistances r_1 and r_2 ($r_1 > r_2$) respectively. When the circuit is closed, the potential difference across the first cell is zero. The value of R is: (2006)

a. $r_1 - r_2$ b. $(r_1 + r_2)/2$
c. $(r_1 - r_2)/2$ d. $r_1 + r_2$

45. Two batteries, one of emf 18 volt and internal resistance 2Ω and the other of emf 12 volt and internal resistance 1Ω , are connected as shown. The voltmeter V will record a reading of: (2001)



a. 30 volt b. 18 volt
c. 15 volt d. 14 volt

46. For a cell terminal P.D. is 2.2 V when circuit is open and reduces to 1.8 V when cell is connected to a resistance of $R = 5 \Omega$. Determine internal resistance of cell (r): (2002)

a. $\frac{10}{9} \Omega$ b. $\frac{9}{10} \Omega$
c. $\frac{11}{9} \Omega$ d. $\frac{5}{9} \Omega$

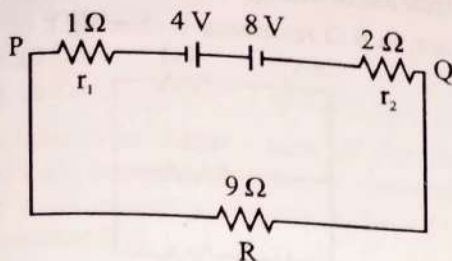
47. A car battery of emf 12 V and internal resistance $5 \times 10^{-2} \Omega$ receives a current of 60 A from external source, then terminal voltage of battery is: (2000)

a. 12 V b. 9 V
c. 15 V d. 20 V

48. The terminal potential difference of a cell is greater than its emf when: (1990)

a. A battery of less emf is connected in its series
b. A battery of higher emf is connected in its series.
c. A battery of higher emf is connected in its parallel.
d. A battery of less emf is connected in its parallel.

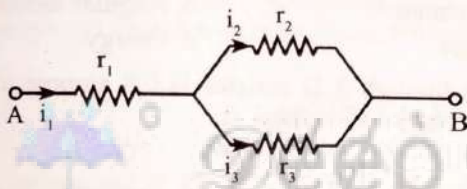
49. Two batteries of emf 4 V and 8 V with internal resistance $1\ \Omega$ and $2\ \Omega$ are connected in a circuit with resistance of $9\ \Omega$ as shown in figure. The current and potential difference between the points P and Q are (1988)



- a. $\frac{1}{3}\text{ A}$ and 3 V
b. $\frac{1}{6}\text{ A}$ and 4 V
c. $\frac{1}{9}\text{ A}$ and 9 V
d. $\frac{1}{12}\text{ A}$ and 12 V

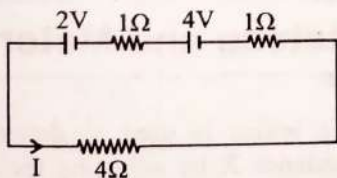
Kirchhoff's Law

50. Three resistors having resistances r_1 , r_2 and r_3 are connected as shown in the given circuit. The ratio $\frac{i_3}{i_1}$ of currents in terms of resistances used in the circuit is: (2021)



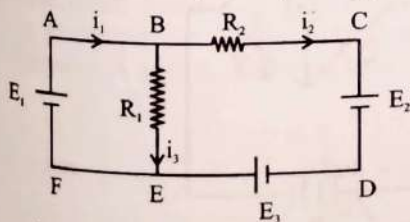
- a. $\frac{r_2}{r_2 + r_3}$
b. $\frac{r_1}{r_1 + r_2}$
c. $\frac{r_2}{r_1 + r_3}$
d. $\frac{r_1}{r_2 + r_3}$

51. For the circuit shown in the figure, the current I will be (2020-Covid)



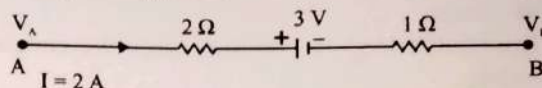
- a. 1 A
b. 1.5 A
c. 0.5 A
d. 0.75 A

52. For the circuit given below, the Kirchhoff's loop rule for the loop BCDEB is given by the equation (2020-Covid)



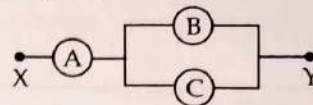
- a. $i_2 R_2 + E_2 - E_3 - i_3 R_1 = 0$
b. $i_2 R_2 + E_2 + E_3 - i_3 R_1 = 0$
c. $-i_2 R_2 + E_2 + E_3 - i_3 R_1 = 0$
d. $-i_2 R_2 + E_2 - E_3 - i_3 R_1 = 0$

53. The potential difference ($V_A - V_B$) between the points A and B in the given figure is: (2016 - II)



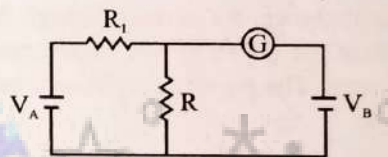
- a. +6 V
b. +9 V
c. -3 V
d. +3 V

54. A, B and C are voltmeters of resistance R , $1.5R$ and $3R$ respectively as shown in the figure. When some potential difference is applied between X and Y, the voltmeter readings are V_A , V_B and V_C respectively, then: (2015)



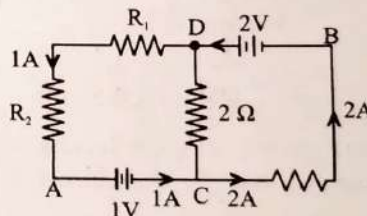
- a. $V_A \neq V_B \neq V_C$
b. $V_A = V_B \neq V_C$
c. $V_A \neq V_B \neq V_C$
d. $V_A = V_B = V_C$

55. In the circuit shown the cells A and B have negligible resistances. For $V_A = 12\text{ V}$, $R_1 = 500\ \Omega$ and $R = 100\ \Omega$ the galvanometer (G) shows no deflection. The value of V_B is: (2012 Pre)



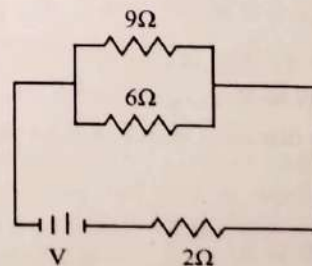
- a. 4 V
b. 2 V
c. 12 V
d. 6 V

56. In the circuit shown in the figure, if the potential at point A is taken to be zero, the potential at point B is: (2011 Mains)



- a. -1 V
b. +2 V
c. -2 V
d. +1 V

57. If power dissipated in the $9\ \Omega$ resistor in the circuit shown is 36 Watt, the potential difference across the $2\ \Omega$ resistor is: (2011 Pre)



- a. 4 Volt
b. 8 Volt
c. 10 Volt
d. 2 Volt

58. Consider the following two statements

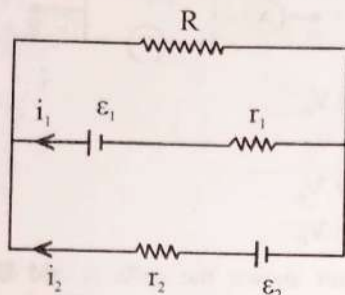
- A. Kirchhoff's junction law follows from the conservation of charge.
B. Kirchhoff's loop law follows from the conservation of energy.

Which of the following is correct?

(2010 Pre)

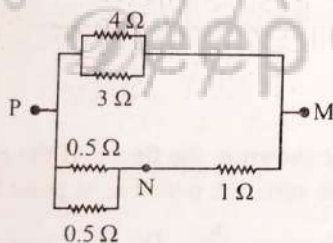
- a. Both (A) and (B) are correct
b. Both (A) and (B) are wrong
c. (A) is correct and (B) is wrong
d. (A) is wrong and (B) is correct

59. See the electric circuit shown in this figure. Which of the following equations is a correct equation for it? (2009)



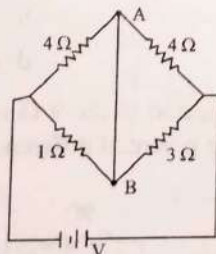
- a. $\varepsilon_2 - i_2 r_2 - \varepsilon_1 - i_1 r_1 = 0$ b. $-\varepsilon_2 - (i_1 + i_2) R + i_2 r_2 = 0$
c. $\varepsilon_1 - (i_1 + i_2) R + i_1 r_1 = 0$ d. $\varepsilon_1 - (i_1 + i_2) R - i_1 r_1 = 0$

60. In the circuit shown, the current through the $4\ \Omega$ resistor is 1 Amp when the points P and M are connected to a d.c. voltage source. The potential difference between the points M and N is: (2008)



- a. 3.2 volt b. 1.5 volt
c. 1.0 volt d. 0.5 volt

61. In the circuit shown, if a conducting wire is connected between points A and B, the current in this wire will: (2006)

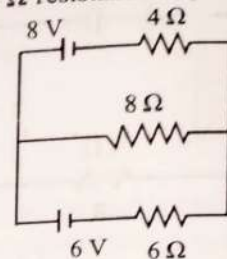


- a. Flow from A to B
b. Flow in the direction which will be decided by the value of V
c. Be zero
d. Flow from B to A

62. Kirchhoff's first and second laws for electrical circuits are consequences of: (2006)

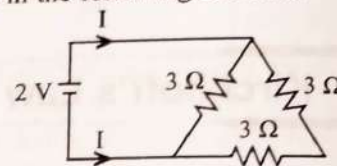
- a. Conservation of energy
b. Conservation of electric charge and energy respectively
c. Conservation of electric charge
d. Conservation of energy and electric charge respectively

63. The current in $8\ \Omega$ resistance is (See fig.): (1999)



- a. 0.69 A b. 0.92 A
c. 1.30 A d. 1.6 A

64. The current in the following circuit is (1997)

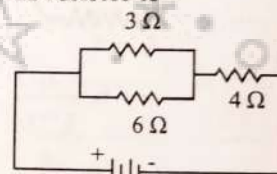


- a. $2/3$ A b. 1 A
c. $1/8$ A d. $2/9$ A

65. Kirchhoff's first law, i.e. $\sum i = 0$ at a junction, deals with the conservation of (1997)

- a. Momentum b. Angular momentum
c. Charge d. Energy

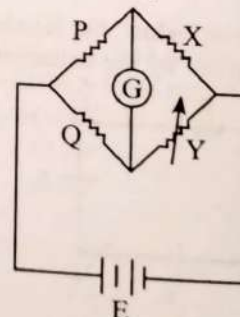
66. Current through $3\ \Omega$ resistor is 0.8 ampere, then potential drop through $4\ \Omega$ resistor is (1993)



- a. 9.6 V b. 2.6 V
c. 4.8 V d. 1.2 V

Wheatstone and Meter Bridge

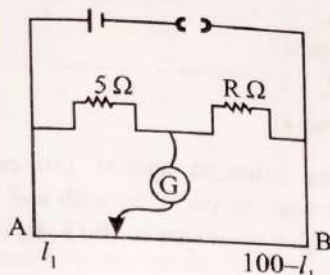
67. A wheatstone bridge is used to determine the value of unknown resistance X by adjusting the variable resistance Y as shown in the figure. For the most precise measurement of X, the resistances P and Q: (2022)



- a. Do not play any significant role
b. Should be approximately equal to $2X$
c. Should be approximately equal and are small
d. Should be very large and unequal

68. A resistance wire connected in the left gap of a metre bridge balances a 10Ω resistance in the right gap at a point which divides the bridge wire in the ratio 3 : 2. If the length of the resistance wire is 1.5 m, then the length of 1Ω of the resistance wire is: (2020)
- a. 1.0×10^{-1} m
b. 1.5×10^{-1} m
c. 1.5×10^{-2} m
d. 1.0×10^{-2} m

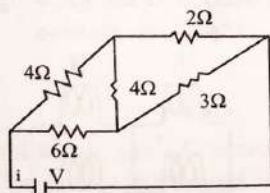
69. The resistances in the two arms of the meter bridge are 5Ω and $R\Omega$, respectively. When the resistance R is shunted with an equal resistance, the new balance point is at $1.6 l_1$. The resistance R is: (2014)



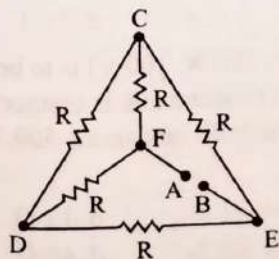
- a. 10Ω
b. 15Ω
c. 20Ω
d. 25Ω
70. The resistances of the four arms P, Q, R and S in a Wheatstone's bridge are 10 ohm, 30 ohm, 30 ohm and 90 ohm, respectively. The e.m.f. and internal resistance of the cell are 7 volt and 5 ohm respectively. If the galvanometer resistance is 50 ohm, the current drawn from the cell will be: (2013)

- a. 2.0 A
b. 1.0 A
c. 0.2 A
d. 0.1 A
71. Three resistances P, Q, R each of 2Ω and an unknown resistance S form the four arms of a Wheatstone bridge circuit. When a resistance of 6Ω is connected in parallel to S the bridge gets balanced. What is the value of S? (2007)

- a. 3Ω
b. 6Ω
c. 1Ω
d. 2Ω
72. For the network shown in the figure, the value of the current I is: (2005)



- a. $8V/7$
b. $5V/18$
c. $9V/7$
d. $18V/5$
73. Five equal resistances each of resistance R are connected as shown in the figure. A battery of V volts is connected between A and B. The current flowing in AFCEB will be: (2004)



a. $\frac{V}{R}$

b. $\frac{V}{2R}$

c. $\frac{2V}{R}$

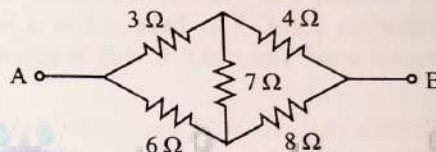
d. $\frac{3V}{R}$

74. A 6 volt battery is connected to the terminals of a three metre long wire of uniform thickness and resistance of 100 ohm. The difference of potential between two points on the wire separated by a distance of 50 cm will be: (2004)

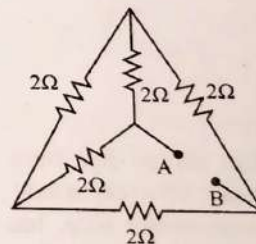
- a. 3 V
b. 1 V
c. 1.5 V
d. 2 V
75. In a Wheatstone's bridge all the four arms have equal resistance R . If the resistance of the galvanometer arm is also R , the equivalent resistance of the combination as seen by the battery is: (2003)

- a. $R/4$
b. $R/2$
c. R
d. $2R$
76. The resistance of each arm of the Wheatstone bridge is 10 ohm. A resistance of 10 ohm is connected in series with galvanometer then the equivalent resistance across the battery will be: (2001)

- a. 10 ohm
b. 15 ohm
c. 20 ohm
d. 40 ohm
77. Calculate the net resistance of the circuit between A and B: (2000)



- a. $8/3\Omega$
b. $14/3\Omega$
c. $16/3\Omega$
d. $22/3\Omega$
78. In the network shown in the figure, each of the resistance is equal to 2Ω . The resistance between the points A and B is (1995)



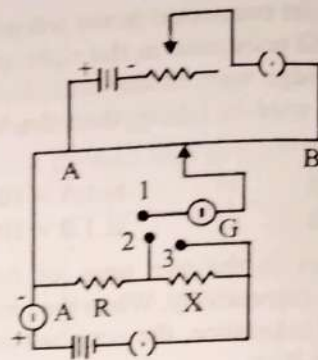
- a. 3Ω
c. 1Ω

- b. 4Ω
d. 2Ω

Potentiometer (i) Comparison of E.M.F. of Two Cells (ii) Internal Resistance of a Cell

79. In a potentiometer circuit a cell of EMF 1.5 V gives balance point at 36 cm length of wire. If another cell of EMF 2.5 V replaces the first cell, then at what length of the wire, the balance point occurs? (2021)
- a. 21.6 cm
b. 64 cm
c. 62 cm
d. 60 cm

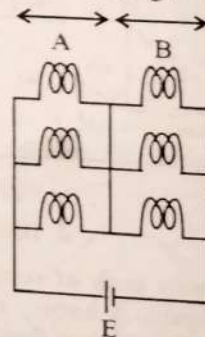
80. A potentiometer is an accurate and versatile device to make electrical measurements of E.M.F. because the method involves: (2017-Delhi)
- Potential gradients
 - A condition of no current flow through the galvanometer
 - A combination of cells, galvanometer and resistances
 - Cells
81. A potentiometer wire is 100 cm long and a constant potential difference is maintained across it. Two cells are connected in series first to support one another and then in opposite direction. The balance points are obtained at 50 cm and 10 cm from the positive end of the wire in the two cases. The ratio of emf's is: (2016 - I)
- 5 : 1
 - 5 : 4
 - 3 : 4
 - 3 : 2
82. A potentiometer wire has length 4 m and resistance 8 Ω . The resistance that must be connected in series with the wire and an accumulator of e.m.f. 2 V, so as to get a potential gradient 1 mV per cm on the wire is: (2015)
- 40 Ω
 - 44 Ω
 - 48 Ω
 - 32 Ω
83. A potentiometer wire of length L and a resistance r are connected in series with a battery of e.m.f. E_0 and a resistance r_1 . An unknown e.m.f. E is balanced at a length l of the potentiometer wire. The e.m.f. E will be given by: (2015 Re)
- $\frac{LE_0 r}{(r + r_1) \ell}$
 - $\frac{LE_0 r}{\ell r_2}$
 - $\frac{E_0 r}{(r + r_1)} \cdot \frac{\ell}{L}$
 - $\frac{E_0 \ell}{L}$
84. A potentiometer circuit has been set up for finding the internal resistance of a given cell. The main battery, used across the potentiometer wire, has an emf of 2.0 V and a negligible internal resistance. The potentiometer wire itself is 4 m long. When the resistance, R , connected across the given cell, has values of (i) Infinity, (ii) 9.5 Ω
- The 'balancing lengths', on the potentiometer wire are found to be 3 m and 2.85 m, respectively. The value of internal resistance of the cell is: (2014)
- 0.25 Ω
 - 0.95 Ω
 - 0.5 Ω
 - 0.75 Ω
85. A potentiometer circuit is set up as shown. The potential gradient, across the potentiometer wire, is k volt/cm and the ammeter, present in the circuit, reads 1.0 A when two way key is switched off. The balance points, when the key between the terminals (i) 1 and 2 (ii) 1 and 3, is plugged in, are found to be at lengths l_1 cm and l_2 cm respectively. The magnitudes, of the resistors R and X , in ohms, are then, equal, respectively, to: (2010 Pre)



- $k l_1$ and $k l_2$
 - $k(l_2 - l_1)$ and $k l_2$
 - $k l_1$ and $k(l_2 - l_1)$
 - $k(l_2 - l_1)$ and $k l_1$
86. A cell can be balanced against 110 cm and 100 cm of potentiometer wire, respectively with and without being short circuited through a resistance of 10 Ω . Its internal resistance is: (2008)
- Zero
 - 1.0 ohm
 - 0.5 ohm
 - 2.0 ohm
87. The potentiometer is best for measuring voltage, as: (2000)
- It has a sensitive galvanometer and gives null deflection
 - It has wire of high resistance
 - It measures p.d. like in closed circuit
 - It measures p.d. like in open circuit

Electrical Power

88. Six similar bulbs are connected as shown in the figure with a DC source of emf E and zero internal resistance. The ratio of power consumption by the bulbs when (i) all are glowing and (ii) in the situation when two from section A and one from section B are glowing, will be: (2019)

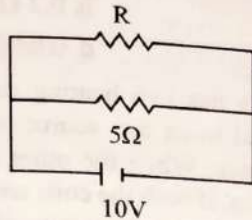


- 4 : 9
 - 1 : 2
 - 9 : 4
 - 2 : 1
89. A filament bulb (500 W, 100 V) is to be used in a 230 V main supply. When a resistance R is connected in series, it works perfectly and the bulb consumes 500 W. The value of R is: (2016 - II)
- 26 Ω
 - 230 Ω
 - 13 Ω
 - 46 Ω

90. Two cities are 150 km apart. Electric power is sent from one city to another city through copper wires. The fall of potential per km is 8 volt and the average resistance per km is 0.5Ω . The power loss in the wire is: (2014)

a. 19.2 W
b. 19.2 kW
c. 19.2 J
d. 12.2 kW

91. The power dissipated in the circuit shown in the figure is 30 watts. The value of R is: (2012 Mains)

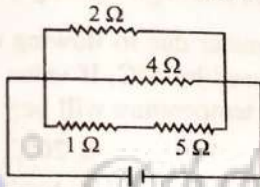


a. 20 Ω
b. 15 Ω
c. 10 Ω
d. 30 Ω

92. If voltage across a bulb rated 220 Volt - 100 Watt drops by 2.5% of its rated value, the percentage of the rated value by which the power would decrease is: (2012 Pre)

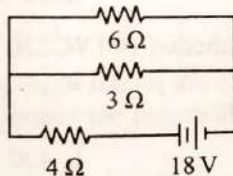
a. 20%
b. 2.5%
c. 5%
d. 10%

93. A current of 3 A flows through the 2Ω resistor shown in the circuit. The power dissipated in the 5Ω resistor is: (2008)



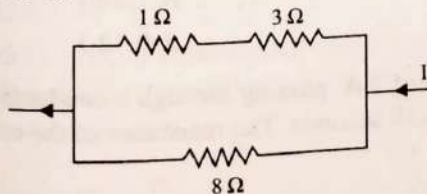
a. 5 watt
b. 4 watt
c. 2 watt
d. 1 watt

94. The total power dissipated in watts in the circuit shown here is (2007)



a. 40
b. 54
c. 4
d. 16

95. Power dissipated across the 8Ω resistor in the circuit shown here is 2 W. The power dissipated in watt units across the 3Ω resistor is: (2006)



a. 6.0
b. 1.5
c. 0.45
d. 3.0

96. When three identical bulbs of 60 watt, 200 volt rating are connected in series to a 200 volt supply, the power drawn by them will be: (2004)

a. 180 watt
b. 10 watt
c. 20 watt
d. 60 watt

97. In India electricity is supplied for domestic use at 220 V. It is supplied at 110 V in USA. If the resistance of a 60 W bulb for use in India is R, the resistance of a 60 W bulb for use in USA will be: (2004)

a. 2R
b. $\frac{R}{4}$
c. $\frac{R}{2}$
d. R

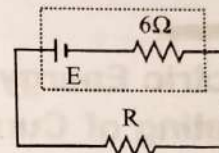
98. Two 220 volt, 100 watt bulbs are connected first in series and then in parallel. Each time the combination is connected to a 220 volt a.c. supply line. The power drawn by the combination in each case respectively will be: (2003)

a. 50 watt, 100 watt
b. 100 watt, 50 watt
c. 200 watt, 150 watt
d. 50 watt, 200 watt

99. Two bulbs of (40 W, 200 V), and (100 W, 200 V). Then correct relation for their resistances: (2000)

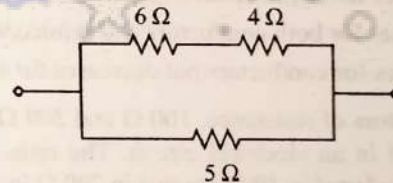
a. $R_{40} < R_{100}$
b. $R_{40} > R_{100}$
c. $R_{40} = R_{100}$
d. No relation can be predicted

100. The value of R for which power in it is maximum: (1999)



a. 3 Ω
b. 6 Ω
c. 12 Ω
d. 9 Ω

101. If the power dissipated in 5Ω is 20 W then power dissipated in 4Ω is: (1999)



a. 4 W
b. 6 W
c. 10 W
d. 20 W

102. When three identical bulbs are connected in series, the consumed power is 10 W. If they are now connected in parallel then the consumed power will be: (1998)

a. 30 W
b. 90 W
c. $\frac{10}{3}$ W
d. 270 W

103. A (100 W, 200 V) bulb is connected to a 160 volts supply. The power consumption would be (1997)

a. 100 W
b. 125 W
c. 64 W
d. 80 W

104. If two bulbs, whose resistances are in the ratio of 1 : 2 are connected in series, the power dissipated in them has the ratio of (1997)

a. 2 : 1
b. 1 : 4
c. 1 : 1
d. 1 : 2

105. An electric bulb is rated 60 W, 220 V. The resistance of its filament is (1994)

a. 870 Ω
b. 780 Ω
c. 708 Ω
d. 807 Ω

106. A battery of e.m.f 10 V and internal resistance 0.5Ω is connected across a variable resistance R . The value of R for which the power delivered in it is maximum is given by (1992)
- 0.5Ω
 - 1.0Ω
 - 2.0Ω
 - 0.25Ω
107. Two identical batteries each of e.m.f. 2 V and internal resistance 1Ω are available to produce heat in an external resistance by passing a current through it. The maximum power that can be developed across R using these batteries is (1990)
- 3.2 W
 - 2.0 W
 - 1.28 W
 - $\frac{8}{9}$ W
108. 40 electric bulbs are connected in series across a 220 V supply. After one bulb is fused the remaining 39 are connected again in series across the same supply. The illumination will be (1989)
- More with 40 bulbs than with 39
 - More with 39 bulbs than with 40
 - Equal in both the cases
 - In the ratio $40^2 : 39^2$

Electric Energy and Heating of Current

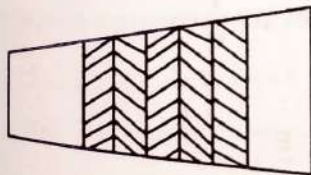
109. As the temperature increases, the electrical resistance: (2022)
- decrease for conductors but increases for semiconductors
 - increases for both conductors and semiconductors
 - decreases for both conductors and semiconductors
 - increases for conductors but decreases for semiconductors
110. Two resistors of resistance, 100Ω and 200Ω are connected in parallel in an electrical circuit. The ratio of the thermal energy developed in 100Ω to that in 200Ω in a given time is: (2022)
- 4 : 1
 - 1 : 4
 - 2 : 1
 - 1 : 2
111. Which of the following acts as a circuit protection device? (2019)
- Conductor
 - Inductor
 - Switch
 - Fuse
112. The charge flowing through a resistance R varies with time t as $Q = at - bt^2$, where a and b are positive constants. The total heat produced in R is: (2016 - I)
- $\frac{a^3 R}{6b}$
 - $\frac{a^3 R}{3b}$
 - $\frac{a^3 R}{2b}$
 - $\frac{a^3 R}{b}$
113. An electric kettle takes 4 A current at 220 V. How much time will it take to boil 1 kg of water from temperature 25°C to the temperature of boiling water is 100°C : (2019)
- 4.2 min
 - 6.3 min
 - 8.4 min
 - 12.6 min
114. A 5-A fuse wire can withstand a maximum power of 1 W in a circuit. The resistance of the fuse wire is: (2016)
- 0.02 Ω
 - 0.2 Ω
 - 0.4 Ω
 - 0.04 Ω
115. An electric kettle has two heating coils. When one of the coils is connected to an a.c. source, the water in the kettle boils in 10 minutes. When the other coil is used the water boils in 40 minutes. If both the coils are connected in parallel the time taken by the same quantity of water to boil will be: (2013)
- 8 min
 - 4 min
 - 25 min
 - 15 min
116. Fuse wire is a wire of: (2013)
- High resistance and high melting point
 - High resistance and low melting point
 - Low resistance and low melting point
 - Low resistance and high melting point
117. In hot wire Ammeter due to flowing of current temperature of wire is increased by 5°C . If value of current is doubled then increase in temperature will be: (1998)
- 15°C
 - 20°C
 - 25°C
 - 30°C
118. One kilowatt hour is equal to (1997)
- $36 \times 10^5 \text{ J}$
 - $36 \times 10^4 \text{ J}$
 - $36 \times 10^3 \text{ J}$
 - $36 \times 10^2 \text{ J}$
119. A heating coil is labelled 100 W, 220 V. The coil is cut in half and the two pieces are joined in parallel to the same source. The energy now liberated per second is (1995)
- 200 W
 - 400 W
 - 25 W
 - 50 W
120. A $4 \mu\text{F}$ capacitor is charged to 400 V. If its plates are joined through a resistance of $2 \text{ k}\Omega$, then heat produced in the resistance is: (1995)
- 0.64 J
 - 1.28 J
 - 0.16 J
 - 0.32 J
121. A current of 2 A, passing through a conductor produces 80 J of heat in 10 seconds. The resistance of the conductor in ohm is: (1990)
- 0.5
 - 2
 - 4
 - 20

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	a	a	a	b	a	c	a	a	a	a	b	c	d	a	d	a
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
b	b	c	a	c	b	c	a	b	d	d	a	d	a	a	a	d
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
d	c	c	c	c	c	c	c	a	a	d	a	c	c	a	a	a
52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68
a	b	d	b	d	c	a	d	a	d	b	a	b	c	c	c	a
69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
b	c	a	b	b	b	c	a	b	d	d	b	d	d	c	c	c
86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102
b	d	b	a	b	c	c	a	b	d	c	b	d	b	b	a	b
103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119
c	d	d	a	b	b	d	c	d	a	b	d	a	b	b	c	b
120	121															
d	b															

Explanations

1. (a) Metallic conductor can be considered as the combination of various conductors connected in series combination. And in series combination the current always remains constant.



2. (a) Flow of electrons, $\frac{n}{t} = 10^7 / \text{sec}$

$$\text{Therefore, current } (I) = \frac{q}{t} = \frac{ne}{t} = \frac{n}{t} \times e$$

$$= 10^7 \times (1.6 \times 10^{-19}) = 1.6 \times 10^{-12} \text{ A}$$

3. (a) The drift velocity is the actual velocity of charge carriers of current inside the metal lattice and its magnitude is near to fractions of mm/sec.

4. (a) Resistance $(R) = \frac{\rho \ell}{A}$

$$R \propto \frac{1}{A}$$

$$\therefore R_1 A_1 = R_2 A_2$$

$$\therefore \frac{A_1}{A_2} = \frac{R_2}{R_1} = 1 \text{ and } (\because R_2 = R_1)$$

5. (b) Volume = constant

$$A_1 L_1 = A_2 L_2$$

$$\pi r_1^2 \times \ell_0 = \pi r_2^2 (n \ell_0)$$

$$r_1^2 = n r_2^2$$

$$R = \rho \frac{\ell_0}{\pi r_1^2} \quad R'' = \rho \frac{n \ell_0}{\pi r_2^2}$$

According to the question,

$$R = \rho \frac{\ell_0}{\pi n r_2^2}$$

$$\therefore \frac{R''}{R} = \frac{\pi r_2^2}{\rho \ell_0} = n^2 \Rightarrow R'' = n^2 R$$

6. (a) Volume is constant

$$V_1 = V_2$$

$$\pi r^2 L = \pi R^2 2L$$

$$r^2 = 2R^2$$

$$R_1 = \rho \frac{L}{\pi r^2}, \quad R_2 = \rho \frac{2L}{\pi R^2}$$

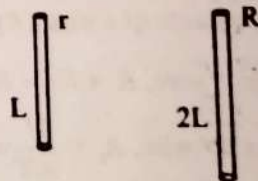
$$R_1 = \rho \frac{L}{\pi (2R^2)}$$

We get

$$R_2 = 4R_1 \quad ; \quad R_2 = 4(4 \Omega)$$

$$R_2 = 16 \Omega$$

$$[\because R_1 = 4 \Omega \text{ (given)}]$$



$$7. (c) R = \frac{\rho \ell}{A} \Rightarrow R = \frac{\rho \ell}{A} \times \frac{\ell}{\ell} = \frac{\rho \ell^2}{A \ell} = \frac{\rho \ell^2}{\text{Volume}}$$

As volume is constant,

$$R \propto \ell^2$$

$\therefore \ell$ is increased by 10 percent

$$\text{Final } \ell_2 = \ell_1 + 10\% \text{ of } \ell_1 = 1.1 \ell_1$$

$$\therefore \frac{R_1}{R_2} = \left(\frac{\ell_1}{\ell_2} \right)^2 = \left(\frac{1.1 \ell_1}{\ell_1} \right)^2 = 1.21$$

$$\therefore R_2 = 1.21 R_1$$

Resistance will become 1.21 times the original resistance.

$\therefore$ Specific resistance depends on the material of the wire, it will not change.

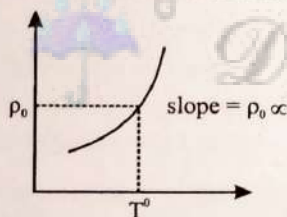
$$8. (a) R_1 = \frac{\rho \ell_1}{A_1} = \frac{\rho \ell_1}{\pi a_1^2}$$

When the length and radius of the wire are doubled, then new length,

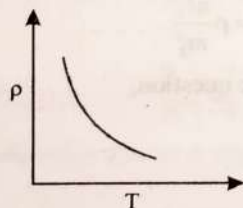
$$\ell_2 = 2\ell_1, a_2 = 2a_1$$

$$\therefore R_2 = \frac{\rho \ell_2}{\pi a_2^2} = \frac{\rho \times 2\ell_1}{\pi (2a_1)^2} = \frac{1}{2} \frac{\rho \ell_1}{\pi a_1^2} = \frac{1}{2} R_1$$

9. (a) With increase in temperature the ions of the conductor vibrate with greater amplitude, and the collisions between electrons and ions become frequent.



While in case of semiconductors (Si)



10. (a) Three wires of lengths and cross-sectional areas (ℓ , A), (2ℓ , $A/2$) and ($\ell/2$, $2A$).

$$\text{Resistance of a wire } (R) \propto \frac{\ell}{A}$$

$$\text{For I}^{\text{st}} \text{ wire, } R_1 \propto \ell/A = R$$

$$\text{For II}^{\text{nd}} \text{ wire, } R_2 \propto \frac{2\ell}{A/2} = 4R$$

$$\text{For III}^{\text{rd}} \text{ wire, } R_3 \propto \frac{\ell/2}{2A} = \frac{R}{4}$$

Therefore resistance of the wire will be minimum for IIIrd wire.

11. (a) For the negative resistance, when we increase the voltage, the current will decrease. Therefore from the graph, we find that the current in CD is decreased when voltage is increased.

12. (b) Length (ℓ) = 50 cm = 0.5 m;

$$\text{Area (a)} = 1 \text{ mm}^2 = 1 \times 10^{-6} \text{ m}^2;$$

$$\text{Current (I)} = 4 \text{ A and voltage (V)} = 2 \text{ volts.}$$

$$\text{Resistance (R)} = \frac{V}{I} = \frac{2}{4} = 0.5 \Omega \text{ and}$$

$$\text{Resistivity } (\rho) = R \times \frac{A}{\ell} = 0.5 \times \frac{1 \times 10^{-6}}{0.5} = 1 \times 10^{-6} \Omega \cdot \text{m.}$$

13. (c) Cu, Ag and Au all are good conductors of electricity.

14. (d) $m = \ell \times \text{area} \times \text{density}$

$$\text{Area} \propto \frac{m}{\ell} \quad (\text{for constant density})$$

$$R \propto \frac{\ell}{\text{Area}} \propto \frac{\ell^2}{m}$$

$$R_1 : R_2 : R_3 = \frac{\ell_1^2}{m_1} : \frac{\ell_2^2}{m_2} : \frac{\ell_3^2}{m_3}$$

$$R_1 : R_2 : R_3 = \frac{25}{1} : \frac{9}{3} : \frac{1}{5} = 125 : 15 : 1$$

15. (a) Here we have,

$$\text{Radius of wire, } r = \frac{10^{-2}}{\sqrt{\pi}}$$

$$\text{Cross sectional area, } A = \pi r^2 = 10^{-4} \text{ m}^2$$

$$\text{Resistance, } R = \frac{\rho \ell}{A} \Rightarrow 10 = \rho \times \frac{10}{10^{-4}} \Rightarrow \rho = 10^{-4}$$

$$\therefore \text{Current density, } J = \frac{1}{\rho} \times E \Rightarrow J = \frac{10}{10^{-4}} = 10^5 \text{ A/m}^2$$

16. (d)

$$A \rightarrow R: \text{Drift velocity, } v_d = \left(\frac{eE}{m} \right) \tau$$

$$B \rightarrow S: \text{Electrical Resistivity, } \rho = \frac{1}{\sigma}$$

$$\therefore J = \sigma E = \sigma \frac{J}{E}$$

$$\text{then, } \rho = \frac{1}{\left(\frac{J}{E} \right)} = \frac{E}{J} \Rightarrow \rho = \frac{E}{J}$$

$$C \rightarrow P: \text{Relaxation Period, } \tau$$

$$\therefore \rho = \frac{E}{J} = \frac{E}{nev_d} \Rightarrow v_d = \frac{E}{nep}$$

$$\Rightarrow \left(\frac{eE}{m} \right) \tau = \frac{E}{nep} \Rightarrow \tau = \frac{m}{ne^2 \rho}$$

$$D \rightarrow Q: \text{Current Density, } J$$

$$\therefore i = neAV_d \Rightarrow \frac{i}{A} = \frac{neAV_d}{A} \Rightarrow J = neV_d$$

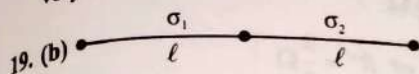
Hence, (A) – (R), (B) – (S), (C) – (P) and (D) – (Q) is the correct answer.

17. (a) Mobility is given as

$$\mu = \frac{v_d}{E}, \text{ where } v_d = \text{drift velocity, } E = \text{Electric field}$$

$$\mu = \frac{7.5 \times 10^{-4}}{3 \times 10^{-10}} = 2.5 \times 10^6 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$$

18. (b) In case of copper, resistivity is nearly proportional to temperature although a non linear region always exists at very low temperature.



$$R_{eq} = R_1 + R_2 \Rightarrow \frac{2l}{\sigma_{eq} A} = \frac{l}{\sigma_1 A} + \frac{l}{\sigma_2 A} \Rightarrow \sigma_{eq} = \frac{2\sigma_1\sigma_2}{\sigma_1 + \sigma_2}$$

20. (c) Energy = 2 eV = eEλ

$$\therefore E = \frac{2}{\lambda} \text{ Volt / metre} = \frac{2}{4 \times 10^{-8}} = 5 \times 10^7 \text{ Volt/meter}$$

21. (a) The value ρ does not depend on geometry but increases with increase in temperature.

22. (c) From colour coding

Yellow 4

Violet 7

Brown 1

Gold 5%

$$R = 47 \times 10^1 \pm 5\%$$

$$R = 470 \Omega, 5\%$$

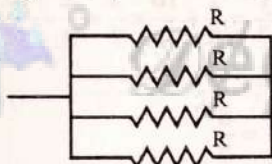
23. (b) $R = (47 \pm 4.7) \text{ k}\Omega$

$$= (47000 \pm 4700) \Omega$$

$$= (47 \times 10^3 \pm 10\%) \Omega$$

$\therefore$ Yellow-violet-orange-silver

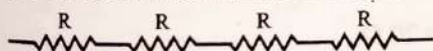
24. (c) Required figure of parallel combination:



Effective resistance = 0.25 Ω.

$$\Rightarrow \frac{R}{4} = 0.25 \Omega \Rightarrow R = 4 \times 0.25 \Omega \Rightarrow R = 1 \Omega$$

All the four resistance are connected in series



Equivalent resistance will be;

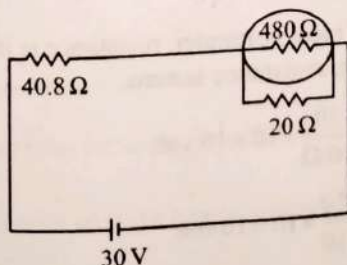
$$R_{eq} = R + R + R + R = 4R = 4(1) \quad [\because \text{all the connected in series}]$$

$$R_{eq} = 4 \Omega$$

25. (a) $R_{eq} = \left(4 + 8 + \frac{12 \times 6}{12 + 6} \right) \Omega$

$$= (12 + 4) \Omega = 16 \Omega$$

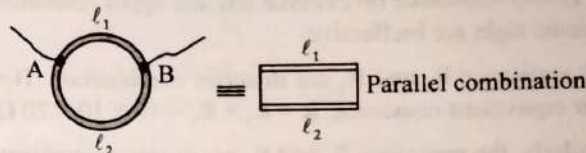
26. (b)



$$R_{eff} = 40.8 + \frac{480 \times 20}{480 + 20} = 40.8 + 19.2 = 60 \Omega$$

$$I = \frac{V_{eff}}{R_{eff}} = \frac{30}{60} = 0.5 \text{ A}$$

27. (d)



Ring has resistance corresponding to l_1 and l_2 .

Ring is divided into l_1 and l_2 .

Let resistance is R_1 for length l_1 and R_2 for length l_2 .

Now, as per the problem

$$\frac{R_1 R_2}{R_1 + R_2} = \frac{8}{3} \text{ and } R_1 + R_2 = 12$$

$$\Rightarrow R_1 R_2 = 32$$

Now, we know that

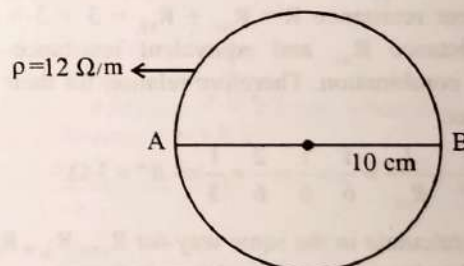
$$R_1 - R_2 = \sqrt{(R_1 + R_2)^2 - 4R_1 R_2}$$

$$= \sqrt{12^2 - 4 \times 32} = 4 \Omega$$

So, $R_1 = 4 \Omega$ and $R_2 = 8 \Omega$

$$\text{Hence, } \frac{l_1}{l_2} = \frac{R_1}{R_2} = \frac{4}{8} = \frac{1}{2}$$

28. (d)

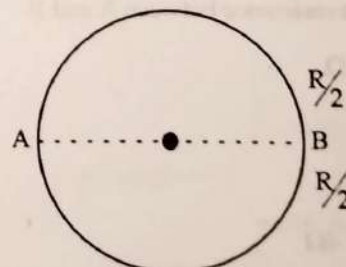


As two parts of circle are in parallel

$$R_{AB} = \frac{R}{2} = \frac{1}{2} (\pi r \rho) \quad [\text{where } R \text{ is resistance of half part}]$$

$$= \frac{1}{2} \times \pi \times 10 \times 10^{-2} \times 12 = 0.6 \pi \Omega$$

29. (a)



Since both are in parallel.

$$1/R' = 2/R + 2/R = 4/R \text{ i.e., } R' = R/4$$

30. (d) According to question $R = \frac{r}{n} \Rightarrow r = nR$

When these resistances are connected in series

$R_{eq} = r + r + \dots + n \text{ times}$

$$= nr = n \cdot nR = n^2 R$$

$$[\because r = nR]$$

31. (a) Lower resistance on extreme left and upper resistance on extreme right are ineffective.

The resistance R_2 and R_1 are in series combination. Hence, their equivalent resistance, $R = R_2 + R_1 = 10 + 10 = 20 \Omega$.

Similarly, the resistance R_5 and R_6 are in series combination. Therefore their equivalent resistance, $R'' = R_5 + R_6 = 10 + 10 = 20 \Omega$. Now the equivalent resistances R' and R'' are in parallel combination.

Therefore their equivalent resistance,

$$R''' = \frac{R' R''}{R' + R''} = \frac{20 \times 20}{20 + 20} = \frac{400}{40} = 10 \Omega$$

Thus equivalent resistance between A and D, $R = R_1 + R''' + R_4 = 10 + 10 + 10 = 30 \Omega$

32. (a) Ratio of cross-sectional areas of the wires, $A_1 : A_2 = 3 : 1$ and resistance of thick wire (R_1) = 10Ω .

$$\text{Resistance } (R) = \rho \frac{l}{A} \propto \frac{1}{A}$$

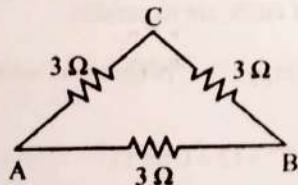
$$\text{Therefore } \frac{R_1}{R_2} = \frac{A_2}{A_1} = \frac{1}{3} \text{ or } R_2 = 3R_1 = 3 \times 10 = 30 \Omega.$$

and equivalent resistance of these two resistances in series combination = $R_1 + R_2 = 30 + 10 = 40 \Omega$.

33. (a) R_{AF} and R_{FE} are in series combination. Therefore their equivalent resistance $R' = R_{AF} + R_{FE} = 3 + 3 = 6 \Omega$. Now the resistance R_{AE} and equivalent resistance R' are in parallel combination. Therefore relation for their equivalent resistance

$$\frac{1}{R''} = \frac{1}{R'} + \frac{1}{R_{AE}} = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3} \Rightarrow R'' = 3 \Omega.$$

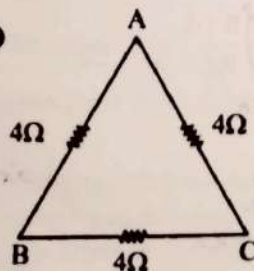
We can calculate in the same way for R_{ED} , R_{AC} , R_{DC} etc. and finally the circuit reduces as shown in figure.



Hence, the equivalent resistance between A and B

$$= \frac{(3+3) \times 3}{(3+3) + 3} = \frac{18}{9} = 2 \Omega.$$

34. (d)

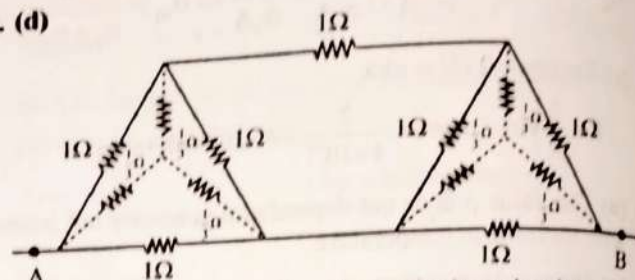


The two resistances are connected in series and the resultant is connected in parallel with the third resistance,

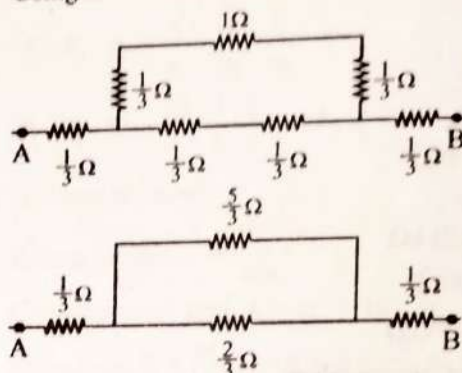
$$\therefore R = 4 \Omega + 4 \Omega = 8 \Omega$$

$$\text{and } \frac{1}{R''} = \frac{1}{8} + \frac{1}{4} = \frac{3}{8} \text{ or } R'' = \frac{8}{3}$$

35. (d)



Using $\Delta - Y$ transformation, the equivalent circuit become,



$$R_{AB} = \frac{1}{\frac{1}{3} + \frac{1}{3} + \frac{5 \times 2}{3 \times 3}} = \frac{1}{\frac{1}{3} + \frac{1}{3} + \frac{10}{9}} = \frac{1}{\frac{14}{9}} = \frac{9}{14}$$

$$= \frac{14 + 10}{21} = \frac{24}{21} = \frac{8}{7}$$

36. (c) For carrying a current of 4 A, we need four path, each carrying a current of one ampere.

Let r be the resistance of each path. These are connected in parallel. Hence their equivalent resistance will be $r/4$.

$$\text{Here, } \frac{r}{4} = 5 \text{ or } r = 20 \Omega.$$

For this purpose two resistances should be connected. There are four such combinations. Hence, the total number of resistance = $4 \times 2 = 8$

37. (c) In series, $R_s = nR$

$$\text{In parallel, } \frac{1}{R_p} = \frac{1}{R} + \frac{1}{R} + \dots n \text{ terms} = \frac{n}{R}$$

$$\Rightarrow R_p = \frac{R}{n}$$

$$\therefore R_s/R_p = n^2/1$$

38. (c) In case of ideal voltmeter, resistance is infinite and for the ideal ammeter, resistance is zero.

$$V_1 = i_1 \times 10 = \frac{10}{10 \Omega} \times 10 = 10 \text{ volt}$$

$$V_2 = i_2 \times 10 = \frac{10}{10} \times 10 = 10 \text{ volt}$$

$$\left[\because i_1 = \frac{V}{R} \right]$$

$$V_1 = V_2$$

$$i_1 = i_2 = \frac{10V}{10\Omega} = 1A$$

39. (c) When 'n' equal resistors of value 'R' are connected in series $I = \frac{E}{nR + R} = \frac{E}{(n+1)R}$... (i)

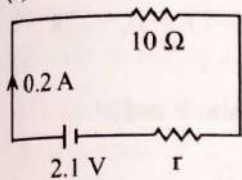
When 'n' equal resistors of value 'R' are connected in parallel. $10I = \frac{E}{\frac{R}{n} + R} = \frac{nE}{(n+1)R}$... (ii)

Solving Eqs. (i) and (ii), $n = 10$

40. (c) $I = \frac{nE}{nr} = \frac{E}{r}$

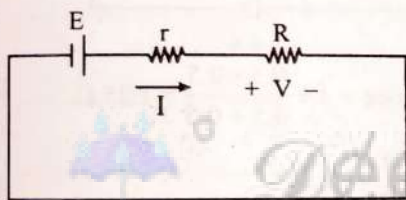
Current I is independent of number of cells 'n'

41. (c)



$$I = \frac{E}{r+R} \Rightarrow 0.2 = \frac{2.1}{r+10} \Rightarrow r = 0.5\Omega$$

42. (c)

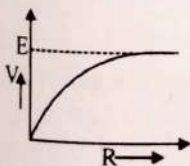


$$V = E - Ir$$

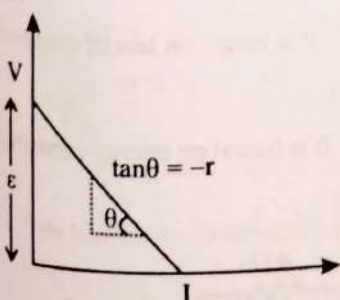
$$V = \frac{ER}{R+r}$$

$$I = \frac{E}{R+r}$$

For $R = 0$, V becomes zero
as $R = \infty$, V becomes E



43. (a)



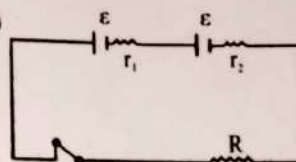
During discharging terminal potential

$$V = \epsilon - Ir$$

Compare this with straight line equation, $y = mx + c$

$$C = \epsilon \text{ and } m = -r$$

44. (a)



When circuit is closed, current in circuit

$$I = \frac{\epsilon + \epsilon}{r_1 + r_2 + R} = \frac{2\epsilon}{r_1 + r_2 + R} \quad \dots (i)$$

Potential difference across first cell

$$\Rightarrow \epsilon - Ir_1 = 0 \text{ [According to question]}$$

$$\Rightarrow I = \frac{\epsilon}{r_1} \quad \dots (ii)$$

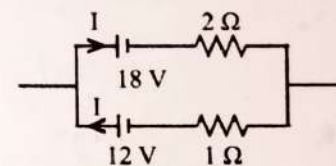
Equating (i) and (ii)

$$\frac{\epsilon}{r_1} = \frac{2\epsilon}{r_1 + r_2 + R}$$

$$\Rightarrow r_1 + r_2 + R = 2r_1 \Rightarrow R = r_1 - r_2$$

45. (d) Current in the circuit

$$I = \frac{(18-12)}{(1+2)} = \frac{6}{3} A = 2 A$$



Voltage drop across 2Ω , $V_1 = 2 \times 2 = 4 V$

Voltmeter reading = $18 - 4 = 14 V$

46. (a) T.P.D (V) = $E - Ir$

$$V = E - \left(\frac{E}{R+r} \right) r = \frac{ER}{R+r}$$

From given conditions, $E = 2.2$ and when $R = 5$ then total potential difference $V = 1.8 V$

$$\therefore 1.8 = \frac{2.2 \times 5}{5+r} \Rightarrow r = \frac{10}{9}$$

47. (c) $V = E + IR$

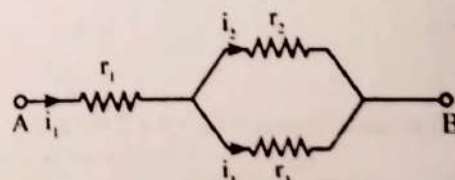
$$= 12 + 60 \times 5 \times 10^{-2} = 12 + 3 = 15 V$$

48. (c) When a battery of higher emf is connected in its parallel then terminal p.d of a cell is greater than its emf.

$$49. (a) I = \frac{8-4}{1+2+9} = \frac{4}{12} = \frac{1}{3} A$$

$$V_P - V_Q = 4 - \frac{1}{3} \times 3 = 3 V$$

50. (a)



Voltage doesn't get divided in parallel connection. So, voltage across r_2 and r_3 remains the same.

$$\Rightarrow V_2 = V_3 \Rightarrow i_2 r_2 = i_3 r_3$$

$$\Rightarrow i_2 = \frac{i_3 r_3}{r_2}$$

According to KCL,

$$\Rightarrow i_1 = i_2 + i_3$$

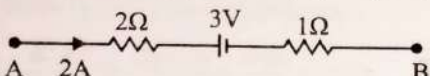
$$\Rightarrow i_1 = \left(\frac{r_3}{r_2}\right)i_3 + i_3 \Rightarrow i_1 = i_3 \left(\frac{r_3}{r_2} + 1\right)$$

$$\Rightarrow \frac{i_1}{i_3} = \frac{r_3 + r_2}{r_2} \Rightarrow \frac{i_3}{i_1} = \left(\frac{r_2}{r_2 + r_3}\right)$$

51. (a) $I = \frac{E_1 + E_2}{R_1 + R_2 + R_3} = \frac{2 + 4}{4 + 1 + 1} = \frac{6}{6} = 1$ ampere

52. (a) By Kirchhoff's voltage law

$$-i_2 R_2 - E_2 + E_3 + i_2 R_1 = 0 \Rightarrow i_2 R_2 + E_2 - E_3 - i_2 R_1 = 0$$

53. (b) 

$$V_B = V_A - (2 \times 2) - 3 - (2 \times 1) \Rightarrow V_A - V_B = 9 \text{ V}$$

54. (d) Effective resistance of B & C = $\frac{(1.5R)(3R)}{1.5R + 3R} = R$

In series sequence $V \propto R$

so voltage across 'A' = voltage across B & C

Now B & C are parallel so $V_B = V_C$

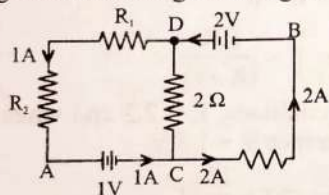
$$\Rightarrow V_A = V_B = V_C$$

55. (b) For no deflection

V_B = potential drop across R

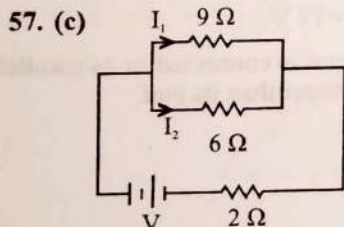
$$\Rightarrow V_B = \left(\frac{R}{R + R_1}\right)V_A = \left(\frac{100}{100 + 500}\right)(12) = 2 \text{ V}$$

56. (d) Applying Kirchhoff voltage law in given path From A to B



Given, $V_A = 0$

$$V_A + 1 + 2 - 2 = V_B \Rightarrow V_B = 1 \text{ V} [\because V_A = 0]$$



$$I_1^2 R = 36 [\because P = I^2 R]$$

Current flows through the 9Ω resistor is

$$I_1^2 = \frac{36}{9} = 4$$

$$\Rightarrow I_1 = 2 \text{ A}$$

$$\text{Potential difference across } 9 \Omega = 9 \times 2 = 18 \text{ V}$$

$$\text{So, current in } 6 \Omega = \frac{18}{6} = 3 \text{ A} = I_2$$

$$\text{The current flowing in } 2 \Omega = I_1 + I_2 = (3 + 2) = 5 \text{ A}$$

$$\therefore \text{Potential difference across } 2 \Omega = 5 \times 2 = 10 \text{ V}$$

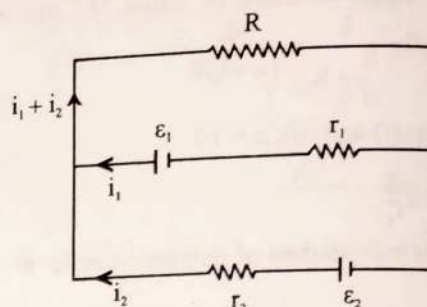
58. (a) **Kirchhoff's junction Law:** The algebraic sum of the current is zero on a node (law of conservation of electric charge)

Kirchhoff's loop law: Sum of the potential difference in any closed loop is equal to zero.

$$\Sigma \Delta V = 0$$

It is based on energy conservation.

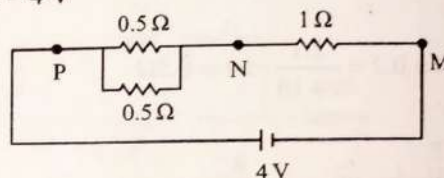
59. (d)



$$\varepsilon_1 - i_1 r_1 = \varepsilon_2 - i_2 r_2 = (i_1 + i_2)R \Rightarrow \varepsilon_1 - i_1 r_1 = (i_1 + i_2)R$$

$$\Rightarrow \varepsilon_1 - (i_1 + i_2)R - i_1 r_1 = 0$$

60. (a) Potential difference between point P and M = $4 \times 1 = 4 \text{ V}$



$$\text{Equivalent resistance} = 1 + \frac{0.5 \times 0.5}{0.5 + 0.5} = 1.25 \Omega$$

$$\text{Current in } 1 \Omega = \frac{4}{1.25} = 3.2 \text{ A}$$

$$\text{So, } V_{NM} = 3.2 \times 1 = 3.2 \text{ volt}$$

61. (d) As we know that current flows from higher potential to lower potential

$$\text{Current in } 4 \Omega = \frac{V}{8} \text{ A and } 1 \Omega = \frac{V}{4} \text{ A}$$

$$V_A (\text{Potential at point A}) = \left[V - \left(\frac{V}{8} \times 4 \right) \right]$$

$$= V - \frac{V}{2} = \frac{V}{2}$$

$$V_B (\text{Potential at point B}) = V - \frac{V}{4} \times 1 = \frac{3V}{4}$$

Since, potential at B is higher than A, current will flow from B to A

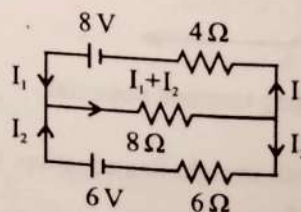
62. (b) **First Law of Kirchhoff:** It is based on law of conservation of electric charge

$$\Sigma I = 0$$

Second Law of Kirchhoff: It is based on energy conservation

$$\Sigma \Delta V = 0$$

63. (a) Method I



$$-8(I_1 + I_2) - 4I_1 + 8 = 0$$

$$-8(I_1 + I_2) - 6I_2 + 6 = 0$$

... (i)

... (ii)

Solving eqⁿ. (i) and (ii), we get

$$I_1 = \frac{8}{13}, I_2 = \frac{1}{13}$$

$$\text{Current in } 8\ \Omega = I_1 + I_2 = 0.69\text{ A}$$

64. (b) From the given circuit, we find that two resistances are in series and third resistance is in parallel. Therefore equivalent resistance for series resistances $3 + 3 \Rightarrow 6\ \Omega$. Now it is connected parallel with $3\ \Omega$ resistance. Therefore

$$R_{eq} = \frac{3 \times 6}{3 + 6} = \frac{18}{9} = 2\ \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{2}{2} = 1\text{ A}$$

65. (c) Kirchhoff's first law is based upon the conservation of charges.

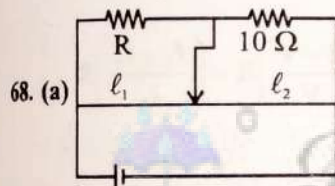
66. (c) Current across $3\ \Omega = 0.8\text{ A}$

$$6\ \Omega \text{ is in parallel, current across } 6\ \Omega = \frac{0.8 \times 3}{6} = 0.4\text{ A}$$

$$\text{Total current} = 1.2\text{ A}$$

$$\therefore \text{Potential difference across } 4\ \Omega \text{ resistor} = 1.2\text{ A} \times 4\ \Omega = 4.8\text{ V.}$$

67. (c) Resistance of P & Q should approximately be equal so as to have precise measurement of the unknown resistance.



$$\frac{R}{10} = \frac{l_1}{l_2} = \frac{R}{10} = \frac{3}{2}; R = 15\ \Omega$$

$$\text{Length of } 1\ \Omega \text{ resistance wire} = \frac{1.5}{15} = 0.1 = 1 \times 10^{-1}\text{ m.}$$

69. (b) According to the question,

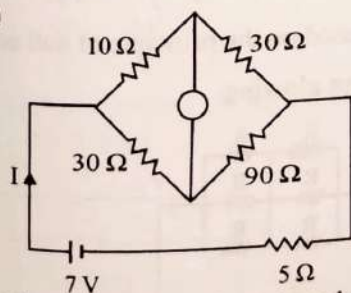
$$\frac{5}{R} = \frac{l_1}{100 - l_1} \quad \dots(i)$$

$$\frac{5}{R/2} = \frac{1.6l_1}{100 - 1.6l_1} \quad \dots(ii)$$

Dividing and solving (i) and (ii), we get.

$$\Rightarrow R = 15\ \Omega$$

70. (c)

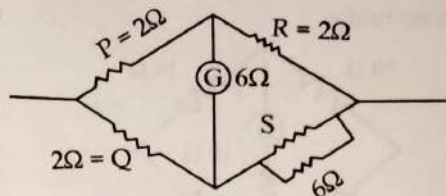


Total resistance of Wheatstone bridge

$$= \frac{(40)(120)}{40 + 120} = 30\ \Omega$$

$$\text{Current through cell} = \frac{7\text{ V}}{(5 + 30)\ \Omega} = \frac{1}{5}\text{ A} = 0.2\text{ A}$$

71. (a)



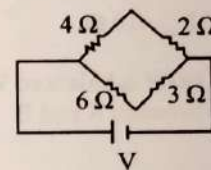
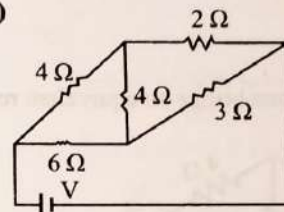
Equivalent resistance of arm $S' = \frac{S \times 6}{6 + S}$ As this wheatstone bridge is balanced

$$\frac{P}{Q} = \frac{R}{S} \Rightarrow \frac{2}{2} = \frac{2}{\left(\frac{6S}{6+S}\right)}$$

$$\Rightarrow \frac{6S}{6+S} = 2$$

$$\Rightarrow 6S = 12 + 2S \Rightarrow 4S = 12 \Rightarrow S = 3\ \Omega$$

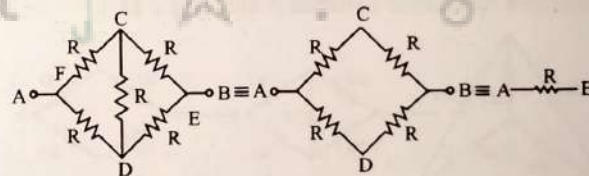
72. (b)



$$R_{eq} = \frac{6 \times 9}{15} = \frac{54}{15} = \frac{18}{5}\ \Omega$$

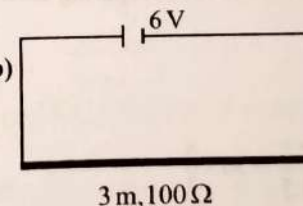
$$i = \frac{V}{R_{eq}} = \frac{5V}{18}\text{ A}$$

73. (b) Given circuit can be reduced to



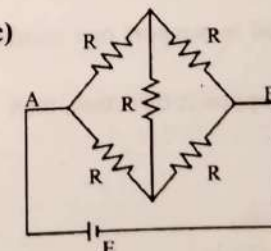
$$\text{Required current} = \frac{V}{2R}$$

74. (b)



$$\text{Voltage on } 50\text{ cm.} = \frac{6}{300} \times 50 = 1\text{ volt.}$$

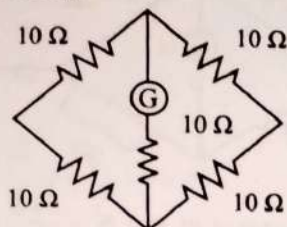
75. (c)



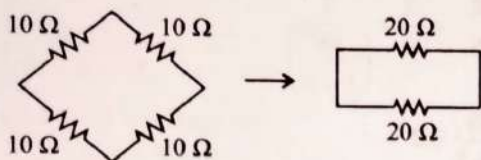
Resistance seen by the battery = Equivalent resistance

$$\text{between A and B} = \frac{2R \times 2R}{2R + 2R} = R$$

76. (a) Wheatstone bridge



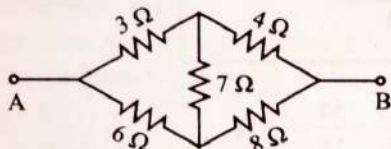
No current passes through galvanometer. Now circuit look like.



$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} = \frac{20 \times 20}{20 + 20}$$

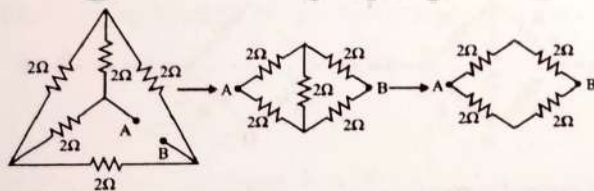
$$R_{eq} = 10 \Omega$$

77. (b) It is balanced Wheatstone bridge so equivalent resistance between A and B



$$R = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{14 \times 7}{21} = \frac{14}{3} \Omega$$

78. (d) The circuit is equivalent to a balanced Wheatstone bridge. Therefore resistance between A and B is 2 Ω.



79. (d) Application of potentiometer for comparing EMF of two cells as follows:

$$\Rightarrow \frac{E_1}{E_2} = \frac{\ell_1}{\ell_2}$$

$$\Rightarrow \frac{1.5}{2.5} = \frac{36}{\ell_2} \Rightarrow \ell_2 = 36 \times \frac{2.5}{1.5} = 36 \times \frac{5}{3}$$

$$= 12 \times 5 = 60 \Rightarrow \ell_2 = 60 \text{ cm}$$

80. (b) In zero deflection condition, potentiometer draws no current.

81. (d) When cells are connected to support one another, then $E_1 + E_2 \propto 50$... (i)

When cells are connected in opposite direction, then

$$E_1 - E_2 \propto 10 \quad \dots (ii)$$

Using (i) and (ii)

$$\frac{E_1 + E_2}{E_1 - E_2} = \frac{50}{10}$$

$$\frac{2E_1}{2E_2} = \frac{50 + 10}{50 - 10} = \frac{E_1}{E_2} = \frac{3}{2}$$

$$82. (d) \text{ Potential gradient} = \frac{1 \text{ mV}}{\text{cm}} = 10^{-3} \text{ V/cm} = 10^{-1} \text{ V/m}$$

$$\text{Let the resistance to be connected is } R \text{ then } I = \frac{2}{8 + R}$$

Potential drop across the potentiometer wire

$$= \frac{8 \times 2}{8 + R} = \frac{16}{8 + R}$$

$$\text{Potential gradient} = \left(\frac{16}{8 + R} \right) \times \frac{1}{4} \text{ V/m}$$

$$= \frac{4}{8 + R} = 0.1 \Rightarrow R = 32 \Omega$$

$$83. (c) i = \frac{E_0}{r + r_1}$$

$$E = \lambda l$$

Where λ = potential gradient

$$\lambda = \frac{E_0 - Ir_1}{L}$$

$$E = \left(\frac{E_0 - Ir_1}{L} \right) l = \left(E_0 - \frac{E_0 r_1}{r + r_1} \right) \frac{l}{L}$$

$$E = \frac{E_0 r l}{(r + r_1) L}$$

$$84. (c) \text{ Internal resistance, } r = \left(\frac{E - V}{V} \right) R = \left(\frac{\ell_1 - \ell_2}{\ell_2} \right) R$$

$$= \left(\frac{3 - 2.85}{2.85} \right) (9.5) \Omega = 0.5 \Omega$$

85. (c) Case I : When switches 1 and 2 are connected $V_1 = \ell_1$

$$IR = k\ell_1$$

$$R = k\ell_1$$

$$(\because I = 1 \text{ A})$$

Case 2 : When switches 1 and 3 are connected

$$V_2 = k\ell_2 \Rightarrow I[R + x] = k\ell_2$$

$$R + x = k\ell_2$$

$$(\because I = 1 \text{ A})$$

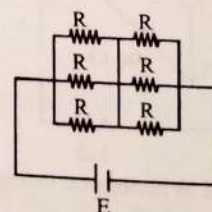
$$k\ell_1 + x = k\ell_2 \Rightarrow x = k[\ell_2 - \ell_1]$$

86. (b) Given, $\ell_1 = 100 \text{ cm}$, $\ell_2 = 110 \text{ cm}$, $R = 10 \Omega$

$$\text{Value of internal resistance} = \left(\frac{\ell_2 - \ell_1}{\ell_1} \right) R = \frac{10}{100} \times 10 = 1 \Omega$$

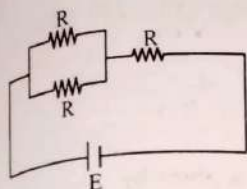
87. (d) Potentiometer is based on the principle of null deflection.

88. (b) Case I: All bulbs are glowing



$$R_{eq} = \frac{R}{3} + \frac{R}{3} = \frac{2R}{3}$$

Case II: Two from section A and one from section B are glowing.

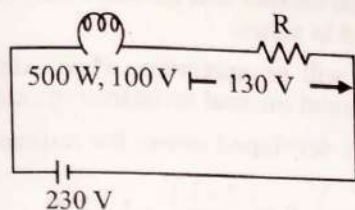


$$R_{eq} = \frac{R}{2} + R = \frac{3R}{2}$$

$$\text{Here } P \propto \frac{1}{R_{eq}}$$

$$\therefore \frac{P_i}{P_f} = \frac{3R/2}{2R/3} = \frac{9}{4}$$

93. (a)



In series combination, current is same

$$\Rightarrow I_b = \frac{500}{100} = 5 \text{ A}$$

$$I_R = I_b = 5 \text{ A}$$

$$R = \frac{V}{I} = \frac{130}{5}$$

$$\therefore R = 26 \Omega$$

$$90. (b) \text{ Resistance} = (0.5 \Omega/\text{km})(150 \text{ km}) = 75 \Omega$$

$$\text{Total voltage drop, } \Delta V = (8 \text{ V/km})(150 \text{ km}) = 1200 \text{ V}$$

$$\text{Power loss} = \frac{(\Delta V)^2}{R} = \frac{(1200)^2}{75} \text{ W} = 19200 \text{ W} = 19.2 \text{ kW}$$

$$91. (c) P_{\text{Total}} = \frac{V^2}{R_{eq}}$$

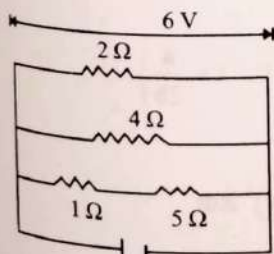
$$\text{So, } R_{eq} = \frac{V^2}{P_{\text{Total}}} = \frac{10 \times 10}{30} = \frac{10}{3}$$

$$\text{As } R_{eq} = \frac{R_1 R_2}{R_1 + R_2} \Rightarrow \frac{10}{3} = \frac{5R}{5+R} \Rightarrow R = 10 \Omega$$

$$92. (c) \therefore P \propto V^2$$

$$\therefore \frac{\Delta P}{P} = 2 \frac{\Delta V}{V} = 5\%$$

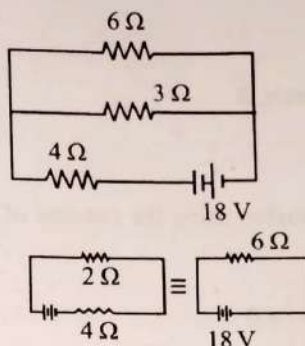
$$93. (a) \text{ As } 3 \text{ A is flowing through } 2 \Omega, \text{ potential difference across } 2 \Omega = 3 \times 2 = 6 \text{ V}$$



$$\text{So, the current flowing through } 5 \Omega = \frac{6}{6} = 1 \text{ A}$$

$$\text{Power dissipated in the } 5 \Omega \text{ resistor} = i^2 R = (1)^2 \times 5 = 5 \text{ W}$$

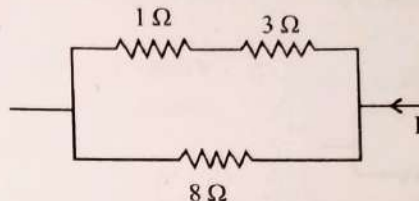
94. (b)



$$I = \frac{V}{R} = \frac{18}{6} = 3 \text{ A}$$

$$P = VI = 18 \times 3 = 54 \text{ W}$$

95. (d)

Power dissipated across 8Ω resistor = 2 W

$$P = \frac{V^2}{R} \Rightarrow 2 = \frac{V^2}{8} \Rightarrow V = 4 \text{ V}$$

$$\text{So, current in } 3 \Omega \text{ resistor} = \frac{4}{1+3} = \frac{4}{4} = 1 \text{ A}$$

$$\text{Power dissipated across } 3 \Omega \text{ resistor} = i^2 R = (1)^2 \times 3 = 3 \text{ W}$$

96. (c) If rated voltage = Supply voltage then in series combination of bulbs for this question

$$\frac{1}{P_{eq}} = \frac{1}{P_1} + \frac{1}{P_2} + \frac{1}{P_3}$$

$$\frac{1}{P_{eq}} = \frac{1}{60} + \frac{1}{60} + \frac{1}{60} = \frac{3}{60} = \frac{1}{20} \Rightarrow P_{eq} = 20 \text{ watt}$$

97. (b) In India : $P = 60 \text{ W}$, $V = 220 \text{ V}$

$$P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P} \Rightarrow R = \frac{220 \times 220}{60} = \frac{2420}{3} \Omega$$

 $\therefore$ In USA: $P = 60 \text{ W}$, $V = 110 \text{ V}$

$$\therefore P = \frac{V^2}{R^2} \Rightarrow R_2 = \frac{V^2}{P} = \frac{110^2}{60} = \frac{1}{4} \times \frac{220^2}{60} = \frac{1}{4} \times \frac{2420}{3} = \frac{R}{4}$$

98. (d) If rated voltage = Supply voltage then use $\frac{1}{P} = \frac{1}{P_1} + \frac{1}{P_2}$ (series connection) & $P = P_1 + P_2$ (parallel connection)

According to the question,

$$P_{\text{series}} = P_1 + P_2 = 100 + 100 = 200 \text{ watt}$$

$$\frac{1}{P_{\text{Parallel}}} = \frac{1}{100} + \frac{1}{100} = \frac{2}{100} = \frac{1}{50}$$

$$\therefore P_{\text{Parallel}} = 50 \text{ watt}$$

$$99. (b) P = \frac{V^2}{R}, P \propto \frac{1}{R} \text{ i.e. } R_{40} > R_{100}$$

100. (b) $I = \frac{E}{R+6} A$

Power consumed in resistor, R

$$P = I^2 R = \frac{E^2}{(R+6)^2} R$$

For maximum power transfer, using the concept of maxima and minima,

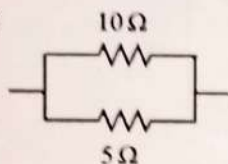
$$\frac{dP}{dR} = 0 \Rightarrow \frac{d}{dR} \left[\frac{E^2}{(R+6)^2} \right] = 0$$

$$\Rightarrow \frac{(R+6)^2 \times E^2 - E^2 R (R+6) \times 2}{(R+6)^4} = 0$$

$$\Rightarrow (R+6)^2 - 2R = 0$$

$$\Rightarrow R = 6$$

101. (a)



Power dissipated in 5Ω ,

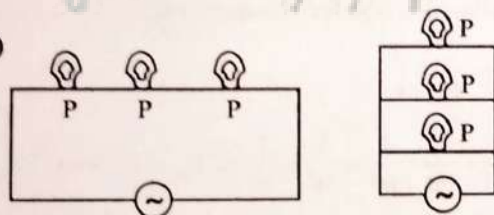
$$P = \frac{V^2}{R} \Rightarrow 20 = \frac{V^2}{5} \Rightarrow V^2 = 100 \Rightarrow V = 10V$$

Current through the 4Ω resistance,

$$I = \frac{V}{R} = \frac{10}{10} = 1A$$

$$\therefore \text{Power dissipated in } 4\Omega \text{ resistance} = I^2 R = (1)^2 \times 4 = 4W$$

102. (b)



$$\frac{1}{P_{eq}} = \frac{1}{P_1} + \frac{1}{P_2} + \frac{1}{P_3} \quad \text{and } P_{eq} = P_1 + P_2 + P_3$$

$$\Rightarrow 10 = \frac{P}{3} \Rightarrow P_{eq} = 3P = 3 \times 30 = 90 \text{ watt}$$

103. (c) Power = 100 W; Voltage of bulb = 200 V and supply voltage (V_s) = 160 V.

$$\text{Therefore resistance of bulb } (R) = \frac{V^2}{P} = \frac{(200)^2}{100} = 400 \Omega$$

$$\text{and power consumption } (P) = \frac{V_s^2}{R} = \frac{(160)^2}{400} = 64 W$$

104. (d) Ratio of resistance $R_1 : R_2 = 1 : 2$ or $\frac{R_1}{R_2} = \frac{1}{2}$. In series combination, power dissipated (P) = $I^2 R$
(as current is same in series)

$$\Rightarrow P \propto R \text{ Therefore } \frac{P_1}{P_2} = \frac{R_1}{R_2} = \frac{1}{2}$$

$$\text{or } P_1 : P_2 = 1 : 2.$$

105. (d) Power (P) = 60 W and voltage (V) = 220 volts.

Resistance of the filament,

$$R = \frac{V^2}{P} = \frac{(220)^2}{60} = 807 \Omega.$$

106. (a) The output power of a cell is given by

$$P = \frac{V^2}{(r+R)^2} R$$

Maximum power is delivered to the load only when the internal resistance of the source is equal to the load resistance (R). Then

$$P_{max} = \frac{V^2}{4R} = \frac{V^2}{4r} \quad (r = R)$$

$$\therefore r = R = 0.5 \Omega$$

107. (b) Maximum current will be when the two batteries should be connected in series.

The current will be maximum when external resistance is equal to the total internal resistance of cells i.e. 2Ω .

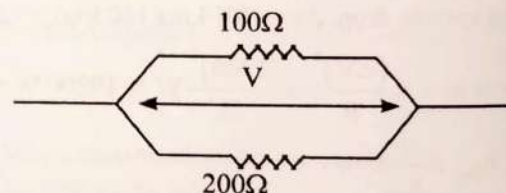
Hence power developed across the resistance R will be

$$I^2 R = \left(\frac{2E}{R+2r} \right)^2 R = \left(\frac{2 \times 2}{2+2} \right)^2 \times 2 = 2 W$$

108. (b) As, the voltage is same for the two combinations, the resistance is less for 39 bulbs. Therefore, the combination of 39 bulbs will glow more as current is more.

109. (d) For conductors, temperature coefficient of resistance α is positive, therefore resistance of the conductor rises with temperature and for semiconductor α is negative, therefore its resistance decreases with temperature.

110. (c)



$$P = \frac{V^2}{R}$$

As both resistors are in parallel so potential drop (V) across both the resistors are same.

$$\Rightarrow P \propto \frac{1}{R} \Rightarrow \frac{P_1}{P_2} = \frac{R_2}{R_1} = \frac{200}{100} = \frac{2}{1} = 2 : 1$$

111. (d) Fuse is a protection device. Fuse wire has less melting point so when excess current flows, due to heat produced in it, it melts.

112. (a) $Q = at - bt^2$

$$i = a - 2bt \quad \left\{ \text{For } i = 0, \quad t = \frac{a}{2b} \right\}$$

From Joules law of heating

$$dH = i^2 R dt \Rightarrow H = \int_0^{a/2b} (a - 2bt)^2 R dt$$

$$H = \left[\frac{(a - 2bt)^3 R}{-3(2b)} \right]_0^{a/2b} = \frac{a^3 R}{6b}$$

113. (b) Now applying Joule's law, heat generated by the conductor,

$$H = I^2 R t = V \times I \times t \text{ (since, } V = IR \text{)}$$

$$\text{or } H = (220) \times 4 \times T \text{ Cal (1 Cal = 4.2 joule)}$$

$$\text{or } H = \frac{880T}{4.2} \text{ Joule.} \quad \dots(i)$$

Again, heat loss,

$$\Delta H = ms \Delta T = 1000 \times 1 \times (100 - 20)$$

$$\text{or } H = 1000 \times 80$$

$$\text{or } H = 80000$$

(specific heat of water is $s = 1 \text{ C/g}^\circ\text{C}$)

Using the property of heat and calorimetry

So, from equations (i) and (ii)

$$\frac{880 \times T}{4.2} = 80000$$

$$\Rightarrow T = \frac{80000}{880} \times 4.2 = 381$$

$$\Rightarrow T = 381 \text{ seconds}$$

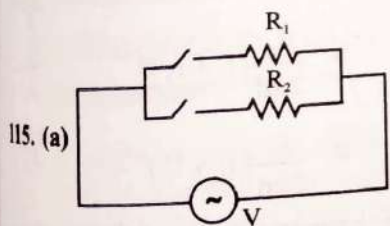
$$\Rightarrow T = \frac{381}{60} = 6.36 \text{ min}$$

$$\Rightarrow T = 6.36 \text{ min.}$$

Therefore, the time taken to increase the temperature of 1 kg of water from 20°C to 100°C is 6.36 min.

114. (d) Since, $P = I^2 R \Rightarrow 1 \text{ W} = (5)^2 R$

$$\Rightarrow R = 0.04 \Omega$$



$$Q = \frac{V^2}{R_1} \times t_1 = \frac{V^2}{R_2} \times t_2 = \frac{V^2}{R} \times t$$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow \frac{Q}{V^2 t} = \frac{Q}{V^2 t_1} + \frac{Q}{V^2 t_2}$$

$$\Rightarrow \frac{1}{t} = \frac{1}{t_1} + \frac{1}{t_2} \Rightarrow t = \frac{t_1 t_2}{t_1 + t_2} = \frac{10 \times 40}{10 + 40} = 8 \text{ min.}$$

116. (b) Fuse wire must have high resistance (per unit length) and low melting point.

117. (b) $I^2 R t = ms \Delta \theta \Rightarrow I^2 \propto \Delta \theta$

$$\frac{\Delta \theta_2}{\Delta \theta_1} = \frac{I_2^2}{I_1^2} \Rightarrow \frac{\Delta \theta_2}{5} = (2)^2 \Rightarrow \Delta \theta_2 = 20^\circ\text{C}$$

118. (c) $1 \text{ kWh} = 1000 \text{ Wh}$

$$= (1000 \text{ W}) \times (3600 \text{ s})$$

$$= 36 \times 10^5 \text{ J}$$

119. (b) Power of heating coil = 100 W and voltage (V) = 220 volts.

When the heating coil is cut into two equal parts and these parts are joined in parallel, the resistance of the coil is reduced to one-fourth of the previous value.

Therefore energy liberated per second becomes 4 times, i.e., $4 \times 100 = 400 \text{ W}$.

120. (d) Capacitance $C = 4 \mu\text{F} = 4 \times 10^{-6} \text{ F}$; Voltage (V) = 400 volts and resistance (R) = $2 \text{ k}\Omega = 2 \times 10^3 \Omega$.

Heat produced = Electrical energy stored

$$= \frac{1}{2} C V^2 = \frac{1}{2} \times (4 \times 10^{-6}) \times (400)^2 = 0.32 \text{ J.}$$

121. (b) $H = I^2 R t$ or $R = \frac{H}{(I^2 t)} = \frac{80}{(2^2 \times 10)} = 2 \Omega$

Moving Charges and Magnetism

Biot Savart's Law and Magnetic Field Due to Current Through the Straight Conductor

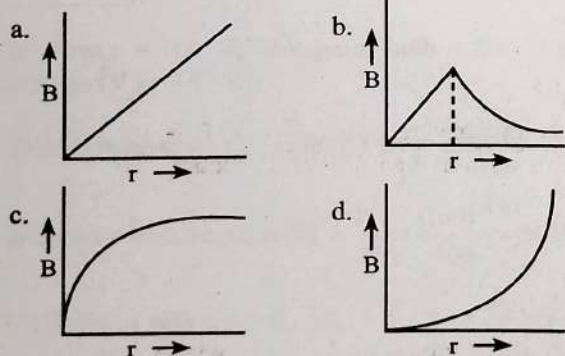
1. Given below are two statements :

Statement I : Biot-Savart's law gives us the expression for the magnetic field strength of an infinitesimal current element ($I d\vec{l}$) of the current carrying conductor only.

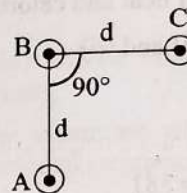
Statement II : Biot-Savart's law is analogous to Coulomb's inverse square law of charge q , with the former being related to the field produced by a scalar source, $I d\vec{l}$, while the latter being produced by a vector source, q .

In light of above statements choose the most appropriate answer from the options given below (2022)

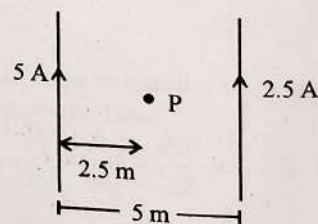
- Statement I is incorrect and statement II is correct
 - Both Statement I and Statement II are correct
 - Both statement I and Statement II are incorrect
 - Statement I is correct and Statement II is incorrect.
2. A thick current carrying cable of radius 'R' carries current 'I' uniformly distributed across its cross-section. The variation of magnetic field $B(r)$ due to the cable with the distance 'r' from the axis of the cable is represented by: (2021)



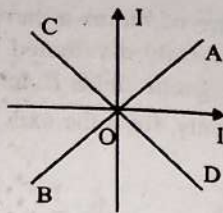
3. An arrangement of three parallel straight wires placed perpendicular to plane of paper carrying same current 'I' along the same direction is shown in figure. Magnitude of force per unit length on the middle wire 'B' is given by: (2017-Delhi)



- $\frac{2\mu_0 I^2}{\pi d}$
 - $\frac{\sqrt{2}\mu_0 I^2}{\pi d}$
 - $\frac{\mu_0 I^2}{\sqrt{2}\pi d}$
 - $\frac{\mu_0 I^2}{2\pi d}$
4. Two identical long conducting wires AOB and COD are placed at right angle to each other, with one above other such that O is their common point for the two. The wires carry I_1 and I_2 currents, respectively. Point P is lying at distance d from O along a direction perpendicular to the plane containing the wires. The magnetic field at the point P will be: (2014)
- $\frac{\mu_0}{2\pi d} \left(\frac{I_1}{I_2} \right)$
 - $\frac{\mu_0}{2\pi d} (I_1 + I_2)$
 - $\frac{\mu_0}{2\pi d} (I_1^2 + I_2^2)$
 - $\frac{\mu_0}{2\pi d} (I_1^2 + I_2^2)^{1/2}$
5. For adjoining figure. The magnetic field point 'P' will be: (2000)



- $\frac{\mu_0}{2\pi} \odot$
 - $\frac{\mu_0}{4\pi} \otimes$
 - $\frac{\mu_0}{2\pi} \otimes$
 - $\frac{\mu_0}{2\pi} \odot$
6. Tesla is the unit of
- Electric field
 - Magnetic field
 - Electric flux
 - None
7. Two equal electric currents are flowing perpendicular to each other as shown in the figure. AB and CD are perpendicular to each other and symmetrically placed with respect to the currents. Where do we expect the resultant magnetic field to be zero? (1997, 88)



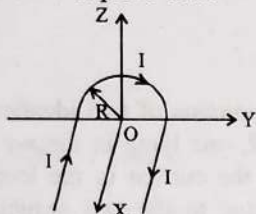
- a. On CD b. On AB
c. On both OD and BO d. On both AB and CD
8. The magnetic field $\vec{dB}$ due to a small current element $d\vec{l}$ at a distance $\vec{r}$ and element carrying current i is (1996)
- a. $d\vec{B} = \frac{\mu_0}{4\pi} i^2 \left(\frac{d\vec{l} \times \vec{r}}{r} \right)$ b. $d\vec{B} = \frac{\mu_0}{4\pi} i \left(\frac{d\vec{l} \times \vec{r}}{r^3} \right)$
c. $d\vec{B} = \frac{\mu_0}{4\pi} i \left(\frac{d\vec{l} \times \vec{r}}{r} \right)$ d. $d\vec{B} = \frac{\mu_0}{4\pi} i^2 \left(\frac{d\vec{l} \times \vec{r}}{r^2} \right)$
9. The magnetic field at a distance r from a long wire carrying current i is 0.4 tesla. The magnetic field at a distance $2r$ is (1992)
- a. 0.2 tesla b. 0.8 tesla
c. 0.1 tesla d. 1.6 tesla
10. The magnetic induction at a point P which is at the distance of 4 cm from a long current carrying wire is 10^{-3} T. The field of induction at a distance 12 cm from the Current will be (1990)
- a. 3.33×10^{-4} T b. 1.11×10^{-4} T
c. 33×10^{-3} T d. 9×10^{-3} T

Magnetic Field in Circular Coil

11. A long wire carrying a steady current is bent into a circular loop of one turn. The magnetic field at the center of the loop is B . It is then bent into a circular coil of n turns. The magnetic field at the center of this coil of n turns will be: (2016 - II)
- a. $2nB$ b. $2n^2B$
c. nB d. n^2B
12. An electron moving in a circular orbit of radius r makes n rotations per second. The magnetic field produced at the center has magnitude: (2015)
- a. Zero b. $\frac{\mu_0 n^2 e}{r}$
c. $\frac{\mu_0 n e}{2r}$ d. $\frac{\mu_0 n e}{2\pi r}$
13. Two similar coils of radius R are lying concentrically with their planes at right angles to each other. The currents flowing in them are I and $2I$, respectively. The resultant-magnetic field induction at the center will be: (2012 Pre)
- a. $\frac{\sqrt{5}\mu_0 I}{2R}$ b. $\frac{3\mu_0 I}{2R}$
c. $\frac{\mu_0 I}{2R}$ d. $\frac{\mu_0 I}{R}$
14. Charge q is uniformly spread on a thin ring of radius R . The ring rotates about its axis with a uniform frequency f Hz. The magnitude of magnetic induction at the center of the ring is: (2011 Mains)
- a. $\frac{\mu_0 q f}{2R}$
b. $\frac{\mu_0 q}{2\pi R}$
c. $\frac{\mu_0 q}{2\pi f R}$
d. $\frac{\mu_0 q f}{2\pi R}$
15. A current loop consists of two identical semicircular parts each of radius R , one lying in the x - y plane and the other in x - z plane. If the current in the loop is i . The resultant magnetic field due to the two semicircular parts at their common center is: (2010 Mains)
- a. $\frac{\mu_0 i}{2\sqrt{2}R}$ b. $\frac{\mu_0 i}{2R}$
c. $\frac{\mu_0 i}{4R}$ d. $\frac{\mu_0 i}{\sqrt{2}R}$
16. A thin ring of radius R metre has charge q coulomb uniformly spread on it. The ring rotates about its axis with a constant frequency of f revolution/s. The value of magnetic induction in Wb/m^2 at the center of the ring is: (2010 Pre)
- a. $\frac{\mu_0 q f}{2R}$ b. $\frac{\mu_0 q f}{2\pi R}$
c. $\frac{\mu_0 q}{2\pi f R}$ d. $\frac{\mu_0 q}{2f R}$
17. Two circular coils 1 and 2 are made from the same wire but the radius of the 1st coil is twice that of the 2nd coil. What is the ratio of potential difference applied across them so that the magnetic field at their centers is the same? (2006)
- a. 5 b. 4
c. 7 d. 2
18. The magnetic field of given length of wire for single turn coil at its centre is ' B ' then its value for two turns coil for the same wire is: (2002)
- a. $\frac{B}{4}$ b. $\frac{B}{2}$
c. $4B$ d. 2
19. A current carrying coil ($I = 5$ A, $R = 10$ cm.) having 50 number of turns find field at its centre: (1999)
- a. 1.57 mT
b. 3.14 mT
c. 1 mT
d. 2 mT
20. A coil of one loop is made from a wire of length L and there after a coil of two loops is made from same wire, if same current is passed in both the cases then the ratio of magnetic field at the centre of coils will be: (1998)
- a. 1 : 4 b. 1 : 1
c. 1 : 8 d. 4 : 1

Magnetic Field Due to Current Through Both Straight Wire and Circular Wire (Mixed Figure)

21. A wire carrying current I has the shape as shown in adjoining figure. Linear parts of the wire are very long and parallel to X-axis while semicircular portion of radius R is lying in Y-Z plane. Magnetic field at point O is: (2015)



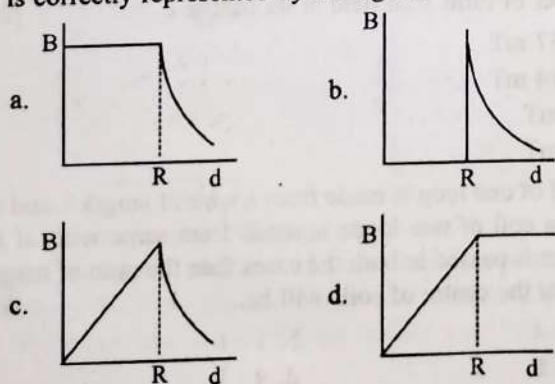
- a. $\vec{B} = -\frac{\mu_0 I}{4\pi R}(\pi\hat{i} - 2\hat{k})$ b. $\vec{B} = -\frac{\mu_0 I}{4\pi R}(\pi\hat{i} + 2\hat{k})$
 c. $\vec{B} = \frac{\mu_0 I}{4\pi R}(\pi\hat{i} - 2\hat{k})$ d. $\vec{B} = \frac{\mu_0 I}{4\pi R}(\pi\hat{i} + 2\hat{k})$

Ampere's Circuital Law

22. From Ampere's circuital law for a long straight wire of circular cross section carrying a steady current, the variation of magnetic field in the inside and outside region of the wire is: (2022)

- a. a linearly decreasing function of distance upto the boundary of the wire and then a linearly increasing one for the outside region.
 b. uniform and remains constant for both the regions.
 c. a linearly increasing function of distance upto the boundary of the wire and then decreasing outside region.
 d. a linearly increasing function of distance r upto the boundary of the wire $1/r$ dependence for the outside region.

23. A cylindrical conductor of radius R is carrying a constant current. The plot of the magnitude of the magnetic field B with the distance d from the centre of the conductor, is correctly represented by the figure: (2019)



24. A long straight wire of radius a carries a steady current I . The current is uniformly distributed over its cross-section. The ratio of the magnetic fields B and B' at radial distances a and $2a$ respectively, from the axis of the wire is: (2016 - I)

- a. $\frac{1}{4}$ b. $\frac{1}{2}$
 c. 1 d. 4

Magnetic Field Due to Solenoid and Toroid

25. A long solenoid of radius 1 mm has 100 turns per mm. If 1 A current flows in the solenoid, the magnetic field strength at the centre of the solenoid is: (2022)

- a. $6.28 \times 10^{-4} \text{ T}$ b. $6.28 \times 10^{-2} \text{ T}$
 c. $12.56 \times 10^{-2} \text{ T}$ d. $12.56 \times 10^{-4} \text{ T}$

26. A long solenoid of 50 cm length having 100 turns carries a current of 2.5 A. The magnetic field at the centre of the solenoid is: (2020)

($\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$)

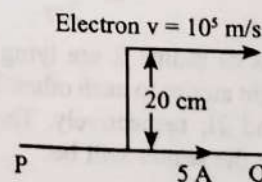
- a. $3.14 \times 10^{-4} \text{ T}$ b. $6.28 \times 10^{-5} \text{ T}$
 c. $3.14 \times 10^{-5} \text{ T}$ d. $6.28 \times 10^{-4} \text{ T}$

27. A long solenoid carrying a current produces a magnetic field B along its axis. If the current is doubled and the number of turns per cm is halved, the new value of the magnetic field is: (2003)

- a. $B/2$ b. B
 c. $2B$ d. $4B$

Motion of Charged Particle in (i) Electric Field and (ii) Magnetic Field

28. An infinitely long straight conductor carries a current of 5 A as shown. An electron is moving with a speed of 10^5 m/s parallel to the conductor. The perpendicular distance between the electron and the conductor is 20 cm at an instant. Calculate the magnitude of the force experienced by the electron at that instant. (2021)



- a. $8\pi \times 10^{-20} \text{ N}$
 b. $4\pi \times 10^{-20} \text{ N}$
 c. $8 \times 10^{-20} \text{ N}$
 d. $4 \times 10^{-20} \text{ N}$

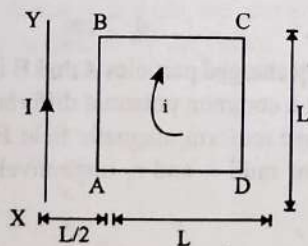
28. Ionized hydrogen atoms and α -particles with same momenta enters perpendicular to a constant magnetic field, B. The ratio of their radii of their paths $r_H : r_\alpha$ will be : (2019)
- 2 : 1
 - 1 : 2
 - 4 : 1
 - 1 : 4
29. An electron is moving in a circular path under the influence of a transverse magnetic field of 3.57×10^{-2} T. If the value of e/m is 1.76×10^{11} C/kg, the frequency of revolution of the electron is: (2016 - II)
- 62.8 MHz
 - 6.28 MHz
 - 1 GHz
 - 100 MHz
30. A proton and an alpha particle both enter a region of uniform magnetic field, B, moving at right angles to the field B. If the radius of circular orbits for both the particles is equal and the kinetic energy acquired by proton is 1 MeV, the energy acquired by the alpha particle will be: (2015 Pre)
- 1 MeV
 - 4 MeV
 - 0.5 MeV
 - 1.5 MeV
32. A proton carrying 1 MeV kinetic energy is moving in a circular path of radius R in uniform magnetic field. What should be the energy of an α -particle to describe a circle of same radius in the same field? (2012 Mains)
- 2 MeV
 - 1 MeV
 - 0.5 MeV
 - 4 MeV
33. Under the influence of a uniform magnetic field, a charged particle moves with constant speed V in a circle of radius R. The time period of rotation of the particle: (2009)
- Depends on R and not on V
 - Is independent of both V and R
 - Depends on both V and R
 - Depends on V and not on R
34. The magnetic force acting on a charged particle of charge $-2 \mu\text{C}$ in a magnetic field of 2 T acting in y direction, when the particle velocity is $(2\hat{i} + 3\hat{j}) \times 10^6 \text{ ms}^{-1}$, is: (2009)
- 4 N in z direction
 - 8 N in y direction
 - 8 N in z direction
 - 8 N in - z direction
35. A particle of mass m, charge Q and kinetic energy T enters a transverse uniform magnetic field of induction $\vec{B}$. After 3 seconds the kinetic energy of the particle will be: (2008)
- 4 T
 - 3 T
 - 2 T
 - T
36. Under the influence of a uniform magnetic field a charged particle is moving in a circle of radius R with constant speed v. The time period of the motion: (2007)
- Depends on both R and v
 - Is independent of both R and v
 - Depends on R and not on v
 - Depends on v and not on R
37. In a mass spectrometer used for measuring the masses of ions, the ions are initially accelerated by an electric potential V and then made to describe semicircular path of radius R using a magnetic field B. If V and B are kept constant, the ratio $\left(\frac{\text{charge on the ion}}{\text{mass of the ion}} \right)$ will be proportional to: (2007)
- $1/R^2$
 - R^2
 - R
 - $1/R$

38. When a charged particle moving with velocity $\vec{V}$ is subjected to a magnetic field of induction $\vec{B}$, the force on it is non-zero. This implies that: (2006)
- Angle between $\vec{V}$ and $\vec{B}$ is necessarily 90°
 - Angle between $\vec{V}$ and $\vec{B}$ can have any value other than 90°
 - Angle between $\vec{V}$ and $\vec{B}$ can have any value other than zero and 180°
 - Angle between $\vec{V}$ and $\vec{B}$ is either zero or 180°
39. An electron moves in a circular orbit with a uniform speed v. It produces a magnetic field B at the center of the circle. The radius of the circle is proportional to: (2005)
- B/v
 - v/B
 - $\sqrt{(v/B)}$
 - $\sqrt{(B/v)}$
40. A charged particle moves through a magnetic field in a direction perpendicular to it. Then the: (2003)
- Speed of the particle remains unchanged
 - Direction of the particle remains unchanged
 - Acceleration remains unchanged
 - Velocity remains unchanged
41. An electron having mass 'm' and kinetic energy E enter in uniform magnetic field B perpendicularly, then its frequency will be: (2001)
- $\frac{eE}{qVB}$
 - $\frac{2\pi m}{eB}$
 - $\frac{eB}{2\pi m}$
 - $\frac{2m}{eBE}$
42. In Thomson mass spectrograph $\vec{E} \perp \vec{B}$ then the velocity of undeflected electron beam will be: (2001)
- $\frac{|\vec{E}|}{|\vec{B}|}$
 - $\vec{E} \times \vec{B}$
 - $\frac{|\vec{B}|}{|\vec{E}|}$
 - $\frac{E^2}{B^2}$
43. A charge having q/m equal to 10^8 C/kg and with velocity 3×10^5 m/s enters into a uniform magnetic field $B = 0.3$ tesla at an angle 30° with direction of field. Then radius of curvature will be: (2000)
- 0.01 cm
 - 0.5 cm
 - 1 cm
 - 2 cm
44. Two identically charged particles A and B initially at rest, are accelerated by a common potential difference V. They enter into a transverse uniform magnetic field B. They describe a circular path of radii r_1 and r_2 respectively then their mass ratio is: (1999)
- $\left(\frac{r_1}{r_2} \right)^2$
 - $\left(\frac{r_2}{r_1} \right)^2$
 - $\left(\frac{r_1}{r_2} \right)$
 - $\left(\frac{r_2}{r_1} \right)$

45. A positively charged particle moving due East enters a region of uniform magnetic field directed vertically upwards. This particle will (1997)
- Move in a circular path with a decreased speed
 - Move in a circular path with a uniform speed
 - Get deflected in vertically upward direction
 - Move in circular path with an increased speed.
46. A 10 eV electron is circulating in a plane at right angles to a uniform field at magnetic induction 10^{-4} Wb/m² ($= 1.0$ gauss), the orbital radius of electron is (1996)
- 11 cm
 - 18 cm
 - 12 cm
 - 16 cm
47. A charge moving with velocity v in X-direction is subjected to a field of magnetic induction in negative X-direction. As a result, the charge will (1993)
- Remain unaffected
 - Start moving in a circular path Y-Z plane
 - Retard along X-axis
 - Moving along a helical path around X-axis
48. A uniform magnetic field acts right angles to the direction of motion of electrons. As a result, the electron moves in a circular path of radius 2 cm. If the speed of electrons is doubled, then, the radius of the circular path will be (1991)
- 2.0 cm
 - 0.5 cm
 - 4.0 cm
 - 1.0 cm
49. A deuteron of kinetic energy 50 keV is describing a circular orbit of radius 0.5 metre in a plane perpendicular to magnetic field B . The kinetic energy of the proton that describes a circular, orbit of radius 0.5 metre in the same plane with the same B is (1991)
- 25 keV
 - 50 keV
 - 200 keV
 - 100 keV

Force in the Presence of Magnetic Field, Lorentz Force and Cyclotron

50. A square loop ABCD carrying a current i , is placed near and coplanar with a long straight conductor XY carrying a current I , the net force on the loop will be: (2016 - I)



- $\frac{2\mu_0 Ii}{3\pi}$
- $\frac{\mu_0 Ii}{2\pi}$
- $\frac{2\mu_0 IiL}{3\pi}$
- $\frac{\mu_0 IiL}{2\pi}$

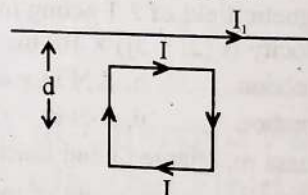
51. When a proton is released from rest in a room, it starts with an initial acceleration a_0 towards west. When it is projected towards north with a speed v_0 it moves with an initial acceleration $3a_0$ toward west. The electric and magnetic fields in the room are: (2013)

- $\frac{ma_0}{e}$ east, $\frac{3ma_0}{ev_0}$ down
- $\frac{ma_0}{e}$ west, $\frac{2ma_0}{ev_0}$ up
- $\frac{ma_0}{e}$ west, $\frac{2ma_0}{ev_0}$ down
- $\frac{ma_0}{e}$ east, $\frac{3ma_0}{ev_0}$ up

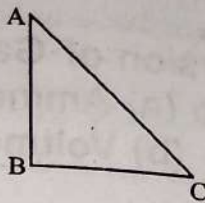
52. An alternating electric field, of frequency ν , is applied across the dees (radius = R) of a cyclotron that is being used to accelerate protons (mass = m). The operating magnetic field (B) used in the cyclotron and the kinetic energy (K) of the proton beam, produced by it, are given by: (2012 Pre)

- $B = \frac{mv}{e}$ and $K = 2m\pi^2\nu^2R^2$
- $B = \frac{2\pi m\nu}{e}$ and $K = m^2\pi\nu^2R^2$
- $B = \frac{2\pi m\nu}{e}$ and $K = 2m\pi^2\nu^2R^2$
- $B = \frac{mv}{e}$ and $K = m^2\pi\nu R^2$

53. A square loop, carrying a steady current I , is placed in a horizontal plane near a long straight conductor carrying a steady current I at a distance d from the conductor as shown in figure. The loop will experience: (2011 Mains)



- A net repulsive force away from the conductor
 - A net torque acting upward perpendicular to the horizontal plane
 - A net torque acting downward normal to the horizontal plane
 - A net attractive force towards the conductor
54. A uniform electric field and uniform magnetic field are acting along the same direction in a certain region. If an electron is projected in the region such that its velocity is pointed along the direction of fields, then the electron: (2011 Pre)
- Will turn towards right of direction of motion
 - Speed will decrease
 - Speed will increase
 - Will turn towards left direction of motion
55. A current carrying loop in the form of a right angle isosceles triangle ABC is placed in a uniform magnetic field acting along AB. If the magnetic force on the arm BC is F the force on the arm AC is: (2011 Pre)



- a. $-\sqrt{2}\vec{F}$ b. $-\vec{F}$
c. $\vec{F}$ d. $\sqrt{2}\vec{F}$

56. A particle having a mass of 10^{-2} kg carries a charge 5×10^{-8} C. The particle is given an initial horizontal velocity of 10^5 ms $^{-1}$ in the presence of electric field $\vec{E}$ and magnetic field $\vec{B}$. To keep the particle moving in a horizontal direction, it is necessary that

1. $\vec{B}$ should be perpendicular to the direction of velocity and $\vec{E}$ should be along the direction of velocity.
2. Both $\vec{B}$ and $\vec{E}$ should be along the direction of velocity.
3. Both $\vec{B}$ and $\vec{E}$ are mutually perpendicular and $\vec{E}$ perpendicular to the direction of velocity.
4. $\vec{B}$ should be along the direction of velocity and $\vec{E}$ should be perpendicular to the direction of velocity.

Which one of the following pairs of statements is possible? (2010 Mains)

- a. 1 and 3 b. 3 and 4
c. 2 and 3 d. 2 and 4

57. A square current carrying loop is suspended in a uniform magnetic field acting in the plane of the loop. If the force on one arm of the loop is $\vec{F}$, the net force on the remaining three arms of the loop is: (2010 Pre)

- a. $\vec{F}$ b. $3\vec{F}$
c. $-\vec{F}$ d. $-3\vec{F}$

58. A beam of cathode rays is subjected to crossed Electric (E) and Magnetic field (B). The fields are adjusted such that the beam is not deflected. The specific charge of the cathode rays is given by (where V is the potential difference between cathode and anode): (2010 Pre)

- a. $\frac{E^2}{2VB^2}$ b. $\frac{B^2}{2VE^2}$
c. $\frac{2VB^2}{E^2}$ d. $\frac{2VE^2}{B^2}$

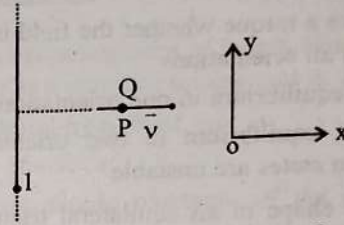
59. A closed loop PQRS carrying a current is placed in a uniform magnetic field. If the magnetic forces on segments PS, SR and RQ are F_1 , F_2 and F_3 respectively and are in the plane of the paper and along the directions shown, the force on the segment QP is: (2008)

- a. $F_3 - F_1 + F_2$ b. $F_3 - F_1 - F_2$
c. $\sqrt{(F_3 - F_1)^2 + F_2^2}$ d. $\sqrt{(F_3 - F_1)^2 - F_2^2}$

60. A beam of electron passes undeflected through mutually perpendicular electric and magnetic fields. If the electric field is switched off, and the same magnetic field is maintained, the electrons move: (2007)

- a. In a circular orbit b. Along a parabolic path
c. Along a straight line d. In an elliptical orbit

61. A very long straight wire carries a current I. At the instant when a charge +Q at point P has velocity as shown, the force on the charge is: (2005)



- a. Opposite to Ox b. Along Ox
c. Opposite to Oy d. Along Oy

62. A charge 'q' moves in a region where electric field and magnetic field both exist, then force on it is: (2002)

- a. $q(\vec{V} \times \vec{B})$ b. $q\vec{E} + q(\vec{V} \times \vec{B})$
c. $q\vec{E} + q(\vec{B} \times \vec{V})$ d. $q\vec{B} + q(\vec{E} \times \vec{V})$

63. Two long parallel wires are at a distance of 1m. If both of them carry one ampere of current in same direction, then the force of attraction on unit length of the wires will be: (1998)

- a. 2×10^{-7} N/m b. 4×10^{-7} N/m
c. 8×10^{-7} N/m d. 10^{-7} N/m

64. Two parallel wires in free space are 10 cm apart and each carries a current of 10 A in the same direction. The force exerted by one wire on the other, per metre length is (1997)

- a. 2×10^{-4} N, repulsive
b. 2×10^{-7} N, repulsive
c. 2×10^{-4} N, attractive
d. 2×10^{-7} N, attractive

65. A beam of electrons is moving with constant velocity in a region having electric and magnetic fields of strength 20 Vm $^{-1}$ and 0.5 T at right angles to the direction of motion of the electrons. What is the velocity of the electrons? (1996)

- a. 8 ms $^{-1}$ b. 5.5 ms $^{-1}$
c. 20 ms $^{-1}$ d. 40 ms $^{-1}$

66. A straight wire of length 0.5 metre and carrying a current of 1.2 ampere is placed in uniform magnetic field of induction 2 tesla. The magnetic field is perpendicular to the length of the wire. The force on the wire is (1992)

- a. 2.4 N b. 1.2 N
c. 3.0 N d. 2.0 N

Torque on a Current Loop

67. A rectangular coil of length 0.12 m and width 0.1 m having 50 turns of wire is suspended vertically in a uniform magnetic field of strength 0.2 Weber/m 2 . The coil carries a current of 2 A. If the plane of the coil is inclined at an angle of 30° with the direction of the field, the torque required to keep the coil in stable equilibrium will be: (2015 Pre)

- a. 0.12 Nm b. 0.15 Nm
c. 0.20 Nm d. 0.24 Nm

68. A current loop in a magnetic field: (2013)
- Can be in equilibrium in two orientations, one stable while the other is unstable
 - Experiences a torque whether the field is uniform or non uniform in all orientations
 - Can be in equilibrium in one orientation
 - Can be in equilibrium in two orientations, both the equilibrium states are unstable

69. A coil in the shape of an equilateral triangle of side L is suspended between the pole pieces of a permanent magnet such that B is in plane of the coil. If due to a current i in the triangle a torque τ acts on it, the side L of the triangle is: (2005)

$$\begin{array}{ll} \text{a. } \frac{2}{\sqrt{3}} \left(\frac{\tau}{Bi} \right) & \text{b. } 2 \left[\frac{\tau}{\sqrt{3}Bi} \right]^{\frac{1}{2}} \\ \text{c. } \frac{2}{\sqrt{3}} \left[\frac{\tau}{Bi} \right]^{\frac{1}{2}} & \text{d. } \frac{1 \times \tau}{\sqrt{3}Bi} \end{array}$$

70. A circular loop of area 0.01 m^2 carrying a current of 10 A , is held perpendicular to a magnetic field of intensity 0.1 T . The torque acting on the loop is (1994)

- 0.001 Nm
- 0.8 Nm
- Zero
- 0.01 Nm

71. A coil carrying electric current is placed in uniform magnetic field (1993)

- Torque is formed
- e.m.f is induced
- Both (a) and (b) are correct
- None of these

72. A current carrying coil is subjected to a uniform magnetic field. The coil will orient so that its plane becomes (1988)

- Inclined at 45° to the magnetic field
- Inclined at any arbitrary angle to dimagnetic field
- Parallel to the magnetic field
- Perpendicular to magnetic field

Galvanometer

73. Current sensitivity of a moving coil galvanometer is 5 div/mA and its voltage sensitivity (angular deflection per unit voltage applied) is 20 div/V . The resistance of the galvanometer is : (2018)

- 250Ω
- 25Ω
- 40Ω
- 500Ω

74. A galvanometer of resistance 50Ω is connected to a battery of 3 V along with a resistance of 2950Ω in series. A full scale deflection of 30 divisions is obtained in the galvanometer. In order to reduce this deflection to 20 divisions, the resistance in series should be: (2008)

- 4450Ω
- 5050Ω
- 5550Ω
- 6050Ω

Conversion of Galvanometer into (a) Ammeter and (b) Voltmeter

75. In an ammeter 0.2% of main current passes through the galvanometer. If resistance of galvanometer is G , the resistance of ammeter will be: (2014)

$$\begin{array}{ll} \text{a. } \frac{1}{499} G & \text{b. } \frac{499}{500} G \\ \text{c. } \frac{1}{500} G & \text{d. } \frac{500}{499} G \end{array}$$

76. A milli voltmeter of 25 milli volt range is to be converted into an ammeter of 25 ampere range. The value (in ohm) of necessary shunt will be: (2012 Pre)

- 0.001
- 0.01
- 1
- 0.05

77. A galvanometer of resistance, G , is shunted by a resistance S ohm. To keep the main current in the circuit unchanged, the resistance to be put in series with the galvanometer is: (2011 Mains)

$$\begin{array}{ll} \text{a. } \frac{S^2}{(S+G)} & \text{b. } \frac{SG}{(S+G)} \\ \text{c. } \frac{G^2}{(S+G)} & \text{d. } \frac{G}{(S+G)} \end{array}$$

78. A galvanometer has a coil of resistance 100 ohm and gives a full scale deflection for 30 mA current. If it is to work as a voltmeter of 30 volt range, the resistance required to be added will be: (2010 Pre)

- 1000Ω
- 900Ω
- 1800Ω
- 500Ω

79. A galvanometer having a coil resistance of 60Ω shows full scale deflection when a current of 1.0 A passes through it. It can be converted into an ammeter to read currents up to 5.0 A by: (2009)

- Putting in series a resistance of 15Ω
- Putting in series a resistance of 240Ω
- Putting in parallel a resistance of 15Ω
- Putting in parallel a resistance of 240Ω

80. The resistance of an ammeter is 13Ω and its scale is graduated for a current upto 100 amps . After an additional shunt has been connected to this ammeter, it becomes possible to measure currents upto 750 amperes by this meter. The value of shunt resistance is (2007)

- 2Ω
- 0.2Ω
- $2 \text{ k}\Omega$
- 20Ω

81. A galvanometer acting as a voltmeter will have: (2004)

- A high resistance in series with its coil
- A low resistance in parallel with its coil
- A low resistance in series with its coil
- A high resistance in parallel with its coil

Magnetic Moment

82. A galvanometer of 50 ohm resistance has 25 divisions. A current of 4×10^{-4} ampere gives a deflection of one division. To convert this galvanometer into a voltmeter having a range of 25 volts, it should be connected with a resistance of:

- 245 Ω as a shunt
- 2550 Ω in series
- 2450 Ω in series
- 2500 Ω as a shunt

(2004)

83. To convert a galvanometer into a voltmeter one should connect a:

- High resistance in series with galvanometer
- Low resistance in series with galvanometer
- High resistance in parallel with galvanometer
- Low resistance in parallel with galvanometer

(2002)

84. Resistance of a galvanometer coil is 8 Ω and 2 Ω shunt resistance is connected with it. If main current is 1 A then the current flowing through 2 Ω resistance will be:

- 0.2 A
- 0.8 A
- 0.1 A
- 0.4 A

(1998)

85. To convert a galvanometer into an ammeter, one needs to connect a

- Low resistance in parallel
- High resistance in parallel
- Low resistance in series
- High resistance in series

(1992)

86. A uniform conducting wire of length 12 a and resistance 'R' is wound up as a current carrying coil in the shape of,

- an equilateral triangle of side 'a'.
- a square of side 'a'.

The magnetic dipole moments of the coil in each case respectively are:

(2021)

- $3 Ia^2$ and Ia^2
- $3 Ia^2$ and $4 Ia^2$
- $4 Ia^2$ and $3 Ia^2$
- $\sqrt{3} Ia^2$ and $3 Ia^2$

87. A wire of length L metre carrying a current of I ampere is bent in the form of a circle. its magnetic moment is.

(2020-Covid)

- $I \pi L^2 / 4 Am^2$
- $2I L^2 / \pi Am^2$
- $I L^2 / 4 \pi Am^2$
- $I L^2 / 4 Am^2$

88. A charged particle (charge q) is moving in a circle of radius R with uniform speed v. The associated magnetic moment μ is given by:

(2007)

- qvR^2
- $qvR^2/2$
- qvR
- $qvR/2$

89. If number of turn, area and current through it is given by n, A and i respectively then its magnetic moment will be:

(2001)

- niA
- $n^2 iA$
- niA^2
- $\frac{ni}{\sqrt{A}}$

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
d	b	c	d	c	b	b	b	a	a	d	c	a	a	a	a	b
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
c	a	a	b	d	c	c	c	d	b	c	a	c	a	b	b	d
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
d	b	a	c	c	a	c	a	b	a	b	a	a	c	d	a	c
52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68
c	d	b	b	c	c	a	c	a	d	b	a	c	d	b	c	a
69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
b	c	a	d	a	a	c	a	c	b	c	a	a	c	a	b	a
86	87	88	89													
d	c	d	a													

Explanations

1. (d) According to Biot Savart's law

$$d\vec{B} = \frac{\mu_0 (I d\vec{\ell} \times \vec{r})}{4\pi r^3}$$

Therefore, the expression for magnetic field depends on current carrying element $I d\vec{\ell}$, which is a vector quantity, therefore, statement-I is correct and statement-II is wrong because charge q is not a vector source.

2. (b) For current carrying cable (i.e., cylindrical) magnetic field are as follows:

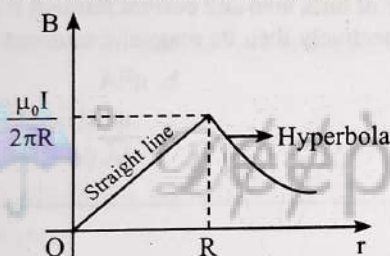
Inside ($r < R$)

$$B = \left(\frac{\mu_0 I}{2\pi R^2} \right) r \text{ i.e., } B_{\text{inside}} \propto r$$

Outside ($r \geq R$)

$$B = \frac{\mu_0 I}{2\pi r} \text{ i.e., } B_{\text{outside}} \propto \frac{1}{r}$$

Both relations gives plot of magnetic field (B) with respect of distance from the axis of the cable as follow:



3. (c) Force per unit length

$$F = \frac{\mu_0 I_1 I_2}{2\pi d} \Rightarrow I_1 = I_2 = i$$

$$F_1 = F_2 = F = \frac{\mu_0 i^2}{2\pi d}$$

$$F_1 \perp F_2, \theta = 90^\circ$$

$$F_{\text{net}} = \sqrt{2} F$$

$$= \frac{\sqrt{2} \mu_0 i^2}{2\pi d} = \frac{\mu_0 i^2}{\sqrt{2} \pi d}$$

4. (d) Net magnetic field, $B = \sqrt{B_1^2 + B_2^2}$

$$= \sqrt{\left(\frac{\mu_0 I_1}{2\pi d} \right)^2 + \left(\frac{\mu_0 I_2}{2\pi d} \right)^2}$$

$$= \frac{\mu_0}{2\pi d} \sqrt{I_1^2 + I_2^2}$$

5. (c) Magnetic field due to a 5A $\rightarrow \frac{5\mu_0}{2\pi \times 2.5} = \frac{2\mu_0}{\pi} \otimes$

$$\text{Magnetic field due to } 2.5A \rightarrow \frac{2.5\mu_0}{2\pi \times 2.5} = \frac{\mu_0}{2\pi} \odot$$

$$\text{Resultant Magnetic field} = \frac{2\mu_0}{2\pi} - \frac{\mu_0}{2\pi} = \frac{\mu_0}{2\pi} \otimes$$

6. (b) The magnetic field is the area around a magnet where the magnetic force can be felt. Magnetic field is measured in units of tesla (T), named after the famous physicist Nikola Tesla.

7. (b) The direction of the magnetic field, due to current, is given by the right-hand rule. At axis AB, the components of magnetic field will cancel each other and the resultant magnetic field will be zero.

8. (b) According to Biot-Savart's law,

$$d\vec{B} \propto i \left(\frac{d\vec{\ell} \times \vec{r}}{r^3} \right) = \frac{\mu_0}{4\pi} i \left(\frac{d\vec{\ell} \times \vec{r}}{r^3} \right)$$

9. (a) $B = \frac{\mu_0 i}{2\pi r}$ or $B \propto \frac{1}{r}$

When r is doubled, the magnetic field becomes halved.
 $\therefore$ Here magnetic field will be

$$\frac{0.4}{2} = 0.2 \text{ T}$$

10. (a) $B \propto 1/r$

As the distance is increased to three times, the magnetic induction will be reduced to one third.

$$\text{Hence } B = \frac{1}{3} \times 10^{-3} \text{ T}$$

$$B = 3.33 \times 10^{-4} \text{ T}$$

11. (d) We know that at the center of the circular loop

$$B = \frac{\mu_0 I}{2R}$$

$$\text{Since } \ell = 2\pi R = n(2\pi r) \Rightarrow r = \frac{R}{n}$$

$$\text{For one turn } B = \frac{\mu_0 i}{2R} \text{ and}$$

$$\text{For } n \text{ turn } B' = \frac{\mu_0 n i}{2r} \Rightarrow B' = \frac{\mu_0 n^2 i}{2R} = n^2 B$$

12. (c) Magnetic field due to a circular loop

$$B = \frac{\mu_0 N I}{2r}$$

where $N \rightarrow$ no. of loops

$$I = \frac{q}{T} = \frac{e}{1/n} = ne$$

$$B = \frac{\mu_0 ne}{2r} \left(\because N = 1 \text{ as } e^- \text{ makes only one loop} \right)$$

13. (a) Here, in first coil current I is flowing.

Magnetic field due to this coil carrying current I at the center

$$B_1 = \frac{\mu_0 I}{2R}$$

Magnetic field due to other coil carrying current $2I$ at the

$$\text{centre } B_2 = \frac{\mu_0 2I}{2R}$$

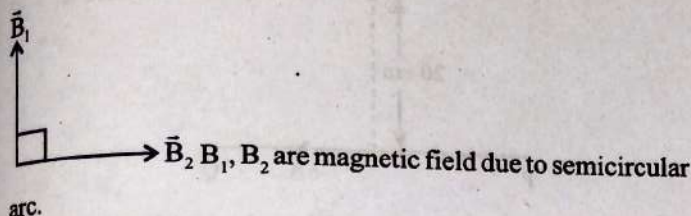
Now, resultant magnetic field $B = \sqrt{B_1^2 + B_2^2}$

$$= \sqrt{\left(\frac{\mu_0 I}{2R}\right)^2 + \left(\frac{\mu_0 2I}{2R}\right)^2} = \frac{\mu_0 I}{2R} \sqrt{1+4} = \sqrt{5} \frac{\mu_0 I}{2R}$$

14. (a) We know that current, $I = \frac{q}{t} = qf$

Magnetic induction at center of the ring $= \frac{\mu_0 I}{2R} = \frac{\mu_0 qf}{2R}$

15. (a)



$$B_1 = B_2 = \frac{\mu_0 i}{4R}$$

$$B_R = \sqrt{B_1^2 + B_2^2} = \sqrt{2} \left(\frac{\mu_0 i}{4R} \right) = \frac{\mu_0 i}{2\sqrt{2}R}$$

$$16. (a) B = \frac{\mu_0 I}{2R} = \frac{\mu_0}{2R} \left(\frac{q}{t} \right) = \frac{\mu_0 qf}{2R}$$

17. (b) Let r_1 and r_2 are the radius of the coil 1 and 2 and Magnetic induction at their center be B_1 and B_2 then,

$$B_1 = \frac{\mu_0 I_1}{2r_1}, B_2 = \frac{\mu_0 I_2}{2r_2}$$

Let R_1 and R_2 are resistance of the coils

$\therefore$ Resistance of coil $\propto$ Length of coil [Circumference $= 2\pi r$]

$$\text{So, } \frac{R_1}{R_2} = \frac{r_1}{r_2} = 2 \quad \dots (i)$$

$$[\because B_1 = B_2; r_1 = 2r_2]$$

$$\Rightarrow \frac{\mu_0 I_1}{2r_1} = \frac{\mu_0 I_2}{2r_2} \quad [\because r_1 = 2r_2]$$

$$\Rightarrow I_1 = 2I_2 \quad \dots (ii)$$

Let potential difference across them be V_1 and V_2 respectively, using eq. (i) & (ii)

$$\frac{V_1}{V_2} = \frac{I_1 R_1}{I_2 R_2} = 2 \times 2 = 4$$

$$18. (c) B_1 = B = \frac{\mu_0 I}{2R}$$

$$B_2 = \frac{\mu_0 (2I)}{2r}$$

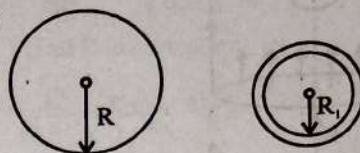
$$\because 2 \times 2\pi r = 2\pi R \quad \therefore r = R/2$$

$$\Rightarrow B_2 = 4 \frac{\mu_0 I}{2R} = 4B$$

19. (a) Magnetic field at the center of coil $B = \frac{\mu_0 i N}{2a}$

$$\frac{4\pi \times 10^{-7} \times 5 \times 50}{2 \times 10/100} = 1.57 \times 10^{-3} \text{ T} = 1.57 \text{ mT}$$

20. (a)

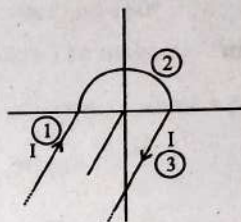


$$2\pi R = 4\pi R_1 = L$$

$$B_1 = \frac{\mu_0 I}{2R} \times N = \frac{\mu_0 I \pi}{L}; B_2 = \frac{\mu_0 I}{2R_1} \times 2 = \frac{4\mu_0 I \pi}{L} \Rightarrow \frac{B_1}{B_2} = \frac{1}{4}$$

21. (b) 'B' due to segment '1'

$$B = \frac{\mu_0 I}{4\pi R} [\sin 90^\circ + \sin \theta] (-\hat{k})$$



Due to wire

$$\vec{B} = \frac{\mu_0 I}{4\pi d} [\sin \theta_1 + \sin \theta_2]$$

$$B_1 = B_3 \Rightarrow \theta_1 = 90^\circ \quad \theta_2 = 0^\circ$$

$$|B_1| = |B_3| = \frac{\mu_0 I}{4\pi R} [\sin 90^\circ + \sin 0^\circ]$$

$$B_1 = \frac{\mu_0 I}{4\pi R} (-\hat{k}) = -\frac{\mu_0 I}{4\pi R} (\hat{k})$$

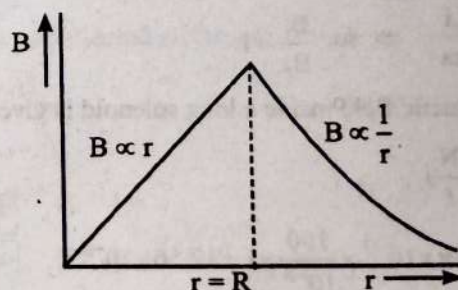
$$B_2 = \frac{1}{2} \frac{\mu_0 I}{2R} (-\hat{i})$$

$$B = -\frac{\mu I}{4R} (\hat{i})$$

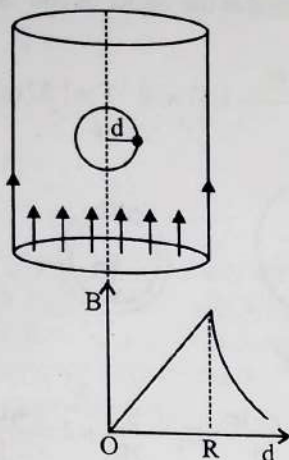
$$\vec{B} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3$$

$$= -\frac{\mu_0 I}{4\pi R} [\pi \hat{i} + 2\hat{k}]$$

22. (d) Inside the wire $B \propto r$ and outside the wire $B \propto \frac{1}{r}$.



23. (c)

Inside ($d < R$)

Magnetic field inside conductor

$$B = \frac{\mu_0 i}{2\pi R^2} d$$

or $B = Kd$

...(1)

Straight line passing through origin

At surface ($d = R$)

$$B = \frac{\mu_0 i}{2\pi R} \quad \dots(2)$$

...(2)

Maximum at surface

Outside ($d > R$)

$$B = \frac{\mu_0 i}{2\pi d}$$

or $B \propto \frac{1}{d}$ (Hyperbolic)24. (c) Current density $J = \frac{I}{\pi a^2}$

From Ampere's circuital

$$\oint B \cdot dl = \mu I_{\text{enclosed}}$$

For $r < a$

$$B \times 2\pi r = \mu_0 J \pi r^2$$

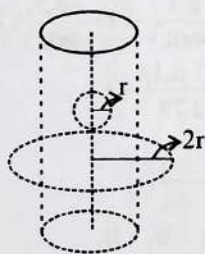
$$\text{at } r = \frac{a}{2}, B_1 = \frac{\mu_0 I}{4\pi a}$$

For $r > a$

$$B \times 2\pi r = \mu_0 I \Rightarrow B = \frac{\mu_0 i}{2\pi r}$$

At $r = 2a$

$$B_2 = \frac{\mu_0 i}{4\pi a} \Rightarrow \text{so, } \frac{B_1}{B_2} = 1$$



25. (c) Magnetic field inside a long solenoid is given by

$$B = \mu_0 \frac{N}{\ell} i$$

$$\therefore B = 4\pi \times 10^{-7} \times \frac{100}{10^{-3}} \times 1 = 12.56 \times 10^{-2} \text{ T}$$

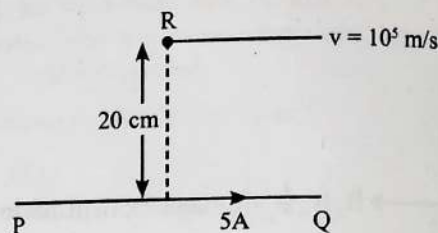
$$26. (d) B = \mu_0 \frac{N}{\ell} I$$

$$= 4\pi \times 10^{-7} \times \frac{100}{0.5} \times 2.5$$

$$= 6.28 \times 10^{-4} \text{ T}$$

$$27. (b) B = \mu_0 n i; n' = \frac{n}{2}; i' = 2i \Rightarrow B' = B$$

28. (c)



A current carrying wire will produce a closed magnetic field lines around itself. So, applying

Biot-savart law for infinite straight current-carrying wire and the magnitude of magnetic field at point R due to infinite wire will be given as follows:

$$B = \frac{\mu_0 I}{2\pi r}$$

...(i)

Given,

$$I = 5A$$

$$r = 20 \text{ cm} = 0.2 \text{ m}$$

Now, putting in eqⁿ (i);

$$|B| = \frac{4\pi \times 10^{-7} \times 5}{2\pi \times 0.2 \text{ m}} = 5 \times 10^{-6} \text{ T}$$

$$\vec{B} = 5 \times 10^{-6} \text{ T}$$

[i.e., directed out of the plane]

The force experienced by electron due to magnetic field produced by current carrying wire is given as below:

$$F = q(\vec{v} \times \vec{B})$$

$$\text{Since, } \vec{v} \perp \vec{B}$$

$$\text{i.e., } F = qvB \sin \theta$$

$$F = qvB \sin 90^\circ$$

$$F = qvB$$

$$F = 1.6 \times 10^{-19} \times 10^5 \times 5 \times 10^{-6}$$

$$= 16 \times 5 \times 10^{-21}$$

$$= 80 \times 10^{-21} \text{ N}$$

$$F = 8 \times 10^{-20} \text{ N}$$

$$29. (a) r_H = \frac{mv}{eB} = \frac{p}{eB} \Rightarrow r_\alpha = \frac{p}{2eB}$$

$$\frac{r_H}{r_\alpha} = \frac{\frac{p}{eB}}{\frac{p}{2eB}} \Rightarrow \frac{r_H}{r_\alpha} = \frac{2}{1}$$

30. (c) We know that

$$f = \frac{qB}{2\pi m}$$

$$f = \frac{eB}{2\pi m}$$

$$f = \frac{1.76 \times 10^{11} \times 3.57 \times 10^{-2}}{2 \times 3.14} \text{ Hz}$$

$$f = 10^9 \text{ Hz or } 1 \text{ GHz}$$

31. (a) $m_a = 4m_p$

$$q_a = 2e$$

$$q_p = e$$

$$R = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$$

$$\therefore R_a = R_p$$

$$\therefore \frac{2m_a K_a}{q_a^2 B^2} = \frac{2m_p K_p}{q_p^2 B^2} \quad (m_a = 4m, q_a = 2e, q_p = e)$$

$$\Rightarrow \frac{4m_p K_a}{4e^2} = \frac{m_p (1 \text{ MeV})}{e^2}$$

$$\Rightarrow K_a = 1 \text{ MeV}$$

32. (b) $\frac{mv^2}{R} = qvB \Rightarrow R = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$

For same R and B, $K \propto \frac{q^2}{m}$

$$\frac{K'}{1 \text{ MeV}} = \frac{(2e)^2}{(4m)} \times \frac{m}{e^2} = 1 \Rightarrow K' = 1 \text{ MeV}$$

33. (b) Since $T = \frac{2\pi m}{qB}$

It is independent of both V and R

34. (d) Magnetic field $(\vec{B}) = 2\hat{j}$

Force on charged particle $= q(\vec{v} \times \vec{B})$

$$= (-2 \times 10^{-6}) [(2\hat{i} + 3\hat{j}) \times 2\hat{j}] \times 10^6$$

$$= -8\hat{k}$$

So, force acting is 8N in -z direction

35. (d) Since the force acts in direction perpendicular to velocity and magnetic field, no work will be done on the body. Hence, magnetic field does not change the kinetic energy of body as it does not change the velocity. K.E. of particle will be T.

36. (b) Speed of particle $(v) = \frac{qBR}{m}$

$$\text{Time period of motion} = \frac{2\pi R}{v}$$

$$= \frac{2\pi R}{qBR} \times m = \frac{2\pi m}{qB}$$

So, time period is independent of both V and R

37. (a) Radius of semicircular path described by the ions $= \frac{mv}{qB}$

And as we know $mv(p) = \sqrt{2m \text{ K.E.}} = \sqrt{2mgV}$

$$r = \frac{\sqrt{2mgV}}{qB} = \sqrt{\frac{m}{q}} \times \sqrt{\frac{2v}{B}}$$

As V and B are constant

$$R \propto \sqrt{\frac{m}{q}} \Rightarrow \text{So, } \frac{q}{m} \propto \frac{1}{R^2}$$

38. (c) Since force on a particle in magnetic field

$$\vec{F} = q(\vec{v} \times \vec{B})$$

$$\vec{F} = qvB \sin \theta \hat{n}$$

Since F is non-zero,

$$\theta \text{ cannot be } 0^\circ \text{ or } 180^\circ \text{ because } \begin{bmatrix} \sin 0^\circ = 0 \\ \sin 180^\circ = 0 \end{bmatrix}$$

Hence, angle between $\vec{v}$ and $\vec{B}$ can have any value other than zero and 180°

39. (c) The magnetic field produced by moving

electron in circular path, $B = \frac{\mu_0 i}{2r}$

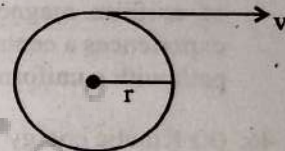
Where $i = \frac{q}{t}$

$$= \frac{q \times v}{2\pi r}$$

Putting value of i in B

$$B = \frac{\mu_0 \times qv}{2r \times 2\pi r} = \frac{\mu_0 qv}{4\pi r^2}$$

$$\Rightarrow r^2 \propto \frac{v}{B} \Rightarrow r \propto \sqrt{\frac{v}{B}}$$



40. (a) When a charged particle moves through a perpendicular magnetic field, then a magnetic force acts on it which changes the direction of particle but does not alter the magnitude of its velocity (i.e. speed).

$$\text{Here } \vec{F} \perp \vec{v} \Rightarrow |\vec{v}| = \text{constant}$$

41. (c) The frequency of revolution of charged particle in a perpendicular magnetic field is

$$v = \frac{1}{T} = \frac{1}{2\pi r/v}$$

$$= \frac{v}{2\pi r} = \frac{v}{2\pi} \times \frac{eB}{mv}$$

$$= \frac{eB}{2\pi m}$$

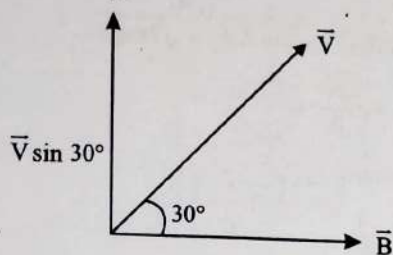
42. (a) The undeflected electrons

$$|\vec{F}|_{\text{elec}} = |\vec{F}|_{\text{magn}}$$

$$e|\vec{E}| = ev|\vec{B}|$$

$$v = \frac{|\vec{E}|}{|\vec{B}|}$$

43. (b) $r = \frac{mV_{\perp}}{qB}$



$$r = \left(\frac{m}{q} \right) \left(\frac{3 \times 10^5 \times \sin 30^\circ}{0.3} \right)$$

$$r = \frac{3 \times 10^5}{10^8 \times 0.3 \times 2} = 0.5 \times 10^{-2} \text{ m} = 0.5 \text{ cm}$$

44. (a) $r = \frac{\sqrt{2mqv}}{qB}$

$$r \propto \sqrt{m}$$

$$\frac{m_1}{m_2} = \left(\frac{r_1}{r_2} \right)^2$$

45. (b) When a positively charged particle enters in a region of uniform magnetic field, directed vertically upwards, it experiences a centripetal force which will move it in circular path with a uniform speed.

46. (a) Kinetic energy of electron $\left(\frac{1}{2} \times mv^2 \right) = 10 \text{ eV}$ and magnetic induction $B = 10^{-4} \text{ Wb/m}^2$.

$$\text{Therefore } \frac{1}{2} (9.1 \times 10^{-31}) v^2 = 10 \times (1.6 \times 10^{-19})$$

$$\text{Or, } v^2 = \frac{2 \times 10 \times (1.6 \times 10^{-19})}{9.1 \times 10^{-31}} = 3.52 \times 10^{12}$$

$$\text{Or, } v = 1.876 \times 10^6 \text{ m/s}$$

$$\text{Centripetal force} = \frac{mv^2}{r} = Bev$$

$$\text{Therefore } r = \frac{mv}{Be} = \frac{(9.1 \times 10^{-31}) \times (1.876 \times 10^6)}{10^{-4} \times (1.6 \times 10^{-19})}$$

$$= 11 \times 10^{-2} \text{ m} = 11 \text{ cm}$$

47. (a) The force acting on a charge is:

$$\vec{F} = q(\vec{v} \times \vec{B}) \text{ or } F = qvB \sin \theta \text{ Here } \theta = 180^\circ$$

$$\therefore F = 0$$

48. (c) The radius of the electron moving in a magnetic field is given by:

$$r = \frac{mv}{qB}$$

$$\Rightarrow r \propto v,$$

So, when the speed of the electron is doubled, the radius will also be doubled for the electron.

$$\Rightarrow r_2 = 2r_1 = 2 \times 2 = 4 \text{ cm}$$

49. (d) If a charged particle is orbiting in a circular path in a magnetic field then,

$$\frac{mv^2}{r} = Bqv$$

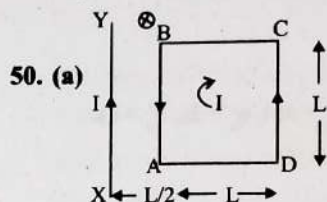
$$mv^2 = Bqvr \Rightarrow v = \frac{Bqr}{m}$$

$$E_{KE} = \frac{1}{2} mv^2 = \frac{1}{2} Bqvr = Bq \frac{r}{2} \cdot \frac{Bqr}{m} = \frac{B^2 q^2 r^2}{2m}$$

$$\text{For deuteron, } E_1 = \frac{B^2 q^2 r^2}{2 \times 2m}$$

$$\text{For proton, } E_2 = \frac{B^2 q^2 r^2}{2m}$$

$$\frac{E_1}{E_2} = \frac{1}{2} \Rightarrow \frac{50 \text{ keV}}{E_2} = \frac{1}{2} \Rightarrow E_2 = 100 \text{ keV}$$



50. (a)

$$F_{AB} = \frac{\mu_0 I i L}{2\pi \left[\frac{L}{2} \right]} \text{ towards XY}$$

$$F_{CD} = \frac{\mu_0 I i L}{2\pi \left[\frac{3L}{2} \right]} \text{ away from XY}$$

$$F_{\text{net}} = F_{AB} - F_{CD}$$

$$= \frac{2 \mu_0 I i}{3 \pi}$$

51. (c) Acceleration of charged particle $\vec{a} = \frac{q}{m} (\vec{E} + \vec{v} \times \vec{B})$

$$\text{Released from rest } \Rightarrow \vec{a} = \frac{q}{m} \vec{E} = a_0 (\text{west})$$

$$\Rightarrow \vec{E} = \frac{ma_0}{e} (\text{west})$$

When it is projected towards north, acceleration due to magnetic force $= 2a_0$

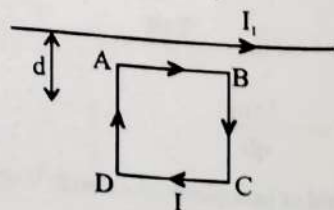
$$\text{Therefore magnetic field} = \frac{2ma_0}{ev_0} (\text{down})$$

52. (c) $\frac{mv^2}{R} = qVB \Rightarrow R = \frac{mv}{qB}$

$$v = \frac{1}{T} = \frac{v}{2\pi R} = \frac{qB}{2\pi m} \Rightarrow B = \frac{2\pi mv}{e}$$

$$\text{Kinetic energy } K = \frac{1}{2} mv^2 = \frac{1}{2} m (2\pi Rv)^2 = 2\pi^2 mv^2 R^2$$

53. (d) Use the formula $\vec{F} = i(\vec{l} \times \vec{B})$



Force on segment AD = BC and force on segment AB is more than DC. And as we know current carrying Wires in same direction attract each other. Hence, a net attractive force is there towards the conductor.

54. (b) Force on electron due to magnetic field

$$\vec{F}_B = -e(\vec{v} \times \vec{B}) \quad [\because \text{Angle between } \vec{v} \text{ and } \vec{B} \text{ is zero}] = 0$$

Force on electron due to electric field

$$\vec{F}_E = -e\vec{E}$$

So, electric field is opposing the motion of electron. Hence, the electron speed will decrease.

55. (b) Net force on a current carrying closed loop in a uniform magnetic field is zero and, the e.m.f is acting along AB so force on part AB is zero

$$\therefore \vec{F}_{AB} + \vec{F}_{AC} + \vec{F}_{BC} = 0$$

$$\Rightarrow 0 + \vec{F}_{AC} + \vec{F}_{BC} = 0$$

$$\Rightarrow \vec{F}_{AC} = -\vec{F}_{BC} = -\vec{F}$$

Force on the arm AC is equal to $(-\vec{F})$

56. (c) If gravity is not considered then option (2) and (3) are correct but if considered then (3), (4) are correct. By considering level of exam we suggest (2), (3) both are correct

57. (c) Net force on loop is zero.

$$\vec{F}_{\text{net}} = 0$$

$$\vec{F} + \vec{F}' = 0$$

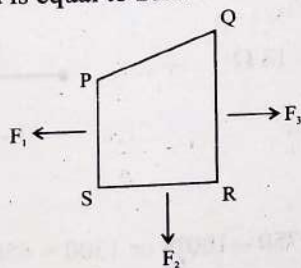
Net force on remaining arm

$$\vec{F}'' = -\vec{F}$$

58. (a) $qvB = qE \Rightarrow v = \frac{E}{B}$

$$\text{But } \frac{1}{2}mv^2 = qV \text{ so } \frac{q}{m} = \frac{v^2}{2V} = \frac{E^2}{2VB^2}$$

59. (c) Force on a current carrying closed loop in a uniform magnetic field is equal to zero.

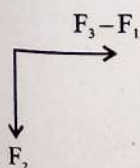


So,

Force on the segment PQ must be equal to resultant force on remaining part.

Net force on remaining part.

$$= \sqrt{(F_3 - F_1)^2 + F_2^2}$$



So, force on segment PQ = $\sqrt{(F_3 - F_1)^2 + F_2^2}$

60. (a) As the direction of magnetic field $\vec{B}$ is perpendicular to the motion of charged particle, the electrons move in a circular orbit.

61. (d) According to Fleming left hand rule, direction of force is along Oy axis which is perpendicular to wire. $\vec{B}$ is along -ve z axis.

$\vec{v}$ is in +ve x axis.

$$\vec{f} = q(\vec{v} \times \vec{B})$$

So, force is along +ve y axis.

62. (b) Lorentz force $\vec{F}_L = \vec{F}_e + \vec{F}_m = q\vec{E} + q(\vec{v} \times \vec{B})$

$$63. (a) F = \frac{\mu_0 i_1 i_2}{2\pi d} = \frac{4\pi \times 10^{-7} \times 1 \times 1}{2\pi \times 1} = 2 \times 10^{-7} \text{ N/m}$$

64. (c) Distance between two parallel wires (x) = 10 cm = 0.1 m; Current in each wire = $I_1 = I_2 = 10$ A and length of wire (l) = 1 m.

$$\text{Force on the wire (F)} = \frac{\mu_0 I_1 I_2 \times l}{2\pi x}$$

$$= \frac{(4\pi \times 10^{-7}) \times 10 \times 10 \times 1}{2\pi \times 0.1} = 2 \times 10^{-4} \text{ N}$$

Since the current is flowing in the same direction, therefore the force will be attractive.

65. (d) Electric field (E) = 20 V/m and magnetic field B = 0.5 T. The force on electron in a magnetic field = $e\vec{v} \times \vec{B}$

Force on electron in an electric field = $e\vec{E}$

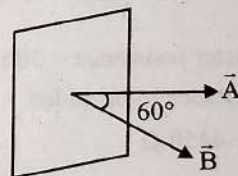
Since the electron is moving with constant velocity.

$\therefore$ Net force on the electron is zero in the direction of its motion

$$eE = evB \Rightarrow v = E/B = 20/0.5 = 40 \text{ ms}^{-1}$$

$$66. (b) F = i(\vec{l} \times \vec{B}) = 1.2 \times 0.5 \times 2 = 1.2 \text{ N}$$

67. (c)



$$\vec{\tau} = \vec{M} \times \vec{B}$$

$$|\vec{\tau}| = MB \sin \theta = NIAB \sin \theta = 0.20 \text{ Nm}$$

68. (a) A current loop in a magnetic field can be in equilibrium in two orientations, one stable while the other is unstable

69. (b) Torque acting on the coil

$$\tau = BiNA \sin \theta \quad \text{where } (M = NiA)$$

Here, N = 1 and $\theta = 90^\circ$

Now, $\tau = BiA$

$$A = \text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} \ell^2$$

$$\Rightarrow \tau = Bi \frac{\sqrt{3}}{4} \ell^2 \Rightarrow \ell^2 = \frac{4\tau}{Bi\sqrt{3}}$$

$$\ell = 2 \left(\frac{\tau}{Bi\sqrt{3}} \right)^{1/2}$$

70. (c) Area (a) = 0.01 m^2 ;

Current (I) = 10 A ;

Angle (ϕ) = 90° and magnetic field (B) = 0.1 T .

Therefore actual angle $\theta = (90^\circ - \phi) = (90^\circ - 90^\circ) = 0^\circ$.

And torque acting on the loop (τ) = $I(\vec{A} \times \vec{B}) = IAB \sin \theta$
 $= 10 \times 0.01 \times 0.1 \times \sin 0^\circ = 0$.

71. (a) A current carrying coil has magnetic dipole moment.
Hence a torque $\vec{M} \times \vec{B}$ acts on it in magnetic field.

72. (d) If the magnetic field is perpendicular to the plane of the coil, the torque acting on the coil becomes zero.

73. (a) Voltage per division = $\frac{1}{20} \text{ V}$

Current per division = $\frac{1}{5000} \text{ A}$

$$R = \frac{V}{I} = \frac{\frac{1}{20}}{\frac{1}{5000}} = 250 \Omega$$

74. (a) Total initial resistance

$$= R_G + R_1 = (50 + 2950) \Omega = 3000 \Omega$$

$$\varepsilon = 3 \text{ V}$$

$$\therefore \text{Current} = \frac{3 \text{ V}}{3000 \Omega} = 1 \times 10^{-3} \text{ mA}$$

If the deflection has to be reduced to 20 divisions,

Current $i = 1 \text{ mA} \times \frac{2}{3}$ as the full deflection scale for 1 mA

= 30 divisions

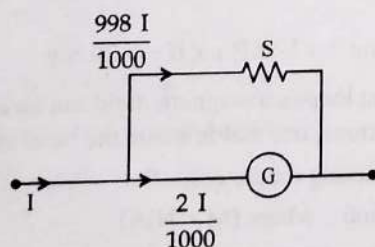
$$3 \text{ V} = 3000 \Omega \times 1 \text{ mA} = x \Omega \times \frac{2}{3} \text{ mA} \Rightarrow x = 3000 \times 1 \times \frac{3}{2} = 4500 \Omega$$

But the galvanometer resistance = 50Ω

Therefore the resistance to be added

$$= (4500 - 50) \Omega = 4450 \Omega$$

75. (c)



$$\left(\frac{2I}{1000}\right)G = \left(\frac{998I}{1000}\right)S \Rightarrow S = \frac{G}{499}$$

Total resistance of Ammeter

$$R = \frac{SG}{S+G} = \frac{\left(\frac{G}{499}\right)G}{\left(\frac{G}{499}\right)+G} = \frac{G}{500}$$

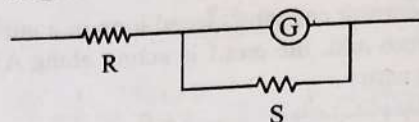
76. (a) $S(I - I_g) = V_g$

$$S \times I = V_g \quad [\text{as } I_g \text{ is very small}]$$

$$S = \frac{V_g}{I}$$

$$S = \frac{25 \times 10^{-3}}{25} = 0.001 \Omega$$

77. (c) Let resistance R is to be put in series with galvanometer G to keep the main current in circuit unchanged.



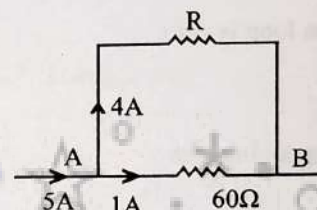
$$\therefore \left(\frac{GS}{G+S} + R\right) = G \Rightarrow R = G - \frac{GS}{G+S} = \frac{G^2}{G+S}$$

78. (b) Let required resistance be R then

$$(R + R_g)I_g = V \Rightarrow (R + 100)(30 \times 10^{-3}) = 30$$

$$\Rightarrow R = 900 \Omega$$

79. (c)



We can put a resistance R in parallel with coil potential difference between A and B will be $4R = 1 \times 60$

$$\Rightarrow R = 15 \Omega$$

80. (a) Let the shunt resistance be S .

Given $I = 750 \text{ A}$,

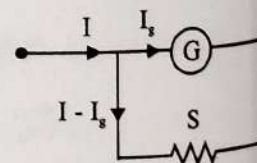
$$I_g = 100 \text{ A}, G = 13 \Omega$$

From the figure,

$$I_g G = (I - I_g)S$$

$$\text{or } 100 \times 13 = [750 - 100]S \text{ or } 1300 = 650 S$$

$$\therefore S = \frac{1300}{650} = 2 \Omega$$



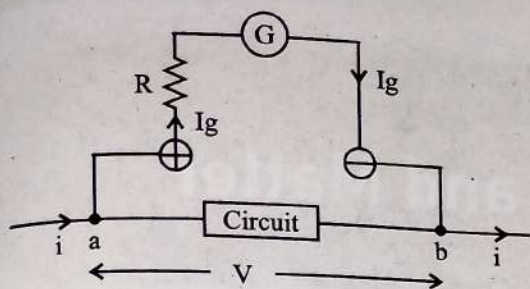
81. (a) Voltmeter $\rightarrow$

82. (c)

According to question $25 = I(R + R_g)$

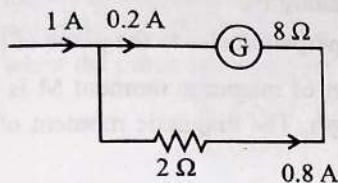
$$= (4 \times 10^{-4} \times 25)(R + 50) \Rightarrow R + 50 = 2500 \Rightarrow R = 2450 \Omega$$

83. (a)



Galvanometer can be converted into a voltmeter by connecting a high resistance (R) in series with it. The whole assembly called voltmeter is connected in parallel between the points where potential difference has to be measured.

$$84. (b) S = \left(\frac{i - i_s}{i_s} \right) G$$



$$\frac{i_s}{i} = \frac{G}{S + G} = \frac{8}{2 + 8} = \frac{8}{10}$$

$$i_s = 0.8i = 0.8 \times 1 = 0.8 \text{ A}$$

85. (a) To convert a galvanometer into ammeter, one needs to connect a low resistance in parallel so that maximum current passes through the shunt and ammeter remains protected.

86. (d) Current flowing in both loops will be same i.e., I .

For magnetic moment of an equilateral triangle.

$$\text{loop} = NIA \Rightarrow M_T = N_T IA_T$$

$$\Rightarrow M_T = \left(\frac{12a}{3a} \right) \times I \times \frac{\sqrt{3}}{4} a^2 \Rightarrow M_T = \sqrt{3} Ia^2$$

For magnetic moment of a square loop $= N_s IA_s$

$$\Rightarrow M_s = N_s IA_s \Rightarrow M_s = \left(\frac{12a}{4a} \right) \times I \times a^2$$

$$\Rightarrow M_s = 3Ia^2$$

87. (c) If r is the radius of the circle,

$$\text{then } L = 2\pi r, \quad r = \frac{L}{2\pi}$$

$$\text{Area} = A = \pi r^2 = \frac{\pi L^2}{4\pi^2} = \frac{L^2}{4\pi}$$

$$\text{Magnetic Moment} = IA = \frac{IL^2}{4\pi}$$

88. (d) Magnetic moment (μ) $= IA$

$$I = \frac{Q}{t} = \frac{qv}{2\pi R}$$

$$A = \pi R^2$$

$$\mu = \frac{qv}{2\pi R} \times \pi R^2 = \frac{q v R}{2}$$

89. (a) Magnetic dipole moment.

$$\vec{M} = n i \vec{A}$$

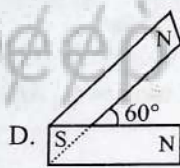
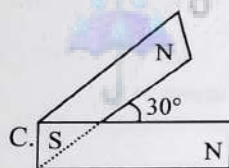
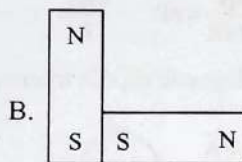
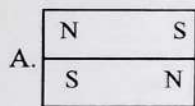


Where,

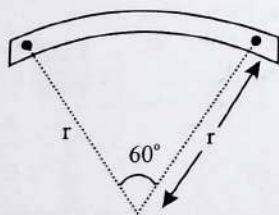
Number of turns $= n$, Area $= A$, Current $= i$

Magnetic Dipole Moment, and Force Between Magnetic Poles

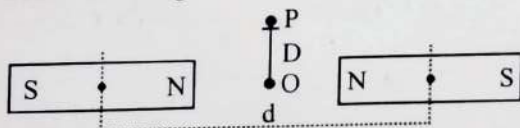
1. Following figures show the arrangement of bar magnets in different configurations. Each magnet has magnetic dipole moment m . Which configuration has highest net magnetic dipole moment? (2014)



- a. A
b. B
c. C
d. D
2. A bar magnet of length l and magnetic dipole moment M is bent in the form of an arc as shown in figure. The new magnetic dipole moment will be: (2013)



- a. $2M$
b. M
c. $3M/\pi$
d. $\pi/2 M$
3. Two identical bar magnets are fixed with their centers at a distance d apart. A stationary charge Q is placed at P in between the gap of the two magnets at a distance D from the center O as shown in the figure



The force on the charge Q is?

(2010 Mains)

- a. Zero
b. Directed along OP
c. Directed along PO
d. Directed perpendicular to the plane of paper
4. A bar magnet of magnetic moment M is cut into two parts of equal length. The magnetic moment of each part will be (1997)
- a. M
b. $2M$
c. Zero
d. $0.5 M$

Torque Acting on Axis of a Solenoid

5. A closely wound solenoid of 2000 turns and area of cross section $1.5 \times 10^{-4} \text{ m}^2$ carries a current of 2.0 A. It is suspended through its centre and perpendicular to its length, allowing it to turn in a horizontal plane in a uniform magnetic field $5 \times 10^{-2} \text{ tesla}$ making an angle of 30° with the axis of the solenoid. The torque on the solenoid will be (2010 Mains)
- a. $3 \times 10^{-3} \text{ Nm}$
b. $1.5 \times 10^{-3} \text{ Nm}$
c. $1.5 \times 10^{-2} \text{ Nm}$
d. $3 \times 10^{-2} \text{ Nm}$

Torque Acting on a Magnetic Dipole and Potential Energy of Dipole in a Magnetic Field

6. A 250 turn rectangular coil of length 2.1 cm and width 1.25 cm carries a current of $85 \mu\text{A}$ and subjected to a magnetic field of strength 0.85 T. Work done for rotating the coil by 180° against the torque is: (2017-Delhi)
- a. $4.55 \mu\text{J}$
b. $2.3 \mu\text{J}$
c. $1.15 \mu\text{J}$
d. $9.1 \mu\text{J}$

7. A bar magnet is hung by a thin cotton thread in a uniform horizontal magnetic field and is in equilibrium state. The energy required to rotate it by 60° is W . Now the torque required to keep the magnet in this new position is: (2016 - II)

- a. $\frac{\sqrt{3}W}{2}$ b. $\frac{2W}{\sqrt{3}}$
c. $\frac{W}{\sqrt{3}}$ d. $\sqrt{3}W$

8. A magnetic needle suspended parallel to a magnetic field requires $\sqrt{3} \text{ J}$ of work to turn it through 60° . The torque needed to maintain the needle in this position will be: (2012 Mains)

- a. $2\sqrt{3} \text{ J}$ b. $\sqrt{3} \text{ J}$
c. 3 J d. $\sqrt{3}/2 \text{ J}$

9. A short bar magnet of magnetic moment 0.4 J T^{-1} is placed in a uniform magnetic field of 0.16 T . The magnet is in stable equilibrium when the potential energy is: (2011 Mains)

- a. -0.064 J b. Zero
c. -0.082 J d. 0.064 J

Magnetic Elements of Earth

10. At a point A on the earth's surface the angle of dip, $\delta = +25^\circ$. At a point B on the earth's surface the angle of dip, $\delta = -25^\circ$. We can interpret that: (2019)

- a. A and B are both located in the northern hemisphere.
b. A is located in the southern hemisphere and B is located in the northern hemisphere.
c. A is located in the northern hemisphere and B is located in the southern hemisphere.
d. A and B are both located in the southern hemisphere.

11. If θ_1 and θ_2 be the apparent angles of dip observed in two vertical planes at right angles to each other, then the true angle of dip θ is given by: (2017-Delhi)

- a. $\tan^2 \theta = \tan^2 \theta_1 + \tan^2 \theta_2$ b. $\cot^2 \theta = \cot^2 \theta_1 - \cot^2 \theta_2$
c. $\tan^2 \theta = \tan^2 \theta_1 - \tan^2 \theta_2$ d. $\cot^2 \theta = \cot^2 \theta_1 + \cot^2 \theta_2$

12. A compass needle which is allowed to move in a horizontal plane is taken to a geomagnetic pole. It: (2012 Pre)

- a. Will become rigid showing no movement
b. Will stay in any position
c. Will stay in north-south direction only
d. Will stay in east-west direction only

13. Due to earth's magnetic field, the charged cosmic rays particles (1997)

- a. Can never reach the pole
b. Can never reach the equator
c. Require greater kinetic energy to reach the equator than pole
d. Require less kinetic energy to reach the equator than pole

Miscellaneous (Work Done by External Force in Magnetic Field and Vibration Magnetometer)

14. A vibration magnetometer placed in magnetic meridian has a small bar magnet. The magnet executes oscillations with a time period of 2 sec in earth's horizontal magnetic field of 24 microtesla. When a horizontal field of 18 microtesla is produced opposite to the earth's field by placing a current carrying wire, the new time period of magnet will be: (2010 Pre)

- a. 4 s b. 1 s
c. 2 s d. 3 s

15. A bar magnet having a magnetic moment of $2 \times 10^4 \text{ JT}^{-1}$ is free to rotate in a horizontal plane. A horizontal magnetic field $B = 6 \times 10^{-4} \text{ T}$ exists in the space. The work done in taking the magnet slowly from a direction parallel to the field to a direction 60° from the field is: (2009)

- a. 12 J b. 6 J
c. 2 J d. 0.6 J

16. A bar magnet is oscillating in the Earth's magnetic field with a period T . What happens to its period and motion if its mass is quadrupled? (2003)

- a. Motion remains S.H.M. with time period = $\frac{T}{2}$
b. Motion remains S.H.M with time period = $2T$
c. Motion remains S.H.M with time period = $4T$
d. Motion remains S.H.M and period remains nearly constant

17. Two bar magnets having same geometry with magnetic moments M and $2M$, are firstly placed in such a way that their similar poles are same side then its time period of oscillation is T_1 . Now the polarity of one of the magnet is reversed then time period of oscillation is T_2 , then: (2002)

- a. $T_1 < T_2$ b. $T_1 = T_2$
c. $T_1 > T_2$ d. $T_2 = \infty$

18. The work done in turning a magnet of magnetic moment M by an angle of 90° from the meridian, is n times the corresponding work done to turn it through an angle of 60° . The value of n is given by (1995)

- a. $1/2$ b. $1/4$
c. 2 d. 1

Magnetic Properties of Materials

19. An iron rod of susceptibility 599 is subjected to a magnetising field of 1200 A m^{-1} . The permeability of the material of the rod is: (2020)

$$(\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1})$$

- a. $8.0 \times 10^{-5} \text{ T m A}^{-1}$ b. $2.4\pi \times 10^{-5} \text{ T m A}^{-1}$
c. $2.4\pi \times 10^{-7} \text{ T m A}^{-1}$ d. $2.4\pi \times 10^{-4} \text{ T m A}^{-1}$

20. A thin diamagnetic rod is placed vertically between the poles of an electromagnet. When the current in the electromagnet is switched on, then the diamagnetic rod is pushed up, out of the horizontal magnetic field. Hence the rod gains gravitational potential energy. The work required to do this comes from: (2018)
- The lattice structure of the material of the rod
 - The magnetic field
 - The current source
 - The induced electric field due to the changing magnetic field
21. The magnetic susceptibility is negative for: (2016 - I)
- Diamagnetic material only
 - Paramagnetic material only
 - Ferromagnetic material only
 - Paramagnetic and ferromagnetic materials
22. There are four light weight rod samples A, B, C, D separately suspended by threads. A bar magnet is slowly brought near each sample and the following observations are noted:
- A is feebly repelled
 - B is feebly attracted
 - C is strongly attracted
 - D remains unaffected
- Which one of the following is true? (2011 Pre)
- B is of a paramagnetic material
 - C is of a diamagnetic material
 - D is of a ferromagnetic material
 - A is of a non magnetic material
23. The magnetic moment of a diamagnetic atom is: (2010 Mains)
- Much greater than one
 - 1
 - Between zero and one
 - Equal to zero
24. If a diamagnetic substance is brought near the north or the south pole of a bar magnet, it is: (2009)
- Repelled by the north pole and attracted by the south pole
 - Attracted by the north pole and repelled by the south pole
 - Attracted by both the poles
 - Repelled by both the poles
25. If the magnetic dipole moment of an atom of diamagnetic material, paramagnetic material and ferromagnetic material are denoted by μ_d , μ_p and μ_f respectively, then (2005)
- $\mu_d = 0$ and $\mu_p \neq 0$
 - $\mu_d \neq 0$ and $\mu_p = 0$
 - $\mu_p = 0$ and $\mu_f \neq 0$
 - $\mu_d \neq 0$ and $\mu_f \neq 0$
26. Diamagnetic material in a magnetic field moves: (2003)
- From stronger to the weaker parts of the field
 - From weaker to the stronger parts of the field
 - Perpendicular to the field
 - In none of the above directions
27. Among which the magnetic susceptibility does not depend on the temperature: (2001)
- Dia-magnetism
 - Para-magnetism
 - Ferro-magnetism
 - Ferrite
28. For protecting a magnetic needle it should be placed: (1998)
- In iron box
 - In wooden box
 - In metallic box
 - None of these

Curie Law and Hysteresis Curve

29. Electromagnets are made of soft iron because soft iron has: (2010 Pre)
- High retentivity and low coercive force
 - Low retentivity and high coercive force
 - High retentivity and high coercive force
 - Low retentivity and low coercive force
30. Curie temperature is the temperature above which: (2008)
- Ferromagnetic material becomes diamagnetic material
 - Ferromagnetic material becomes paramagnetic material
 - Paramagnetic material becomes diamagnetic material
 - Paramagnetic material becomes ferromagnetic
31. Nickel shows ferromagnetic property at room temperature. If the temperature is increased beyond Curie temperature, then it will show: (2007)
- Anti ferro-magnetism
 - No magnetic property
 - Diamagnetism
 - Paramagnetism
32. Above Curie temperature: (2006)
- A ferromagnetic substance becomes paramagnetic
 - A paramagnetic substance becomes diamagnetic
 - A diamagnetic substance becomes paramagnetic
 - A paramagnetic substance becomes ferromagnetic
33. According to Curie's law, the magnetic susceptibility of a substance at an absolute temperature T is proportional to: (2003)
- $1/T$
 - T
 - T^3
 - T^2

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
c	c	a	d	c	d	d	c	a	c	d	b	c	a	b	b	a
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	
c	d	c	a	a	d	d	a	a	a	a	d	b	d	a	a	

Explanations

1. (c) Net magnetic moment

$$m = \sqrt{m_1^2 + m_2^2 + 2m_1m_2 \cos \theta}$$

$m_1 = m_2 = m$ will be maximum if $\cos \theta$ is maximum. $\cos \theta$ will be maximum when θ will be minimum.

$$\therefore \theta = 30^\circ$$

2. (c) Let magnetic pole strength be m then in new situation

$$M' = (m) \left(2r \sin \frac{60^\circ}{2} \right) \text{ where } r = \left(\frac{\pi}{3} \right) \ell$$

$$M' = 2m \left(\frac{2\ell}{\pi} \right) \left(\frac{1}{2} \right) = \frac{3m\ell}{\pi} = \frac{3M}{\pi}$$

3. (a) Magnetic field due to bar magnets exert force on moving charges only, but q is at rest hence net $F = 0$

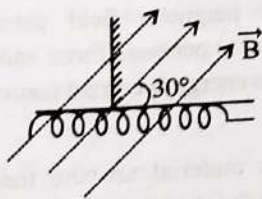
4. (d) Magnetic moment = Pole strength $\times$ length

$$\therefore M \propto l \quad (\text{for constant pole strength})$$

$\therefore$ As we cut magnet into two equal pieces then length becomes half so new Magnetic Moment, $M' = M/2 = 0.5 M$

5. (c) Magnetic moment of the loop,

$$M = NIA = 2000 \times 2 \times 1.5 \times 10^{-4} = 0.6 \text{ J/T}$$



$$\text{Torque, } \tau = MB \sin 30^\circ$$

$$= 0.6 \times 5 \times 10^{-2} \times \frac{1}{2} = 1.5 \times 10^{-2} \text{ Nm}$$

6. (d) Work = $MB [\cos \theta_1 - \cos \theta_2]$

$$= MB [\cos 0^\circ - \cos 180^\circ]$$

$$= NiAB [1 - (-1)]$$

$$= 2NiAB$$

$$= 2 \times 250 \times 85 \times 10^{-6} \times 2.1 \times 10^{-2} \times 1.25 \times 10^{-2} \times 0.85$$

$$= 9.12 \mu\text{J}$$

7. (d) We know that

$$\tau = M \times B \Rightarrow \tau = MB \sin \theta$$

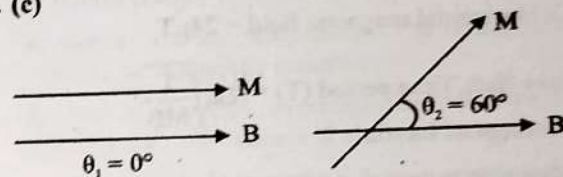
$$\tau = MB \sin 60^\circ$$

$$W = MB(1 - \cos 60^\circ)$$

From Eqs. (1) and (2)

$$\frac{\tau}{W} = \frac{\sqrt{3}/2}{1/2} \Rightarrow \tau = W\sqrt{3}$$

8. (c)



$$W = MB(\cos \theta_1 - \cos \theta_2) = MB(\cos 0^\circ - \cos 60^\circ)$$

$$= MB \left(1 - \frac{1}{2} \right) = \frac{MB}{2}$$

$$MB = 2\sqrt{3} \text{ J}$$

$$\tau = MB \sin 60^\circ = (2\sqrt{3}) \left(\frac{\sqrt{3}}{2} \right) \text{ J} = 3 \text{ J}$$

9. (a) Here, $M = 0.4 \text{ JT}^{-1}$

Magnetic field, $B = 0.16 \text{ T}$

When bar magnet is placed in uniform external magnetic field $\vec{B}$, its potential energy

$$U = -\vec{M} \cdot \vec{B} = -MB \cos \theta$$

For stable equilibrium, $\theta = 0^\circ$

$$\therefore U = -MB = (-0.4) 0.16 = -0.064 \text{ J}$$

10. (c) The dip is called positive if the north pole of the magnetic needle faces downward, negative if it faces upward. The magnetic dip is equal to 0° at the equator and increase positively in the northern hemisphere, negatively in the southern hemisphere.

11. (d) The first apparent angle of dip = θ_1

The second apparent angle of dip = θ_2

The angle between the vertical planes = 90°

Let, V = Vertical component

H = Horizontal component

$$\text{then, Angle of dip} = \tan \theta = \frac{V}{H}$$

...(1)

...(2)

$$\therefore \tan \theta_1 = \frac{V}{H_1}, \quad \tan \theta_2 = \frac{V}{H_2}$$

H_1 and H_2 are horizontal planes at 90° .

$$\text{So, } H^2 = H_1^2 + H_2^2$$

$$\therefore \left(\frac{V}{\tan \theta} \right)^2 = \left(\frac{V}{\tan \theta_1} \right)^2 + \left(\frac{V}{\tan \theta_2} \right)^2$$

$$(V \cot \theta)^2 = (V \cot \theta_1)^2 + (V \cot \theta_2)^2$$

$$\therefore \cot^2 \theta = \cot^2 \theta_1 + \cot^2 \theta_2$$

12. (b) As the magnetic field is in the vertical direction and the magnetic needle rotates in the horizontal plane, therefore it will come to rest in any direction.

13. (c) Due to earth's magnetic field, the charged cosmic rays particles, require greater kinetic energy to reach the equator than pole.

14. (a) Time period of magnet's oscillation = 2 sec.

Earth's Horizontal magnetic field = $24 \mu\text{T}$

We know that, Time period (T) = $2\pi \sqrt{\frac{I}{MB}}$

I : Moment of Inertia

M : Magnetic moment

B : Magnetic field

$$\text{Now, } T \propto \frac{1}{\sqrt{B}}$$

$$\therefore \frac{T_1}{T_2} = \sqrt{\frac{B_2}{B_1}}$$

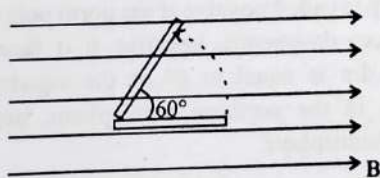
$$\therefore B_1 = 24 \mu\text{T}$$

$$B_2 = 24 - 18 = 6 \mu\text{T}$$

$$\therefore T_2 = T_1 \sqrt{\frac{B_1}{B_2}} = 2 \sqrt{\frac{24}{6}}$$

$$T_2 = 4 \text{ sec}$$

15. (b)



Magnetic moment (M) = $2 \times 10^4 \text{ J/T}$

Magnetic field (B) = $6 \times 10^{-4} \text{ T}$

$$W = MB(\cos \theta_1 - \cos \theta_2) \quad \begin{matrix} \theta_1 = 0^\circ \\ \theta_2 = 60^\circ \end{matrix}$$

$$= 2 \times 10^4 \times 6 \times 10^{-4} \left(1 - \frac{1}{2} \right)$$

$$= 12 \times 0.5 = 6 \text{ J}$$

16. (b) Use $T = 2\pi \sqrt{\frac{I}{MB}}$

$$\therefore I \propto \text{mass} \Rightarrow T' = 2\pi \sqrt{\frac{4I}{MB}} = 2T$$

17. (a) $T = 2\pi \sqrt{\frac{I}{MB}} \Rightarrow T \propto \frac{1}{\sqrt{M}}$

case I : $M_1 = 2M + M = 3M$

case II : $M_2 = 2M - M = M$

$$\frac{T_1}{T_2} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{M}{3M}} = \frac{1}{\sqrt{3}} \Rightarrow T_2 = \sqrt{3} T_1$$

$$\therefore T_1 < T_2$$

18. (c) Work done in turning the magnet through 60° .

$$(W_1) = MB(\cos 0^\circ - \cos 60^\circ) = MB \left(1 - \frac{1}{2} \right) \Rightarrow \frac{MB}{2}$$

Work done in turning the magnet through 90° .

$$W_2 = MB(\cos 0^\circ - \cos 90^\circ) \\ = MB$$

According to the question

$$W_2 = nW_1$$

$$\therefore \text{Here } \frac{W_2}{W_1} = \frac{MB}{\frac{MB}{2}} = 2$$

$$W_2 = 2W_1$$

Comparing (i) and (ii)

$$n = 2$$

19. (d) $\mu_r = X_m + 1 = 599 + 1 = 600$

$$\mu = \mu_0 \mu_r = 4\pi \times 10^{-7} \times 600$$

$$= 2.4\pi \times 10^{-4} \frac{\text{T.m}}{\text{A}}$$

20. (c) Due to the magnetic field generated by current in electromagnet rod experience force and gain potential energy and source of this energy is current source.

21. (a)

- Diamagnetic material are the magnetic materials that align against the magnetic field.
- The magnetic susceptibility of diamagnetic material is less than zero. This means that a diamagnetic material always has a negative value of magnetic susceptibility.
- Paramagnetic material are the magnetic materials that align with the magnetic field. The magnetic susceptibility of paramagnetic material is always greater than zero.
- Ferromagnetic material are the highly magnetized materials. Ferromagnetic material do not have constant permeability and they differ from 1000 to 100000.
- Both ferromagnetic and paramagnetic material have positive magnetic susceptibility. Diamagnetic material always has a negative magnetic susceptibility.

22. (a) Diamagnetic materials are feebly repelled by bar magnet
paramagnetic materials are feebly attracted by bar magnet
ferromagnetic materials are strongly attracted by bar magnet
So,

A - Diamagnetic B - Paramagnetic
C - Ferromagnetic D - Non magnetic

23. (d) Diamagnetic substances are those substances that get weakly magnetized in the presence of an external magnetic field, in a direction opposite to that of the external magnetic field.

- Each atom of the material has an orbital and spin magnetic moment caused by the orbital motion and spin of electrons respectively. The vector sum of the spin and orbital moments gives the net magnetic field of the atom.
- In diamagnetic substances, the net magnetic moment of an atom is zero because of all the electrons are paired so there is no permanent net magnetic moment per atom.

24. (d) The diamagnetic substance is weakly magnetised in a direction opposite to that of applied magnetic field. So, in an external magnetic field it has a tendency to move from high to low field region, irrespective to the poles.

25. (a) Material with no unpaired, or isolated electrons are considered diamagnetic. Diamagnetic substances do not have magnetic dipole moments and have negative susceptibilities. However, materials having unpaired electrons whose spins do not cancel each other are called paramagnetic. These substances have positive magnetic moments and susceptibilities.

$$\mu_d = 0, \mu_p \neq 0.$$

26. (a) When diamagnetic materials are placed in the external magnetic field they get magnetized in the direction opposite to that of the external magnetic field. Since, the diamagnetic material is magnetized in the opposite direction of the magnetic field so they will try to reduce the magnetic field in the direction of the magnetic field hence, they start to move from stronger to weaker part of the magnetic field.

27. (a) The susceptibility of diamagnetic substances is always negative. It is constant and does not vary with field or the temperature.

28. (a) An iron box serves as a magnetic shield and should be used to keep magnetic needles safe.

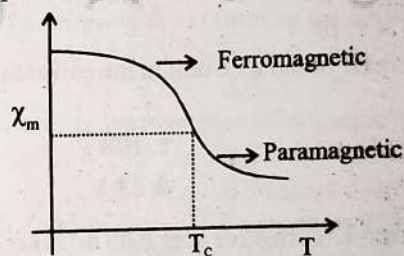
29. (d)

- An electromagnet is a soft metal core made into a magnet by allowing electric current to pass through a coil surrounding it. The magnetic behavior of an electromagnet depends on the time of flow of current through the coil surrounding it.
- After the removal of the externally applied field, the magnetic field of an electromagnet vanishes.
- Retentivity is the ability to retain a magnetic field after the removal of the externally applied magnetic field.
- Coercivity is the amount of external magnetic field to be applied to remove the persisting magnetic field in an object after the removal of the external field.
- If an object has retentivity and retentivity that means magnetic field persist inside the material and another external field is to be applied in order to remove the persisting field which violates the basic definition of an electromagnet
- Hence the basic requirement of an object to become an electromagnet is low retentivity and low coercive force or coercivity.

30. (b) Above Curie temperature, ferromagnetic material becomes paramagnetic material.

31. (d) If the temperature is increased beyond curie temperature, ferromagnetic substance will show paramagnetism.

32. (a) Above Curie Temperature, ferromagnetic substance becomes paramagnetic



Ferromagnetic

$$\chi_m \propto \frac{1}{(T - T_c)}$$

$$\chi_m \propto \frac{1}{T}$$

33. (a) Curie law $\chi_m \propto \frac{1}{T}$

Electromagnetic Induction

Magnetic Flux, Faraday's Law of E.M.F.

1. A square loop of side 1 m and resistance 1Ω is placed in a magnetic field of 0.5 T . If the plane of loop is perpendicular to the direction of a magnetic field, the magnetic flux through the loop is: (2022)

a. zero weber
b. 2 weber
c. 0.5 weber
d. 1 weber

2. The magnetic flux linked with a coil (in Wb) is given by the equation

$$\phi = 5t^2 + 3t + 16$$

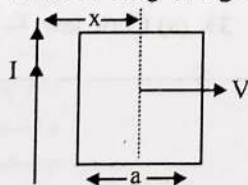
The magnitude of induced emf in the coil at the fourth second will be (2020-Covid)

a. 43 V
b. 108 V
c. 10 V
d. 33 V

3. A 800 turn coil of effective area 0.05 m^2 is kept perpendicular to a magnetic field of $5 \times 10^{-5} \text{ T}$. When the plane of the coil is rotated by 90° around any of its coplanar axis in 0.1 s , the emf induced in the coil will be: (2019)

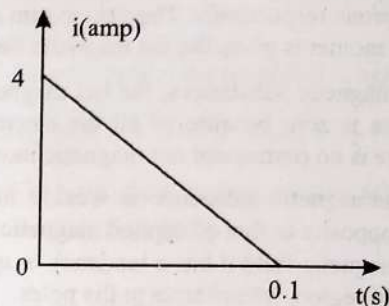
a. 2 V
b. 0.2 V
c. $2 \times 10^{-3} \text{ V}$
d. 0.02 V

4. A conducting square frame of side a and a long straight wire carrying current I are located in the same plane as shown in the figure. The frame moves to the right with a constant velocity V . The emf induced in the frame will be proportional to: (2015)



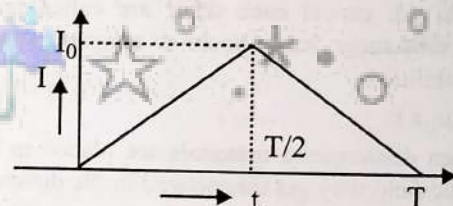
a. $\frac{1}{(2x-a)^2}$
b. $\frac{1}{(2x-a)(2x+a)}$
c. $\frac{1}{(2x+2a)(2x+a)}$
d. $\frac{1}{x^2}$

5. In a coil of resistance 10Ω , the induced current developed by changing magnetic flux through it, is shown in figure as a function of time. The magnitude of change in flux through the coil in Weber is: (2012 Mains)

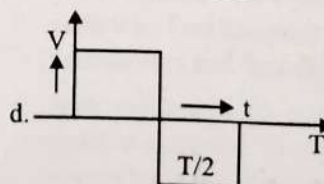
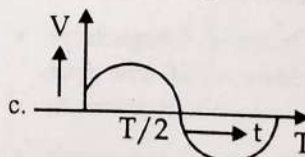
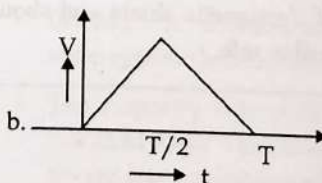
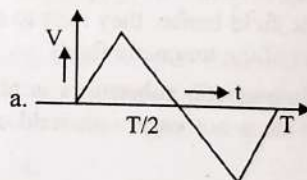


a. 8
b. 2
c. 6
d. 4

6. The current (I) in the inductance is varying with time according to the plot shown in figure.

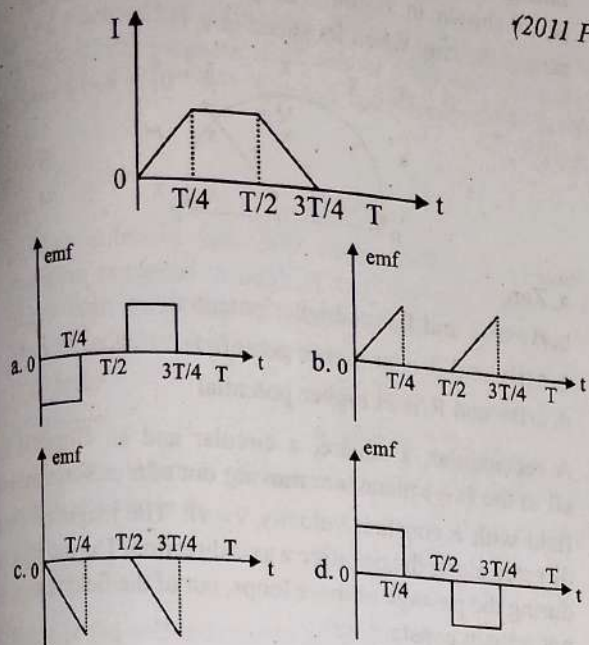


Which one of the following is the correct variation of voltage with time in the coil? (2012 Pre)



7. The current i in a coil varies with time as shown in the figure. The variation of induced emf with time would be:

(2011 Pre)



8. A conducting circular loop is placed in a uniform magnetic field, $B = 0.025 \text{ T}$ with its plane perpendicular to the loop. The radius of the loop is made to shrink at a constant rate of 1 mm s^{-1} . The induced emf when the radius is 2 cm is:

(2010 Pre)

- a. $2 \mu\text{V}$ b. $2\pi \mu\text{V}$
c. $\pi \mu\text{V}$ d. $\pi/2 \mu\text{V}$

9. A conducting circular loop is placed in a uniform magnetic field 0.04 T with its plane perpendicular to the magnetic field. The radius of the loop starts shrinking at 2 mm/s . The induced emf in the loop when the radius is 2 cm is: (2009)

- a. $4.8\pi \mu\text{V}$ b. $0.8\pi \mu\text{V}$
c. $1.6\pi \mu\text{V}$ d. $3.2\pi \mu\text{V}$

10. A circular disc of radius 0.2 meter is placed in a uniform magnetic field of induction $\frac{1}{\pi} \left(\frac{\omega b}{\text{m}^2} \right)$ in such a way that its axis makes an angle of 60° with the magnetic flux linked with the disc is: (2008)

- a. 0.01 Wb b. 0.02 Wb
c. 0.06 Wb d. 0.08 Wb

11. As a result of change in the magnetic flux linked to the closed loop shown in the figure, an emf V volt is induced in the loop. The work done (joules) in taking a charge Q coulomb once along the loop is: (2005)



- a. QV b. Zero
c. $2QV$ d. $QV/2$

12. The magnetic flux through a circuit of resistance R changes by an amount $\Delta\phi$ in a time Δt . Then the total quantity of electric charge Q that passes any point in the circuit during the time Δt is represented by: (2004)

- a. $Q = \frac{\Delta\phi}{R}$ b. $Q = \frac{\Delta\phi}{\Delta t}$
c. $Q = R \cdot \frac{\Delta\phi}{\Delta t}$ d. $Q = \frac{1}{2} \cdot \frac{\Delta\phi}{\Delta t}$

13. For a coil having $L = 2 \text{ mH}$, current flow through it is $I = t^2 e^{-t}$ then the time at which emf become zero: (2001)

- a. 2 s b. 1 s
c. 4 s d. 3 s

14. Initially plane of coil is parallel to the uniform magnetic field B . In time Δt it makes to perpendicular to the magnetic field, then charge flows in Δt depends on this time as: (1999)

- a. $\propto \Delta t$ b. $\propto \frac{1}{\Delta t}$
c. $\propto (\Delta t)^0$ d. $\propto (\Delta t)^2$

15. The total charge, induced in a conducting loop when it is moved in magnetic field depend on (1992)

- a. The rate of change of magnetic flux
b. Initial magnetic flux only
c. The total change in magnetic flux
d. Final magnetic flux only

16. A rectangular coil of 20 turns and area of cross-section 25 sq.cm has a resistance of 100Ω . If a magnetic field which is perpendicular to the plane of coil changes at a rate of $1000 \text{ tesla per second}$, the current in the coil is: (1992)

- a. 1 A b. 50 A
c. 0.5 A d. 5 A

17. Faraday's laws are consequence of conservation of (1991)

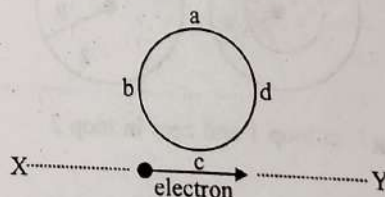
- a. Energy b. Energy and magnetic field
c. Charge d. Magnetic field

18. A magnetic field of $2 \times 10^{-2} \text{ T}$ acts at right angles to a coil of area 100 cm^2 , with 50 turns. The average e.m.f. induced in the coil is 0.1 V , when it is removed from the field in $t \text{ sec}$. The value of t is (1991)

- a. 10 s b. 0.1 s
c. 0.01 s d. 1 s

Lenz's Law

19. An electron moves on a straight line path XY as shown. The $abcd$ is a coil adjacent to the path of electron. What will be the direction of current, if any, induced in the coil? (2015 Pre)



- a. No current induced
b. $abcd$
c. $adcb$
d. The current will reverse its direction as the electron goes past the coil

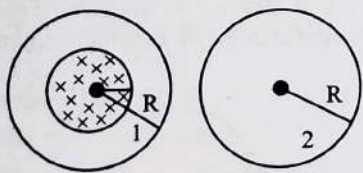
20. A wire loop is rotated in a magnetic field. The frequency of change of direction of the induced e.m.f. is: (2013)
 a. Six times per revolution b. Once per revolution
 c. Twice per revolution d. Four times per revolution
21. A metal ring is held horizontally and bar magnet is dropped through the ring with its length along the axis of the ring. The acceleration of the falling magnet is (1996)
 a. More than g b. Equal to g
 c. Less than g d. Either a or c

Motional E.M.F.

22. A big circular coil of 100 turns and average radius 10 m is rotating about its horizontal diameter at 2 rad s^{-1} . If the vertical component of earth's magnetic field at that place is $2 \times 10^{-5} \text{ T}$ and electrical resistance of the coil is 12.56Ω , then the maximum induced current in the coil will be: (2022)
 a. 2A b. 0.25 A
 c. 1.5 A d. 1 A
23. A wheel with 20 metallic spokes each 1 m long is rotated with a speed of 120 rpm in a plane perpendicular to a magnetic field of 0.4 G. The induced emf between the axle and rim of the wheel will be. ($1 \text{ G} = 10^{-4} \text{ T}$) (2020-Covid)
 a. $2.51 \times 10^{-5} \text{ V}$ b. $4.0 \times 10^{-5} \text{ V}$
 c. 2.51 V d. $2.51 \times 10^{-4} \text{ V}$
24. A metallic rod of mass per unit length 0.5 kg m^{-1} is lying horizontally on a smooth inclined plane which makes an angle of 30° with the horizontal. The rod is not allowed to slide down by flowing a current through it when a magnetic field of induction 0.25 T is acting on it in the vertical direction. The current flowing in the rod to keep it stationary is (2018)
 a. 14.76 A b. 5.98 A
 c. 7.14 A d. 11.32 A

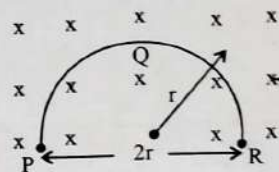
25. A uniform magnetic field is restricted within a region of radius r . The magnetic field changes with time at a rate $\frac{d\vec{B}}{dt}$.

Loop 1 of radius $R > r$ encloses the region r and loop 2 of radius R is outside the region of magnetic field as shown in the figure below. Then the e.m.f generated is: (2016 - II)



- a. $-\frac{d\vec{B}}{dt} \pi R^2$ in loop 1 and zero in loop 2
 b. $-\frac{d\vec{B}}{dt} \pi r^2$ in loop 1 and zero in loop 2
 c. Zero in loop 1 and zero in loop 2
 d. $-\frac{d\vec{B}}{dt} \pi r^2$ in loop 1 and $-\frac{d\vec{B}}{dt} \pi R^2$ in loop 2

26. A thin semicircular conducting ring (PQR) of radius r is falling with its plane vertical in a horizontal magnetic field B , as shown in figure. The potential difference developed across the ring when its speed is v , is: (2014)



- a. Zero
 b. $Bv\pi r^2/2$ and P is at higher potential
 c. $\pi r Bv$ and R is at higher potential
 d. $2r Bv$ and R is at higher potential
27. A rectangular, a square, a circular and an elliptical loop, all in the (x-y) plane, are moving out of a uniform magnetic field with a constant velocity, $\vec{v} = v\hat{i}$. The magnetic field is directed along the negative z axis direction. The induced emf, during the passage of these loops, out of the field region, will not remain constant for: (2009)
 a. The circular and the elliptical loops
 b. Only the elliptical loop
 c. Any of the four loops
 d. The rectangular, circular and elliptical loops
28. A straight line conductor of length 0.4 m is moved with a speed of 7 m/s perpendicular to a magnetic field of intensity 0.9 Wb/m^2 . The induced e.m.f. across the conductor is (1995)
 a. 5.04 V b. 25.2 V
 c. 1.26 V d. 2.52 V
29. In a region of magnetic induction $B = 10^{-2} \text{ tesla}$, a circular coil of radius 30 cm and resistance $\pi^2 \text{ ohm}$ is rotated about an axis which is perpendicular to the direction of B and which forms a diameter of the coil. If the coil rotates at 200 rpm the amplitude of the alternating current induced in the coil is (1988)
 a. $4 \pi^2 \text{ mA}$ b. 30 mA
 c. 6 mA d. 200 mA

Eddy Currents

30. In which of the following devices, the eddy current effect is not used? (2019)
 a. Induction furnace
 b. Magnetic braking in train
 c. Electromagnet
 d. Electric heater
31. Eddy currents, are produced when (1988)
 a. A metal is kept in varying magnetic field
 b. A metal is kept in steady magnetic field
 c. A circular coil is placed in a magnetic field
 d. Through a circular coil, current is passed

Self Induction

32. A long solenoid has 1000 turns. When a current of 4 A flows through it, the magnetic flux linked with each turn of the solenoid is 4×10^{-3} Wb. The self inductance of the solenoid is: (2016 - I)
- a. 4 H b. 3 H
c. 2 H d. 1 H
33. A long solenoid has 500 turns. When a current of 2 ampere is passed through it, the resulting magnetic flux linked with each turn of the solenoid is 4×10^{-3} Tm². The self inductance of the solenoid is: (2008)
- a. 4.0 henry b. 2.5 henry
c. 2.0 henry d. 1.0 henry
34. If N is the number of turns in a coil, the value of self inductance varies as (1993)
- a. N⁰ b. N
c. N² d. N⁻²
35. What is the self-inductance of a coil which produces 5 V when the current changes from 3 ampere to 2 ampere in one millisecond? (1993)
- a. 5000 henry b. 5 mili-henry
c. 50 henry d. 5 henry
36. If the number of turns per unit length of a coil of solenoid is doubled, the self-inductance of the solenoid will: (1991)
- a. Remain unchanged b. Be halved
c. Be doubled d. Become four times
37. The current in self inductance $L = 40$ mH is to be increased uniformly from 1 amp to 11 amp in 4 milliseconds. The e.m.f. induced in inductor during process is (1990)
- a. 100 volt b. 0.4 volt
c. 4.0 volt d. 440 vol

Energy Stored or Work Done in Inductor

38. The magnetic potential energy stored in a certain inductor is 25 mJ, when the current in the inductor is 60 mA. This inductor is of inductance: (2018)
- a. 1.389 H b. 138.88 H
c. 0.138 H d. 13.89 H

39. For a inductor coil $L = 0.04$ H, then work done by source to establish a current of 5 A in it is: (1999)
- a. 0.5 J b. 1.00 J
c. 100 J d. 20 J
40. A 100 millihenry coil carries a current of 1 A. Energy stored in its magnetic field is (1991)
- a. 0.5 J b. 1 A
c. 0.05 J d. 0.1 J
41. An inductor may store energy in (1990)
- a. Its electric field
b. Its coils
c. Its magnetic field
d. Both in electric and magnetic fields

Mutual Induction

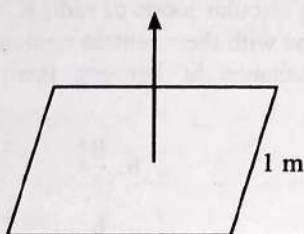
42. Two conducting circular loops of radii R_1 and R_2 are placed in the same plane with their centres coinciding. If $R_1 \gg R_2$, the mutual inductance M between them will be directly proportional to: (2021)
- a. $\frac{R_2}{R_1}$ b. $\frac{R_1^2}{R_2}$
c. $\frac{R_2^2}{R_1}$ d. $\frac{R_1}{R_2}$
43. A long solenoid of diameter 0.1 m has 2×10^4 turns per metre. At the centre of solenoid, a coil of 100 turns and radius 0.01 m is placed with its axis coinciding with the solenoid axis. The current in the solenoid reduces at a constant rate to 0 A from 4 A in 0.05 s. If the resistance of the coil is $10\pi^2\Omega$, the total charge flowing through the coil during this time is: (2017-Delhi)
- a. 16 μ C b. 32 μ C
c. 16 π μ C d. 32 π μ C
44. Two coils of self-inductances 2 mH and 8 mH are placed so close together that the effective flux in one coil is completely linked with the other. The mutual inductance between these coils is: (2006)
- a. 8 mH b. 12 mH
c. 4 mH d. 16 mH
45. Two coil have a mutual inductance 0.005 H. The current changes in first coil according to equation $I = I_0 \sin \omega t$ where $I_0 = 2$ A and $\omega = 100\pi$ rad/sec. The maximum value of emf in second coil is: (1998)
- a. 4 π b. 3 π
c. 2 π d. π

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
c	a	d	b	b	d	a	c	d	b	a	a	a	c	c	c	a
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
b	d	c	c	d	d	d	b	d	a	d	c	d	a	d	d	c
35	36	37	38	39	40	41	42	43	44	45						
b	d	a	d	b	c	c	c	b	c	d						

Explanations

1. (c) $B = 0.5 \text{ T}$



As the angle between the magnetic field and normal to the plane of the coil is zero

$$\Rightarrow \theta = 0^\circ$$

$$\begin{aligned} \therefore \phi &= B.A. \cos \theta \\ &= 0.5 \times (1) \times 1 \\ \phi &= 0.5 \text{ Wb} \end{aligned}$$

2. (a) Magnitude of induced emf $|\varepsilon| = \left| \frac{d\phi}{dt} \right| = \frac{d}{dt}(5t^2 + 3t + 16)$

$$= 10t + 3$$

$$\left(\frac{d\phi}{dt} \right)_{t=4} = 40 + 3 = 43 \text{ volt}$$

3. (d) Number of turns in coil $N = 800$

$$\text{Area of coil } A = 0.05 \text{ m}^2$$

$$\text{Magnetic field } B = 5 \times 10^{-5} \text{ T}$$

$$\text{Time taken to rotate } \Delta t = 0.1 \text{ s}$$

$$\text{Initial angle } \theta_1 = 0^\circ$$

$$\text{Final angle } \theta_2 = 90^\circ$$

$$\text{Change in magnetic flux } \Delta \phi$$

$$= N(BA \cos 90^\circ - BA \cos 0^\circ)$$

$$= -NBA$$

$$= -800 \times 5 \times 10^{-5} \times 0.05$$

$$= -2 \times 10^{-3} \text{ weber}$$

$$\text{So, } \varepsilon = -\frac{\Delta \phi}{\Delta t} = \frac{(-)2 \times 10^{-3} \text{ Wb}}{0.1 \text{ s}} = 0.02 \text{ V}$$

4. (b) Emf Induced in side (1)

$$\varepsilon_1 = B_1 V \ell$$

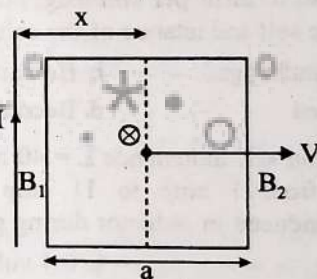
$$\text{Emf Induced in side (2)}$$

$$\varepsilon_2 = B_2 V \ell$$

$$\text{Emf in the frame} = B_1 V \ell - B_2 V \ell$$

$$\varepsilon = V \ell [B_1 - B_2]$$

$$\Rightarrow \varepsilon \propto B_1 - B_2 \quad \text{Since } B \propto \frac{1}{r}$$



$$\text{So, } \varepsilon \propto \left[\frac{1}{x - \frac{a}{2}} - \frac{1}{x + \frac{a}{2}} \right]$$

$$\Rightarrow \varepsilon \propto \left[\frac{1}{(2x - a)} - \frac{1}{(2x + a)} \right] \Rightarrow \varepsilon \propto \frac{1}{(2x - a)(2x + a)}$$

5. (b) We know that, $q = \frac{\Delta \phi}{R} \Rightarrow \Delta \phi = qR$

$$\Delta \phi = (\text{area under the curve}) R$$

$$\Delta \phi = \frac{4 \times 0.1}{2} \times 10 = 2 \text{ Weber}$$

6. (d) Voltage across the inductor,

$$V = L \frac{di}{dt}$$

$$\text{For } t = 0 \text{ to } t = T/2$$

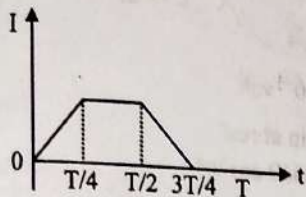
$$V = L \frac{d}{dt} \left(\frac{2I_0 t}{T} \right) = \text{constant (+ve)}$$

for $t = T/2$ to $t = T$

$$V = L \frac{d}{dt} \left(\frac{-2I_0 t}{T} \right) = \text{constant} (-ve)$$

Hence graph of option (d) is correct.

7. (a) Induced emf, $\varepsilon = -L \frac{di}{dt}$



ε will depend on $\frac{di}{dt}$

If slope of graph $\left(\frac{di}{dt} \right)$ is +ve, ε will be negative

For $0 \leq t \leq \frac{T}{4}$;

$\frac{di}{dt} = \text{Constant and +ve}$, So $\varepsilon = -ve$ and constant

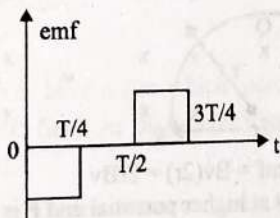
For $\frac{T}{4} \leq t \leq \frac{T}{2}$;

$\frac{di}{dt} = 0$, So $\varepsilon = 0$

For $\frac{T}{2} \leq t \leq \frac{3T}{4}$

$\frac{di}{dt} = \text{Constant and -ve}$, So, $\varepsilon = +ve$ and constant

Hence, graph



8. (c) We know that,

$$\text{Flux } \phi = B.A \cos \theta$$

$$\because \theta = 90^\circ$$

$$\therefore \phi = B.A$$

$$\phi = B.(\pi r^2)$$

$$\varepsilon = \frac{d\phi}{dt} = (B)(2\pi r) \left(\frac{dr}{dt} \right)$$

$$= (0.025)(2\pi)(2 \times 10^{-2})(10^{-3}) = \pi \mu V$$

$$9. (d) \text{ Induced emf } (\varepsilon) = \frac{d\phi}{dt} = \frac{d(BS \cos \theta)}{dt} = \frac{d[B\pi r^2]}{dt}$$

$$= 2\pi r B \frac{dr}{dt}$$

$$= 2\pi (2 \times 10^{-2}) \times 0.04 \times 2 \times 10^{-3}$$

$$= 3.2\pi \mu V$$

10. (b) Magnetic flux (ϕ) linked with the disc = $BA \cos \theta$

where B = Magnetic field

A = Area of disc

θ = Angle between magnetic field and area vector

$$\therefore \phi = \frac{1}{\pi} \times \pi \times r^2 \times \cos 60^\circ$$

$$\phi = (0.2)^2 \times \frac{1}{2}$$

$$\phi = 0.02 \text{ Wb}$$

11. (a) Work done will be $(Q \times V)$ in taking a charge Q coulomb once along the loop which has induced emf (V).

12. (a) $\because$ we know that $\text{emf} = \frac{\Delta \phi}{\Delta t}$

$$IR = \frac{\Delta \phi}{\Delta t} \quad \{ \because \text{emf} = IR \}$$

$$I dt R = \Delta \phi$$

$$Q.R = \Delta \phi \quad \{ \because I dt = Q \}$$

$$\therefore Q = \frac{\Delta \phi}{R}$$

13. (a) Given, $I = e^{-t} t^2$... (1)

when $\frac{dI}{dt} = 0$ then $\text{emf} = 0$

Differentiating (1) w.r.t.

$$\frac{dI}{dt} = 2te^{-t} + (-1)e^{-t}t^2$$

$$\frac{dI}{dt} = 2te^{-t} - t^2e^{-t}$$

$$0 = (2t - t^2) e^{-t}$$

$$t[2 - t] = 0 \Rightarrow t = 0, t = 2 \text{ sec}$$

14. (c) We know that, $\because q = \frac{\Delta \phi}{R}$

Hence charge is independent of time.

$$\therefore q \propto (\Delta t)^0$$

$$15. (c) q = \int idt = \frac{1}{R} \int \varepsilon dt = \left(\frac{-d\phi}{dt} \right) \frac{1}{R} \int dt = \frac{1}{R} \int d\phi$$

As the emf or iR depends on rate of change of ϕ , charge induced depends on change of flux.

Hence total charge induced in the conducting loop depend upon the total change in magnetic flux.

16. (c) We know that, Current $i = \frac{\varepsilon}{R}$

where, R : Resistance

$$\varepsilon : \text{Induced emf} = \frac{d\phi}{dt} = \frac{NAB}{dt}$$

$$\therefore i = \frac{\varepsilon}{R} = \frac{\frac{d\phi}{dt}}{R} = \frac{NAdB}{R dt}$$

$$= \frac{20 \times (25 \times 10^{-4}) \times 1000}{100} = 0.5 \text{ A}$$

17. (a) Faraday's laws are based on conservation of energy. This law states that total energy of an isolated object can never be lost or gained but can be transfer.

18. (b) By using, $\varepsilon = \frac{-(\phi_2 - \phi_1)}{t} = \frac{-(0 - NBA)}{t} = \frac{NBA}{t}$

$$\therefore t = \frac{NBA}{\varepsilon} = \frac{50 \times 2 \times 10^{-2} \times 10^{-2}}{0.1} = 0.1 \text{ s}$$

19. (d) First current develops in direction of abcd but when electron moves away, then magnetic field inside loop decreases and current will change its direction.

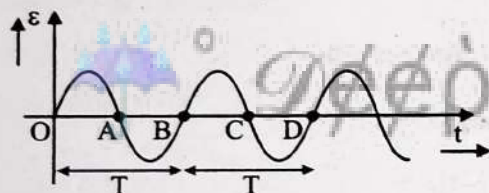
20. (c) We know that,

$$\text{Flux } \phi = BA \cos \theta = BA \cos(\omega t)$$

where, B: magnetic field, A: Area of loop

$\theta = \omega t$ (Angle made by area vector with magnetic field)

$$\text{So, induced emf } \varepsilon = -\frac{d\phi}{dt} = BA\omega \sin(\omega t)$$



From graph, we have four location (A, B, C and D) of change of direction in emf induced, in two time periods (T each). So there are two changes in one time period.

21. (c) When the magnet is dropped through the ring an induced current is developed into the ring in the direction opposing the motion of magnet (Lenz's law). Therefore this induced current decreases the acceleration of bar magnet.

Hence, acceleration of the falling magnet is less than g.

22. (d) Maximum emf induced in a circular coil having N turns, area A and angular speed ω is given by

$$E_{\max} = NBA\omega$$

$\therefore$ Maximum current induced in the coil is given by

$$i_{\max} = \frac{E_{\max}}{R}$$

$$i_{\max} = \frac{NBA\omega}{R}$$

$$i_{\max} = \frac{1000 \times 2 \times 10^{-5} \times \pi(10^2) \times 2}{12.56}$$

$$i_{\max} = 1 \text{ A}$$

23. (d) Induced emf, $\varepsilon = \frac{1}{2} B\omega r^2$

where, B : Magnetic field

ω : Angular speed

r : Radius

$$\varepsilon = \frac{1}{2} \times (0.4 \times 10^{-4}) \times \left(2\pi \left[\frac{120}{60} \right] \right) (1)^2$$

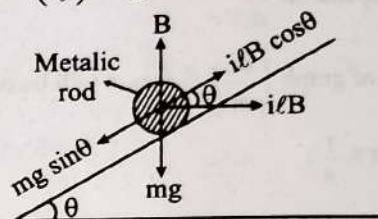
$$\varepsilon = 0.8\pi \times 10^{-4}$$

$$\varepsilon = 2.512 \times 10^{-4} \text{ volt}$$

24. (d) To remain at rest

$$mg \sin 30^\circ = i\ell B \cos 30^\circ$$

$$i = \frac{mg \sin 30^\circ}{\ell B \cos 30^\circ} = \left(\frac{m}{\ell} \right) \frac{g \tan 30^\circ}{B}$$



$$i = \frac{\frac{1}{2} \times 9.8 \times \frac{1}{\sqrt{3}}}{\frac{1}{4}} = 11.32 \text{ A}$$

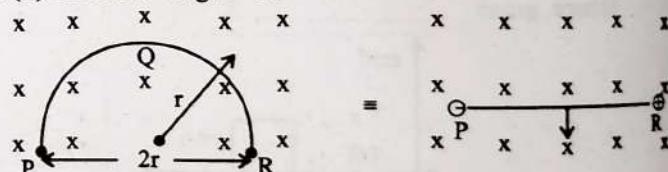
25. (b) For Loop 1

$$\varepsilon_{\text{ind}} = -\frac{d\phi}{dt} = -A \left(\frac{dB}{dt} \right) \cos 0^\circ = -\pi r^2 \left(\frac{dB}{dt} \right)$$

$A = \pi r^2$ as B is restricted within radius r.

For Loop 2, $\varepsilon_{\text{ind}} = 0$ as no flux linkage.

26. (d) Effective length = $2r$



$$\text{Induced emf} = Bv(2r) = 2rBv$$

where R is at higher potential and P is at lower potential.

27. (a) Induced emf $\varepsilon = -\frac{d\phi}{dt}$

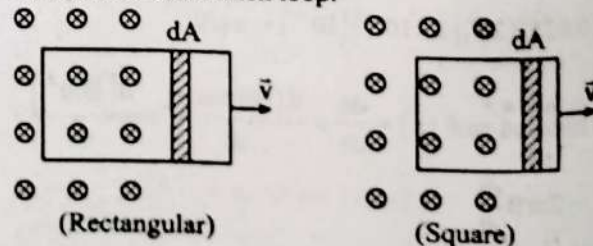
where flux $\phi = BA$

$$\therefore \varepsilon = -\frac{d}{dt}(BA)$$

$\therefore$ Magnetic field (B) is constant and area (A) is varying.

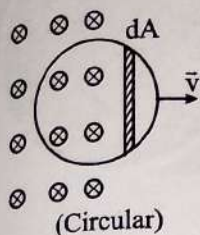
$$\therefore \varepsilon = -B \left(\frac{dA}{dt} \right)$$

Now, let consider each loop:



$\therefore \frac{dA}{dt}$ is constant.

So, emf is constant.

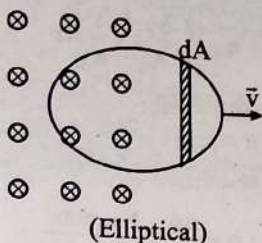


$\therefore \frac{dA}{dt}$ is variable.

so emf is not constant.

$\therefore \frac{dA}{dt}$ is constant.

$\therefore$ emf is constant.



$\therefore \frac{dA}{dt}$ is variable.

So, emf is not constant.

28. (d) Length of conductor (l) = 0.4 m;

Speed = 7 m/s and magnetic field $B = 0.9 \text{ Wb/m}^2$

Induced e.m.f. (ε) = $B/v = 0.9 \times 0.4 \times 7 = 2.52 \text{ V}$

29. (c) $\therefore I_0 = \frac{E_0}{R} = N \frac{BA\omega}{R}$

Given, $N = 1$, $B = 10^{-2} \text{ T}$,

$A = \pi (0.3)^2 \text{ m}^2$, $R = \pi^2 \Omega$

$f = (200/60)$ and $\omega = 2\pi (200/60)$

Substituting these values and solving, we get

$$I_0 = (1) \frac{10^{-2} \times \pi (0.3)^2 \times 2\pi \left(\frac{200}{60}\right)}{\pi^2}$$

$$I_0 = 10^{-2} \times (0.3)^2 \times 2\pi \left(\frac{200}{60}\right)$$

$$I_0 = 6 \times 10^{-3} \text{ A} = 6 \text{ mA}$$

30. (d)

- Magnetic brakes have a disc which rotates perpendicular to the magnetic field. As disc starts rotate, Eddy current are introduced.
- In an induction furnace, a coil carrying AC surrounds the chamber of metal and Eddy currents are introduced in the metal.
- Electric heater does not involve Eddy currents. It uses Joule's heating effect. Heater takes all the energy and supply it in the form of heat.

31. (a) When a metal is kept in a varying magnetic field, Eddy currents are produced. This is also called Foucault's current. It has 100% efficiency.

32. (d) Number of turn = 1000

Net magnetic flux = $N\phi = 4 \times 10^{-3} \times 10^3 = 4 \text{ Wb}$

Current = 4A

$\therefore \phi = LI$

$4 = L \times 4$

$L = 1 \text{ H}$

33. (d) Flux linked with each turn of Solenoid

$$= 4 \times 10^{-3} \text{ Wb}$$

Total flux linked with Solenoid

$$= 500 \times 4 \times 10^{-3} = 2 \text{ Wb}$$

As we know,

$$\phi = Li \Rightarrow 2 = 2 \times L$$

$$\Rightarrow L = 1 \text{ H}$$

34. (c) Inductance, $L = \frac{N\phi}{i}$;

Where flux $\phi = BA$

$$\text{Magnetic field } B = \mu_0 ni = \frac{\mu_0 Ni}{l}$$

$$\text{So, } L = \frac{\mu_0 N^2}{l} A = \mu_0 n^2 Al$$

where n is for number of turns/unit length

$$\therefore L \propto n^2$$

35. (b) Induced emf $\varepsilon = -L \frac{di}{dt}$

$$\therefore L = \frac{-\varepsilon}{\frac{di}{dt}}$$

$$\therefore dt = 10^{-3} \text{ and } di = 2 - 3$$

$$\text{So, } L = \frac{-5 \times 10^{-3}}{(2-3)} \text{ H} = 5 \text{ mH}$$

36. (d) Self inductance of a solenoid = $\mu_0 n^2 Al$

Where n is the number of turns/length.

So self induction $\propto n^2$

So inductance becomes 4 times when n is doubled.

37. (a) Induced emf, $|\varepsilon| = L \frac{di}{dt} = L \frac{\Delta i}{\Delta t}$

Given that, $L = 40 \times 10^{-3} \text{ H}$,

$$\Delta i = 11 \text{ A} - 1 \text{ A} = 10 \text{ A}$$

$$\text{And } \Delta t = 4 \times 10^{-3} \text{ s}$$

$$\therefore |\varepsilon| = 40 \times 10^{-3} \times \left(\frac{10}{4 \times 10^{-3}} \right) = 100 \text{ V}$$

38. (d) Potential energy stored in inductor,

$$U = \frac{1}{2} LI^2$$

$$\therefore L = \frac{2U}{I^2} = \frac{2 \times 25 \times 10^{-3}}{(60 \times 10^{-3})^2} = 13.89 \text{ H}$$

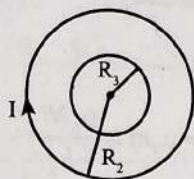
39. (b) Work done by source (W),

$$W = E \times q = E \left(\frac{\Delta \phi}{R} \right) = E \frac{LI_0}{R}$$

$$W = \left(\frac{E}{R} \right) LI_0 = (I_0) LI_0 = LI_0^2 = 0.04 \times (5)^2 = 1.0 \text{ J}$$

40. (c) Stored energy $E = \frac{1}{2} Li^2 = \frac{1}{2} (100 \times 10^{-3}) \times 1^2 = 0.05 \text{ J}$
41. (c) Electric current produces a strong magnetic field around the coil, this field flux equates to a storage of energy representing the kinetic motions of electrons through the coil. The more the current in the coil, the stronger the magnetic field will be, and the more energy the inductor will store.

42. (c)



For magnetic flux, $\phi = B.S$

$$\Rightarrow \phi = \left(\frac{\mu_0 I}{2R_1} \right) (\pi R_2^2)$$

and also, $\phi = MI$

Where, M is mutual inductance between the two loops.

$$\Rightarrow M = \frac{\phi}{I} \Rightarrow M = \frac{\left(\frac{\mu_0 I}{2R_1} \right) (\pi R_2^2)}{I}$$

$$\Rightarrow M = \left(\frac{\mu_0 \pi}{2} \right) \left(\frac{R_2^2}{R_1} \right) \Rightarrow M \propto \frac{R_2^2}{R_1}$$

43. (b) Given, $N = 100$

$$n = 2 \times 10^4$$

$$r = 10^{-2}$$

$$R = 10\pi^2$$

$$i = 4$$

$$\therefore q = \left[\left(\frac{\Delta\phi}{\Delta t} \right) \cdot \frac{1}{R} \right] \Delta t$$

$$\Delta\phi = NBA$$

$$B = \mu_0 ni$$

$$A = \pi r^2$$

$$\therefore q = \frac{\Delta\phi}{R} = \frac{N\mu_0 ni\pi r^2}{R}$$

$$q = \left[4\pi \times 10^{-7} \times 2 \times 10^4 \times 10^2 \pi \times (10^{-2})^2 \times 4 \right] \frac{1}{10\pi^2}$$

$$q = 32 \mu\text{C}$$

44. (c) As effective flux through one coil is completely linked with the other, $k = 1$.

$$\text{Mutual inductance } (M) = k\sqrt{L_1 L_2}$$

where, k = coupling constant.

$$\Rightarrow M = 1\sqrt{2 \times 8} \Rightarrow M = 4 \text{ mH}$$

45. (d) $\text{emf}, \varepsilon = M \frac{di}{dt} = 0.005 \times \frac{d}{dt} (i_0 \sin \omega t)$

where m is mutual inductance.

$$\varepsilon = 0.005 i_0 \omega \cos \omega t = \varepsilon_0 \cos \omega t$$

$$\therefore \varepsilon_{\max} = 0.005 \times 2 \times 100\pi = \pi \text{ volt}$$

Maximum value of $\text{emf} = \pi \text{ volt}$.

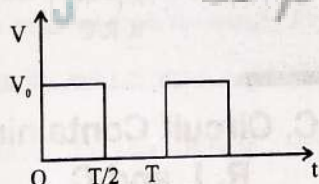
Alternating Current

Mean Value and Virtual Value of A.C.

1. The peak voltage of the ac source is equal to (2022)

- $\frac{1}{\sqrt{2}}$ times the rms value of the source
- the value of voltage supplied to the circuit
- the rms value of the ac source
- $\sqrt{2}$ times the rms value of the ac source

2. The r.m.s. value of potential difference V shown in the figure is: (2011 Mains)



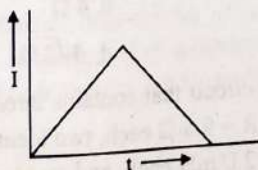
- V_0
- $V_0/\sqrt{2}$
- $V_0/2$
- $V_0/\sqrt{3}$

3. In an A.C. circuit, I_{rms} and I_0 are related as (1994)

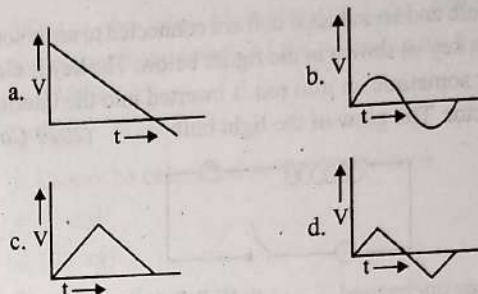
- $I_{rms} = \pi I_0$
- $I_{rms} = \sqrt{2} I_0$
- $I_{rms} = I_0/\pi$
- $I_{rms} = I_0/\sqrt{2}$

Self Inductance

4. The current I in an A.C. circuit inductance coil varies with time according to the graph given below.



Which one of the following graphs gives the variation of voltage with time? (1994)



A.C. Circuit Containing Capacitance Only

5. A capacitor of capacitance ' C ', is connected across an ac source of voltage V , given by $V = V_0 \sin \omega t$. The displacement current between the plates of the capacitor, would then be given by: (2021)

- $I_d = \frac{V_0}{\omega C} \cos \omega t$
- $I_d = \frac{V_0}{\omega C} \sin \omega t$
- $I_d = V_0 \omega C \sin \omega t$
- $I_d = V_0 \omega C \cos \omega t$

6. A $40 \mu\text{F}$ capacitor is connected to a 200V , 50Hz ac supply. The rms value of the current in the circuit is, nearly: (2020)

- 2.05 A
- 2.5 A
- 25.1 A
- 1.7 A

7. A small signal voltage $V(t) = V_0 \sin \omega t$ is applied across an ideal capacitor C : (2016 - I)

- Current $I(t)$ lags voltage $V(t)$ by 90°
- Over a full cycle the capacitor C does not consume any energy from the voltage source
- Current $I(t)$ is in phase with voltage $V(t)$
- Current $I(t)$ leads voltage $V(t)$ by 180°

8. In an AC circuit an alternating voltage $E = 200\sqrt{2} \sin 100t$ volts is connected to a capacitor of capacity $1 \mu\text{F}$. The r.m.s value of the current in the circuit is: (2011 Pre)

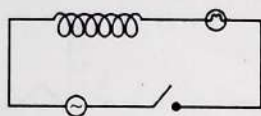
- 10 mA
- 100 mA
- 200 mA
- 20 mA

9. A capacitor of capacity C and reactance X if capacitance and frequency become double then reactance will be: (2001)

a. $4X$ b. $\frac{X}{2}$
c. $\frac{X}{4}$ d. $2X$

A.C. Circuit Containing R and L Only

10. A light bulb and an inductor coil are connected to an ac source through a key as shown in the figure below. The key is closed and after sometime an iron rod is inserted into the interior of the inductor. The glow of the light bulb (2020-Covid)



- a. Remains unchanged b. Will fluctuate
c. Increases d. Decreases
11. A resistance R draws power P when connected to an AC source. If an inductance is now placed in series with the resistance, such that the impedance of the circuit becomes Z , the power drawn will be: (2015)

a. $P\sqrt{\frac{R}{Z}}$ b. $P\left(\frac{R}{Z}\right)$
c. P d. $P\left(\frac{R}{Z}\right)^2$

12. A coil of self-inductance L is connected in series with a bulb B and an AC source. Brightness of the bulb decreases when: (2013)

a. An iron rod is inserted in the coil
b. Frequency of the AC source is decreased
c. Number of turns in the coil is reduced
d. A capacitance of reactance $X_C = X_L$ is included in the same circuit

13. A coil has resistance 30 ohm and inductive reactance 20 ohm at 50 Hz frequency. If an AC source, of 200 volt, 100 Hz, is connected across the coil, the current in the coil will be: (2011 Mains)

a. 4.0 A b. 8.0 A
c. $\frac{20}{\sqrt{13}}$ A d. 2.0 A

14. An AC voltage is applied to a resistance R and an inductor L in series. If R and the inductive reactance are both equal to $3\ \Omega$, the phase difference between the applied voltage and the current in the circuit is: (2011 Pre)

a. $\pi/6$ b. $\pi/4$
c. $\pi/2$ d. Zero

A.C. Circuit Containing R and C Only

15. A series RC circuit is connected to an alternating voltage source. Consider two situations

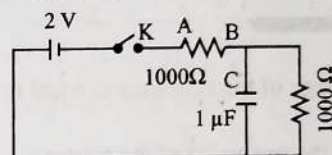
A. When capacitor is air filled.

B. When capacitor is mica filled.

Current through resistor is i and voltage across capacitor is V then: (2015 Re)

a. $V_a = V_b$ b. $V_a < V_b$
c. $V_a > V_b$ d. $I_a > I_b$

16. When the key K is pressed at time $t = 0$, then which of the following statement about the current I in the resistor AB of the given circuit is true? (1995)



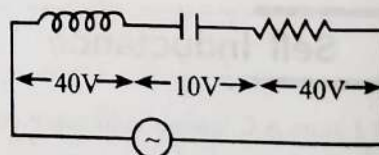
a. I Oscillates between 1 mA and 2 mA
b. at $t = 0$, $I = 2$ mA and with time it goes to 1 mA
c. $I = 1$ mA at all t
d. $I = 2$ mA at all t

17. The time constant of C-R circuit is (1992)

a. $1/CR$ b. C/R
c. CR d. R/C

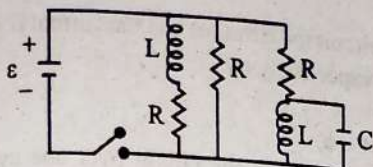
A.C. Circuit Containing R, L and C

18. An inductor of inductance L , a capacitor of capacitance C and a resistor of resistance ' R ' are connected in series to an ac source of potential difference ' V ' volts as shown in figure. Potential difference across L , C and R is 40V, 10V and 40V, respectively. The amplitude of current flowing through LCR series circuit is $10\sqrt{2}$ A. The impedance of the circuit is: (2021)



a. $5/\sqrt{2}\ \Omega$ b. $4\ \Omega$
c. $5\ \Omega$ d. $4\sqrt{2}\ \Omega$

19. Figure shows a circuit that contains three identical resistors with resistance $R = 9.0\ \Omega$ each, two identical inductors with inductance $L = 2.0$ mH each, and an ideal battery with emf $\mathcal{E} = 18$ V. The current ' i ' through the battery just after the switch closed is: (2017-Delhi)



- a. 0.2 A
c. 0 ampere

- b. 2 A
d. 2 mA

20. In a circuit L, C and R are connected in series with an alternating voltage source of frequency f . The current leads the voltage by 45° . The value of C is: (2005)

- a. $\frac{1}{2\pi f(2\pi fL + R)}$
c. $\frac{1}{2\pi f(2\pi fL - R)}$

- b. $\frac{1}{\pi f(2\pi fL + R)}$
d. $\frac{1}{\pi f(2\pi fL - R)}$

Electric Resonance

21. A series LCR circuit with inductance 10 H, capacitance $10 \mu\text{F}$, resistance 50Ω is connected to an ac source of voltage, $V = 200 \sin(100t)$ volt. If resonant frequency of the LCR circuit is ν_0 and the frequency of the ac source is ν , then: (2022)

- a. $\nu = 100\text{Hz}; \nu_0 = \frac{100}{\pi}\text{Hz}$ b. $\nu_0 = \nu = 50\text{Hz}$

- c. $\nu_0 = \nu = \frac{50}{\pi}\text{Hz}$ d. $\nu_0 = \frac{50}{\pi}\text{Hz}, \nu = 50\text{Hz}$

22. A series LCR circuit is connected to an ac voltage source. When L is removed from the circuit, the phase difference between current and voltage is $\frac{\pi}{3}$. If instead C is removed

from the circuit, the phase difference is again $\frac{\pi}{3}$ between current and voltage. The power factor of the circuit is: (2020)

- a. 0.5
c. -1.0

- b. 1.0
d. Zero

23. Which of the following combinations should be selected for better tuning of an L-C-R circuit used for communication? (2016 - II)

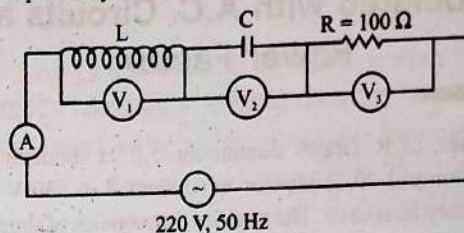
- a. $R = 15 \Omega, L = 3.5 \text{ H}, C = 30 \mu\text{F}$
b. $R = 25 \Omega, L = 1.5 \text{ H}, C = 45 \mu\text{F}$
c. $R = 20 \Omega, L = 1.5 \text{ H}, C = 35 \mu\text{F}$
d. $R = 25 \Omega, L = 2.5 \text{ H}, C = 45 \mu\text{F}$

24. In an electrical circuit R, L, C and an AC voltage source are all connected in series. When L is removed from the circuit, the phase difference between the voltage the current in the circuit is $\pi/3$. If instead, C is removed from the circuit, the phase difference is again $\pi/3$. The power factor of the circuit is: (2012 Pre)

- a. $1/2$
c. 1

- b. $\frac{1}{\sqrt{2}}$
d. $3/2$

25. In the given circuit the reading of voltmeter V_1 and V_2 are 300 volts each. The reading of the voltmeter V_3 and ammeter A are respectively: (2010 Pre)



- a. 100 V, 2.0 A
c. 220 V, 2.2 A

- b. 150 V, 2.2 A
d. 220 V, 2.0 A

26. What is the value of inductance L for which the current is maximum in a series LCR circuit with $C = 10 \mu\text{F}$ and $\omega = 1000 \text{ s}^{-1}$: (2007)

- a. 1 mH
b. Cannot be calculated unless R is known
c. 10 mH
d. 100 mH

27. The value of quality factor is: (2000)

- a. $\frac{\omega L}{R}$
c. $\sqrt{LC}$

- b. $\frac{\omega}{RC}$
d. L/R

28. An series L-C-R circuit is connected to a source of A.C. current. At resonance, the phase difference between the applied voltage and the current in the circuit, is: (1994)

- a. π
c. $\pi/4$

- b. Zero
d. $\pi/2$

Energy Stored in Condenser, Inductor and LC Oscillations

29. A condenser of capacity C is charged to a potential difference of V_1 . The plates of the condenser are then connected to an ideal inductor of inductance L. The current through the inductor when the potential difference across the condenser reduces to V_2 is: (2010 Mains)

a. $\left(\frac{C(V_1 - V_2)^2}{L} \right)^{1/2}$

b. $\frac{C(V_1^2 - V_2^2)}{L}$

c. $\frac{C(V_1^2 + V_2^2)}{L}$

d. $\left[\frac{C(V_1^2 - V_2^2)}{L} \right]^{1/2}$

30. A transistor-oscillator using a resonant circuit with an inductor L (of negligible resistance) and a capacitor C in series produce oscillations of frequency f . If L is doubled and C is changed to $4C$, the frequency will be: (2006)

- a. $f/4$
c. $f/2\sqrt{2}$

- b. $8f$
d. $2f$

Average Power and Power Associated with A.C. Circuits and Power Factor

31. A series LCR circuit containing 5.0 H inductor, 80 μF capacitor and 40 Ω resistor is connected to 230 V variable frequency ac source. The angular frequencies of the source at which power transferred to the circuit is half the power at the resonant angular frequency are likely to be: (2021)

a. 50 rad/s and 25 rad/s
b. 46 rad/s and 54 rad/s
c. 42 rad/s and 58 rad/s
d. 25 rad/s and 75 rad/s

32. An inductor 20mH, a capacitor 100 μF and a resistor 50 Ω are connected in series across a source of emf, $V = 10 \sin 314 t$. The power loss in the circuit is (2018)

a. 2.74 W b. 0.43 W
c. 0.79 W d. 1.13 W

33. The potential differences across the resistance, capacitance and inductance are 80 V, 40 V and 100 V respectively in an LCR circuit. The power factor of this circuit is: (2016 - II)

a. 0.8 b. 1.0
c. 0.4 d. 0.5

34. An inductor 20 mH, a capacitor 50 μF and a resistor 40 Ω are connected in series across a source of emf $V = 10 \sin 340t$. The power loss in A.C. circuit is: (2016 - I)

a. 0.51 W b. 0.67 W
c. 0.76 W d. 0.89 W

35. The instantaneous values of alternating current and voltages in a circuit are given as

$$i = \frac{1}{\sqrt{2}} \sin(100\pi t) \text{ ampere}$$

$$e = \frac{1}{\sqrt{2}} \sin\left(100\pi t + \frac{\pi}{3}\right)$$

The average power in watts consumed in the circuit is:

(2012 Mains)

a. 1/4 b. $\frac{\sqrt{3}}{4}$
c. 1/2 d. 1/8

36. Power dissipated in an LCR series circuit connected to an AC source of emf ϵ is: (2009)

a. $\frac{\epsilon^2 \sqrt{R^2 + \left(L\omega - \frac{1}{C\omega}\right)^2}}{R}$

b. $\frac{\epsilon^2 \left[R^2 + \left(L\omega - \frac{1}{C\omega}\right)^2 \right]}{R}$

c. $\frac{\epsilon^2 R}{\sqrt{R^2 + \left(L\omega - \frac{1}{C\omega}\right)^2}}$

d. $\frac{\epsilon^2 R}{\left[R^2 + \left(L\omega - \frac{1}{C\omega}\right)^2 \right]}$

37. In an AC circuit the e.m.f (e) and the current (i) at any instant are given respectively by

$$e = E_0 \sin \omega t$$

$$i = I_0 \sin(\omega t - \phi)$$

The average power in the circuit over one cycle of a.c. is: (2008)

a. $E_0 I_0$ b. $\frac{E_0 I_0}{2}$
c. $\frac{E_0 I_0}{2} \sin \phi$ d. $\frac{E_0 I_0}{2} \cos \phi$

38. A coil of inductive reactance 31 Ω has a resistance of 8 Ω . It is placed in series with a condenser of 25 Ω . The combination is connected to an AC source of 110 V. The power factor of the circuit is: (2006)

a. 0.40 b. 0.128
c. 0.80 d. 0.66

39. For a series LCR circuit the power loss at resonance is: (2002)

a. $\frac{V^2}{\omega L - \frac{1}{\omega C}}$ b. $I^2 L \omega$
c. $I^2 R$ d. $\frac{V^2}{C \omega}$

40. In an a.c. circuit with phase voltage V and current I, the power dissipated is (1997)

a. V.I
b. Depends on phase angle between V and I

c. $\frac{1}{2} \times V.I$
d. $\frac{1}{\sqrt{2}} \times V.I$

41. In an A.C. circuit, the current flowing is $I = 5 \sin(100t - \pi/2)$ ampere and the potential difference is $V = 200 \sin(100t)$ volts. The power consumption is equal to (1995)

a. 20 W b. 0 W
c. 1000 W d. 40 W

42. Two cables of copper are of equal lengths. One of them has a single wire of area of cross-section A, while other has 10 wires of cross-sectional area A/10 each. Give their suitability for transporting A.C. and D.C. (1994)

a. Only multiple strands for A.C., either for D.C.
b. Only multiple strands for A.C., only single strand for D.C.
c. Only single strand for D.C., either for A.C.
d. Only single strand for A.C., either for D.C.

Transformer

43. A step down transformer connected to an ac mains supply of 220 V is made to operate at 11 V, 44 W lamp. Ignoring power losses in the transformer, what is the current in the primary circuit? (2021)

a. 0.4 A b. 2 A
c. 4 A d. 0.2 A

44. A transformer having efficiency of 90% is working on 200 V and 3 kW power supply. If the current in the secondary coil is 6 A, the voltage across the secondary coil and the current in the primary coil respectively are: (2014)
- a. 300 V, 15 A
b. 450 V, 15 A
c. 450 V, 13.5 A
d. 600 V, 15 A
45. A 220 volt input is supplied to a transformer. The output circuit draws a current of 2.0 ampere at 440 volts. If the efficiency of the transformer is 80%, the current drawn by the primary windings of the transformer is: (2010 Pre)
- a. 5.0 ampere
b. 3.6 ampere
c. 2.8 ampere
d. 2.5 ampere
46. A transformer is used to light a 100 W and 110 V lamp from a 220 V mains. If the main current is 0.5 amp, the efficiency of the transformer is approximately: (2007)
- a. 50%
b. 90%
c. 10%
d. 30%
47. The primary and secondary coil of a transformer have 50 and 1500 turns respectively. If the magnetic flux ϕ linked with the primary coil is given by $\phi = \phi_0 + 4t$, where ϕ is in Webers, t is time in seconds and ϕ_0 is a constant, the output voltage across the secondary coil is: (2007)
- a. 120 volts
b. 220 volts
c. 30 volts
d. 90 volts
48. The core of a transformer is laminated because: (2006)
- a. Energy losses due to Eddy currents may be minimised
b. The weight of the transformer may be reduced
c. Rusting of the core may be prevented
d. Ratio of voltage in primary and secondary may be increased
49. Turn ratio of a step-up transformer is 1 : 25. If current in load coil is 2 A, then the current in primary coil will be: (1998)
- a. 25 A
b. 50 A
c. 0.25 A
d. 0.5 A
50. The primary winding of a transformer has 500 turns whereas its secondary has 5000 turns. The primary is connected to an A.C. supply 20 V, 50 Hz. The secondary will have an output of (1997)
- a. 2 V, 50 Hz
b. 2 V, 5 Hz
c. 200 V, 50 Hz
d. 200 V, 500 Hz

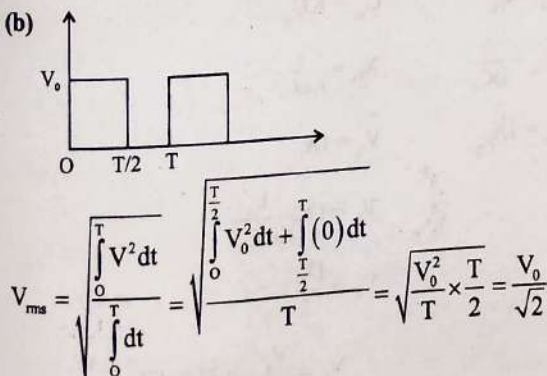
Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
d	b	d	a	d	b	b	d	c	d	d	a	a	b	c	b	c
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
c	None	a	c	b	a	c	c	d	a	b	d	c	b	c	a	a
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	
d	d	d	c	c	b	b	a	d	b	a	b	a	a	b	c	

Explanations

1. (d) Peak or maximum is $\sqrt{2}$ times the rms voltages in ac.

2. (b)



3. (d) $I_{rms} = \frac{I_0}{\sqrt{2}}$

4. (a) In an Alternating Current circuit with inductance coil, the voltage V leads the current I by a phase difference of 90° . Or the current lags behind the voltage V by a phase difference of 90° . Thus the voltage goes on decreasing with the increase in time as shown in the graph (a).

5. (d) Charge stored by capacitor is given by, $q = CV$

Differentiating both sides w.r.t time,

$$\Rightarrow \frac{dq}{dt} = C \frac{dV}{dt} \Rightarrow I_d = C \frac{d}{dt}(V)$$

$$\Rightarrow I_d = C \frac{d}{dt}(V_0 \sin \omega t) \quad [\text{Given, } V = V_0 \sin \omega t]$$

$$\Rightarrow I_d = CV_0 \omega \cos \omega t \Rightarrow I_d = V_0 \omega C \cos \omega t$$

$$6. (b) I = \frac{V}{X_C} = \frac{V}{1/C\omega}$$

$$= 200 \times 40 \times 10^{-6} \times 2\pi \times 50$$

$$= 2.5 \text{ A}$$

$$7. (b) \text{ Power} = V_{\text{rms}} \times I_{\text{rms}} \times \cos\phi$$

As for the circuit containing capacitor only power factor,
 $\cos\phi = 0^\circ$, since $\phi = 90^\circ$

Hence power consumed in one complete cycle is zero.

$$8. (d) E = 200\sqrt{2} \sin 100t \Rightarrow E = E_0 \sin \omega t$$

Comparing these, $E_0 = 200\sqrt{2}$, $\omega = 100 \text{ rads}^{-1}$

$$X_C = \frac{1}{\omega C} = \frac{1}{100 \times 1 \times 10^{-6}} = 10^4 \Omega$$

The r.m.s. value of current in circuit is

$$i_{\text{rms}} = \frac{V_{\text{rms}}}{X_C} = \frac{E_0/\sqrt{2}}{10^4} = \frac{200}{10^4} = 0.02 \text{ A} = 20 \text{ mA}$$

$$9. (c) X_C = \frac{1}{C\omega} = \frac{1}{C \times 2\pi f}$$

When both capacitance and frequency are doubled, then new capacitive reactance,

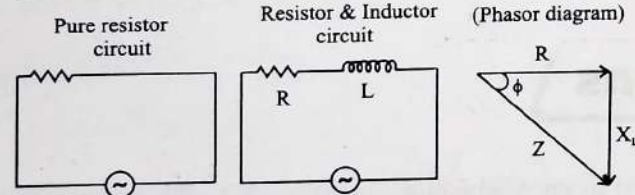
$$X_C' = \frac{1}{C'\omega'} = \frac{1}{2C \times 2\pi \times 2f} = \frac{1}{4 \times C \times 2\pi f} = \frac{X_C}{4}$$

$$10. (d) \text{ Impedance } (Z) = \sqrt{R^2 + X_L^2}$$

Current, $I \propto \frac{1}{Z} \propto \text{glow of bulb}$

thus $X_L \uparrow$, $Z \uparrow$, $I \downarrow$, glow of bulb decreases

11. (d)



AC source

Impedance = Z

$$P' = V.I.\cos\phi$$

$$P = \frac{V^2}{R}$$

$$P' = V \cdot \left[\frac{V}{Z} \right] \cdot \cos\phi$$

$$\Rightarrow V^2 = PR$$

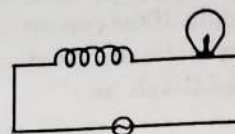
$$P' = \frac{V^2}{Z} \cdot \frac{R}{Z}$$

(from phasor diagram)

$$P' = \frac{(PR)R}{Z^2}$$

$$P' = \left(\frac{R}{Z} \right)^2 P$$

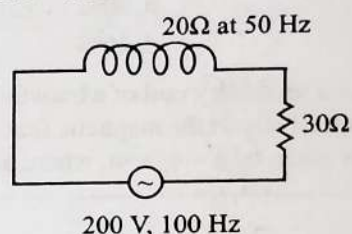
12. (a)



Brightness of the bulb

- Decreases when an iron rod is inserted in the coil as impedance of circuit increases.
- Increases when frequency of the AC source is decreased as impedance of circuit decreases.
- Increases when number of turns in the coil is reduced as impedance of circuit decreases.
- Increases when a capacitance of reactance $X_C = X_L$ is included in the circuit as impedance of circuit decreases.

$$13. (a) X_L \text{ at } 50\text{Hz} = 20 \Omega$$



As we know $X_L \propto \omega$ [$X_L = \omega L$]

$$\frac{X_L}{X'_L} = \frac{\omega}{\omega'} \Rightarrow \frac{20}{X'_L} = \frac{50}{100} \Rightarrow X'_L = 40 \Omega$$

$$\text{So, impedance} = Z = \sqrt{(X'_L)^2 + (R)^2} = \sqrt{(40)^2 + (30)^2}$$

$$= 50 \Omega$$

$$\text{Current in circuit (i)} = \frac{V}{Z} = \frac{200}{50} = 4 \text{ A}$$

$$14. (b) \text{ Here, } R = 3 \Omega, X_L = 3 \Omega$$

Phase difference between applied voltage and the current in the circuit, $\tan\phi = \frac{X_L}{R} = 1 \Rightarrow \phi = \tan^{-1}(1) \Rightarrow \phi = \pi/4$

15. (c)

Air

$$C_a = C_0$$

$$X_{C_a} = \frac{1}{\omega C_0}$$

$$V_a = IX_{C_a}$$

Mica

$$C_m = KC_0$$

$$X_{C_m} = \frac{1}{K\omega C_0}$$

$$V_b = IX_{C_m}$$

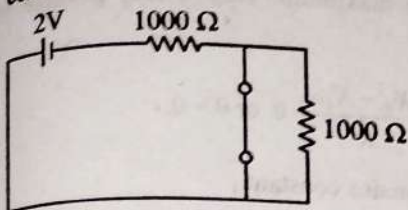
$$V_b = I \times \frac{1}{K\omega C_0}$$

$$V_b = \frac{IX_{C_a}}{K}$$

$$\therefore V_b = \frac{V_a}{K} \Rightarrow V_a = KV_b$$

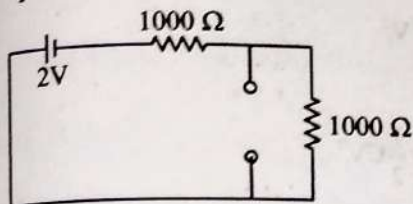
$$V_a > V_b [\because K > 1]$$

16. (b) At, $t = 0$, capacitor behaves like short circuit. Under that condition.



$$i(0) = \frac{2}{1000} = 2\text{mA}$$

Under steady state condition ($t = \infty$), capacitor behaves like open circuit,



$$i(\infty) = \frac{2}{1000 + 1000} = 1\text{mA}$$

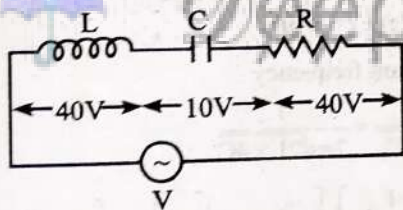
Hence, option (b) is the correct answer.

17. (c) For R and C circuit, the time constant, $\tau = CR$
Growth of charge in a circuit containing capacitance and resistance is given by the formula,

$$q = q_0(1 - e^{-t/CR})$$

CR is known as time constant in this formula.

18. (c)



Given, $V_L = 40\text{ V}$, $V_C = 10\text{ V}$, $V_R = 40\text{ V}$, $I_0 = 10\sqrt{2}\text{ A}$

$$I_{\text{RMS}} = \frac{I_0}{\sqrt{2}} = \frac{10\sqrt{2}}{\sqrt{2}} = 10\text{A}$$

$$I_{\text{RMS}} = 10\text{A}$$

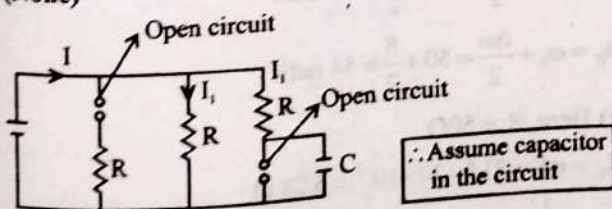
$$V_{\text{RMS}} = \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{(40)^2 + (40 - 10)^2}$$

$$= \sqrt{1600 + 900} = \sqrt{2500}$$

$$\therefore V_{\text{RMS}} = 50\text{ V}$$

$$\therefore V_{\text{RMS}} = I_{\text{RMS}} \times Z \Rightarrow Z = \frac{V_{\text{RMS}}}{I_{\text{RMS}}} = \frac{50}{10} = 5\Omega$$

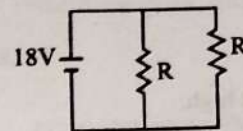
19. (None)



At $t = 0$ just after the switch closed.

Inductor opposes flow of current hence branch containing inductor do not allow current pass through it.

While capacitor allow flow of current pass through



Note: If capacitor is not considered, current (I_1) through

$$= 0 \text{ then } I = \frac{18}{9} = 2\text{ A}$$

If capacitor is considered then,

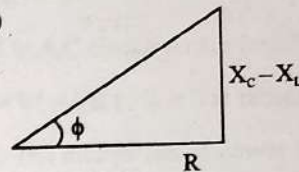
$$R_{\text{eq}} = \frac{R}{2} = \frac{9}{2}\Omega, \quad \varepsilon = 18\text{ V}$$

$$I = \frac{\varepsilon}{R} = \frac{18}{9} \times 2$$

$$\therefore I = 4\text{ A}$$

None of the given options are correct.

20. (a)



$$\Rightarrow \tan \phi = \frac{X_C - X_L}{R} \Rightarrow \tan\left(\frac{\pi}{4}\right) = \frac{\frac{1}{\omega C} - L\omega}{R}$$

$$\Rightarrow R = \frac{1}{\omega C} - L\omega \Rightarrow (R + 2\pi fL) = \frac{1}{2\pi fC}$$

$$\Rightarrow C = \frac{1}{2\pi f(R + 2\pi fL)}$$

21. (c) Given, $\omega = 100$

$$\therefore \omega = 2\pi v$$

$$\Rightarrow v = \frac{\omega}{2\pi} = \frac{50}{\pi}\text{ Hz}$$

Resonance frequency,

$$v = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{10 \times 10 \times 10^{-6}}} = \frac{50}{\pi}\text{ Hz}$$

22. (b) When L removed, $\tan \phi = \frac{X_C}{R}$

$$\text{When L removed, } \tan \phi = \frac{X_L}{R}$$

According to the question,

$$\frac{X_C}{R} = \frac{X_L}{R} \Rightarrow X_L = X_C$$

Resonance condition occurs.

$$\therefore \text{Power factor, } \cos \phi = \frac{R}{Z} = \frac{R}{R} = 1$$